

Compact Spectral Imaging: A Review of Miniaturized and Integrated Systems

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The push for portable and cost-effective spectral imaging has accelerated the transition from traditional bulky instruments to compact, integrated systems. This review outlines recent advancements in compact spectral imaging systems (SIS) and their applications in environmental monitoring, precision agriculture, and medical diagnostics. It examines four major design trends: Do-It-Yourself (DIY) approaches, freeform optics, integrated filter-on-chip technologies, and metasurface-based solutions. DIY systems have made spectral imaging more accessible by offering customizable, low-cost platforms for researchers and hobbyists. Freeform optics enhance miniaturization by merging multiple optical functions into a single, non-axisymmetric surface, thereby reducing size and complexity. Filter-on-chip designs, where spectral filters are fabricated directly on imaging detectors, further shrink device dimensions while retaining high spectral resolution and sensitivity. Metasurface-based designs leverage ultrathin, nanostructured materials to enable advanced functionalities, such as tunable spectral and polarization responses, in extremely compact formats. Together, these innovations not only fulfill practical requirements across diverse fields but also extend the potential of spectral imaging into new and challenging application domains. This review aims to provide an in-depth understanding of these trends, their applications, and the future direction of miniaturized and compact spectral imaging technology.

spatially resolved spectral acquisition across a broad range of wavelengths. Unlike traditional imaging systems, which record information in a limited number of broadband channels—typically red, green, and blue—SIS capture data across multiple narrow spectral bands, yielding rich spectral profiles for each spatial pixel. This process results in the generation of a three-dimensional datasets, commonly referred to as a “spectral data cubes”, comprising two spatial dimensions and one spectral dimension (Figure 1). The spectral signature derived from each pixel encapsulates the intrinsic optical properties of the underlying material, thereby facilitating a wide array of applications, including material identification, target classification, tissue diagnostics, quantitative environmental analysis, among others.^[1] Fundamentally, SIS can be broadly categorized into three modalities: panchromatic imaging (PI), multispectral imaging (MSI), and hyperspectral imaging (HSI).^[2,3] PI captures a broad spectrum in a single channel, producing a high-resolution grayscale image with limited spectral fidelity. In contrast,

MSI operates across a modest number of discrete spectral bands—typically 3 to 25—offering intermediate spectral resolution.^[3] HSI, on the other hand, extends this concept by acquiring data in hundreds of contiguous spectral channels, enabling precise discrimination of minute spectral variations and subtle material contrasts. Moreover, HSI systems employ different acquisition methods—namely: spatial scanning technique (whiskbroom or pushbroom approaches), spectral scanning technique (staring or tunable filter-based approaches), and snapshot acquisition technique, which allows simultaneous acquisition of the entire 3-D data cube within a single integration period.^[3,4]

Recently, SIS have transitioned from bulky laboratory setups to compact, portable, and miniaturized designs driven by advancements in sensor technology, miniaturization of optical components, improved computational algorithms, and increased demand in fields like agriculture, environmental monitoring, and healthcare. One key element of this transition has been the proliferation of DIY schemes that use commercially available components, such as monochrome cameras paired with off-the-shelf optics, filters, and LED arrays, democratizing the development of SIS and enabling rapid prototyping of application-specific solutions that would otherwise be costly or

1. Introduction

Spectral imaging systems are advanced optical platforms that synergize conventional imaging and spectroscopy to enable

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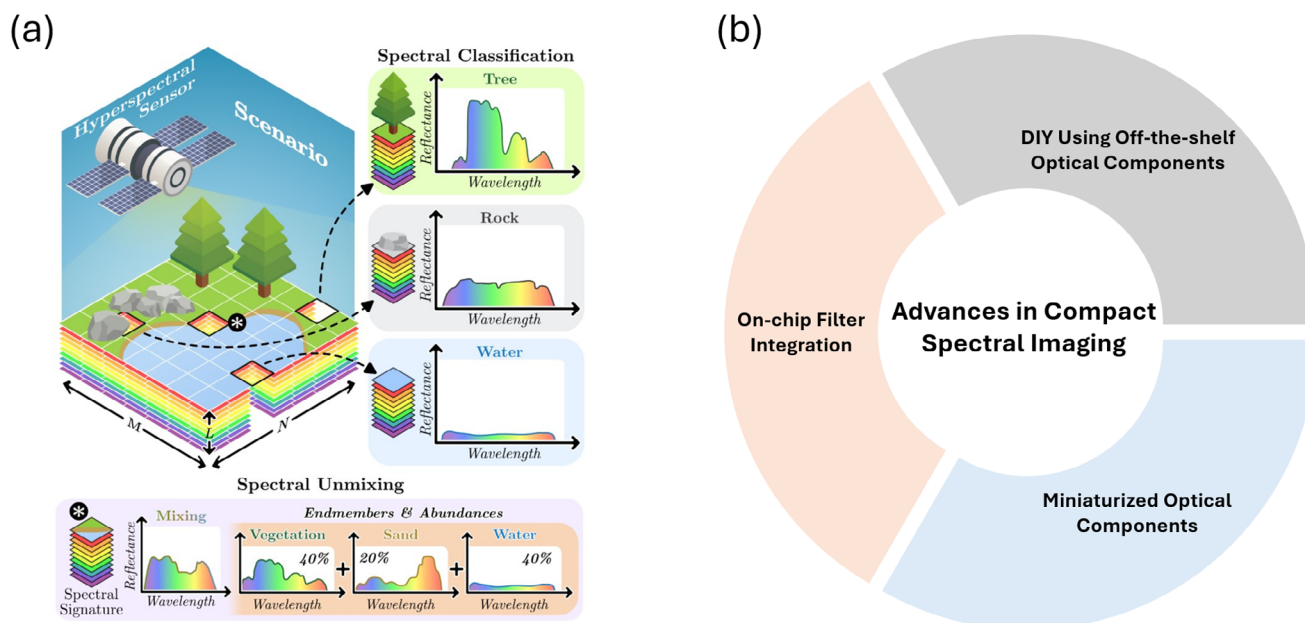


Figure 1. a) Illustration of spectral imaging where an airborne SIS generates 3D data cubes with each spectral signature designating materials such as trees, rocks, and water. Spectral unmixing is employed to analyze the mixture of different materials, where the observed spectral signature is represented as a linear combination of the spectral contributions of the individual constituent materials. Reproduced with permission.^[19] Copyright 2023, Optica Publishing Group. b) Summarized advances in compact and miniaturized SIS enabled by the progress in do-it-yourself schemes using the off-the-shelf optical components, miniaturization of optical components, and on-chip filter integration.

impractical with traditional commercial systems. Freeform optical elements—characterized by non-axisymmetric, highly aspheric geometries—have emerged as a pivotal innovation by integrating multiple optical functionalities into a single element, drastically reducing system size and alignment complexity while maintaining high optical performance.^[5]

The advent of filter-on-chip architecture represents another milestone, where spectral filters are monolithically integrated onto the sensor arrays. This integration eliminates the need for external dispersive elements or tunable filters, yielding compact systems with enhanced robustness and manufacturability. Additionally, metasurface-based optical components—comprising subwavelength nanostructures designed to manipulate phase, amplitude, and polarization of light—offer unprecedented opportunities for miniaturization and multifunctional integration in SIS design. These flat, ultrathin elements can perform complex optical operations such as beam steering, spectral splitting, and wavefront shaping, effectively replacing bulky refractive components with sub-micron alternatives. This technology paves the way for the development of even more compact and lightweight SIS.^[6] Simultaneously, the integration of computational imaging and machine learning algorithms has further catalyzed the miniaturization of SIS. Compressive sensing, sparsity-driven reconstruction, and deep learning enabled spectral demultiplexing allow for high-fidelity image reconstruction from significantly reduced measurement sets, relaxing hardware requirements without compromising spectral resolution.^[7] These advancements collectively enable the realization of SIS that are not only compact and energy-efficient but also capable of delivering high-throughput, real-time spectral analysis across diverse application domains such as precision agriculture, environ-

mental monitoring, biomedical diagnostics, and mobile sensing platforms.

Prior reviews on compact and miniaturized SIS have focused on specific technological domains. Several works emphasized computational spectral imaging, detailing reconstruction algorithms and architectures such as coded apertures, single-pixel imaging, and deep learning frameworks.^[7–10] Others centered on nano-photonics and metamaterials, exploring how engineered optical components like metasurfaces or photonic crystals reduce device size while maintaining or enhancing performance.^[11–13] Reviews on spectrometer classification analyzed architectures such as dispersive, filter-based, Fourier-transform, and computational systems, highlighting trade-offs in resolution and integration potential.^[14]

Additional studies targeted application-specific developments, particularly in LIDAR, biosensing, and biomedical imaging, often within hybrid or sensor-fusion frameworks.^[15–17] For instance, in 2020, Song et al. explored computational spectral imaging, emphasizing reconstruction algorithms that enable miniaturization, cost reduction, and rapid data acquisition.^[9] Their review covered grating-based, filter-based, coded aperture, and quantum dot spectrometers, highlighting the integration of computational methods with optical hardware for efficient, compact devices across diverse applications.

Similarly, Gao et al.^[14] and Yang et al.^[8] examined computational spectrometers, emphasizing nanophotonic hardware and machine learning-based reconstruction for miniaturization and integration into portable devices. Complementing these, Bacca et al.^[7] provided a detailed analysis of computational spectral imaging (CSI), covering compressive sensing, deep learning-based reconstruction, and end-to-end optical-computational system

design. They emphasized snapshot techniques—including coded aperture snapshot spectral imaging (CASSI), single-pixel imaging, and diffractive optical elements—as key enablers of reduced acquisition time and system complexity, with applications in spectral classification, unmixing, and anomaly detection directly from compressed data. Guan et al.^[10] provided a broad overview of miniaturized computational spectrometers, detailing diverse hardware platforms and reconstruction methods across biomedical and consumer applications. N. Li et al.^[17] offers a thorough examination of nano-photonics-based sensing systems, integrating SIS with spectral LIDAR technologies. Their review, oriented towards applications such as autonomous driving and biomedical imaging, emphasized LIDAR-centric approaches progressing toward chip-scale implementation. A. Li et al.^[16] reviewed the developments in miniaturized, chip-scale spectrometers designed for consumer and biomedical applications. Their work highlighted the growing demand for portable, low-cost spectrometers driven by the Internet of Things (IoT) and wearable devices. Recently, Tran and Fei^[18] concentrated on biomedical imaging, surveying compact SIS implementations and comparing device specifications for clinical use. Meanwhile, Hu et al.^[11] and Zou et al.^[12] reviewed metasurface-enabled imaging systems, emphasizing their multifunctionality, compactness, and integration with computational algorithms. Overall, these reviews provide valuable insights into algorithmic innovations, nanophotonic components, and application-specific implementations. However, these works mostly address SIS in isolation, with limited synthesis across optical design, fabrication, and system-level integration.

In contrast, our review offers a comprehensive, application-oriented synthesis of emerging compact and miniaturized SIS design technologies, with particular emphasis on integrated optical solutions tailored for environmental monitoring, precision agriculture, medical diagnostics, and remote sensing. Rather than focusing solely on isolated components or computational framework, this work highlights the convergence of key enabling technologies that collectively redefine SIS design:

- i. DIY platforms, built from commercial cameras, filters, and 3D-printed structures, enable low-cost, customizable solutions ideal for rapid prototyping and educational use.
- ii. Freeform optics, with their non-rotationally symmetric geometries, consolidate multiple imaging functions into a single, compact element, supporting lightweight implementations for drones, CubeSats, and handheld devices.
- iii. Filter-on-chip technologies embed spectral discrimination directly onto sensor arrays using nanofabricated multilayer structures or plasmonic materials, enabling ultra-compact, mechanically robust systems suitable for wearable or point-of-care diagnostics.
- iv. Metasurface-based optics—composed of engineered sub-wavelength nanostructures—offer unprecedented control over phase, dispersion, and polarization, making them well-suited for spectral splitting, beam shaping, and multifunctional integration within flat, planar devices.

By offering detailed analyses of performance metrics and practical trade-offs, this review addresses gaps in prior work, targeting

researchers, engineers, and practitioners seeking insights into current innovations and future trends in compact SIS designs.

2. Compact Spectral Imaging Applications

Recent advancements in photonic integration, sensor miniaturization, and computational imaging have paved the way for compact SIS to move beyond the laboratory and into real-world applications. Their lightweight and integrated designs make them particularly well-suited for deployment on platforms with strict size, weight, and power constraints, such as UAVs, CubeSats, and portable medical devices. These systems offer the ability to collect rich spectral and spatial data in a compact form factor, significantly broadening the scope and accessibility of compact SIS technologies. This section highlights key application areas where compact SIS has made substantial progress, with a focus on UAV and CubeSat platforms as well as emerging roles in medical diagnostics. In the context of remote sensing, CubeSats equipped with SIS enables efficient Earth observation for monitoring atmospheric phenomena, assessing water quality, and tracking land-use changes—all while offering lower-cost, rapid-deployment alternatives to traditional satellite missions. Similarly, UAV-based systems support high-resolution, real-time imaging for use in precision agriculture, environmental monitoring, surveillance, and reconnaissance. In the medical domain, compact SIS are increasingly being explored for diagnostic purposes, offering non-invasive assessments of tissue health, skin abnormalities, wound healing processes, and other applications.

2.1. Small Satellites and UAVs

Over the past decade, the development of high-performance, miniaturized, and compact SIS has been driven by the growing demand for lightweight instruments suitable for deployment on UAVs and small satellites such as CubeSats.^[20–24] CubeSats are small, standardized satellite platforms, typically with dimensions in multiples of $10 \times 10 \times 10$ cm units (1U, 2U, 3U, etc.) (Figure 2). These compact satellites are increasingly equipped with advanced instruments for various scientific and commercial missions, including miniaturized SIS, which have become cost-effective tools for remote sensing and Earth observation from space.^[20,21] CubeSats offers broader and more cost-effective capabilities and have gained rapid momentum as an attractive technology in the industry, scientific research, governmental, and educational settings due to their lighter weight, and faster development timeline compared with conventional satellites.^[25,26] For instance, compact SIS that are well-matched with the design specifications of small satellites have been demonstrated by VTT Technical Research Centre of Finland,^[27,28] followed by the development of a handheld version for terrestrial applications.^[29] These systems employ Fabry-Perot Interferometer (FPI) and are designed to capture 2D spatial images across multiple wavelength bands simultaneously. Their architecture offers a low-cost approach for compact SIS, making such systems more accessible for applications previously constrained by expensive instrumentation. However, despite detailed calibration protocols, the process remains complex and time-consuming, requiring precise

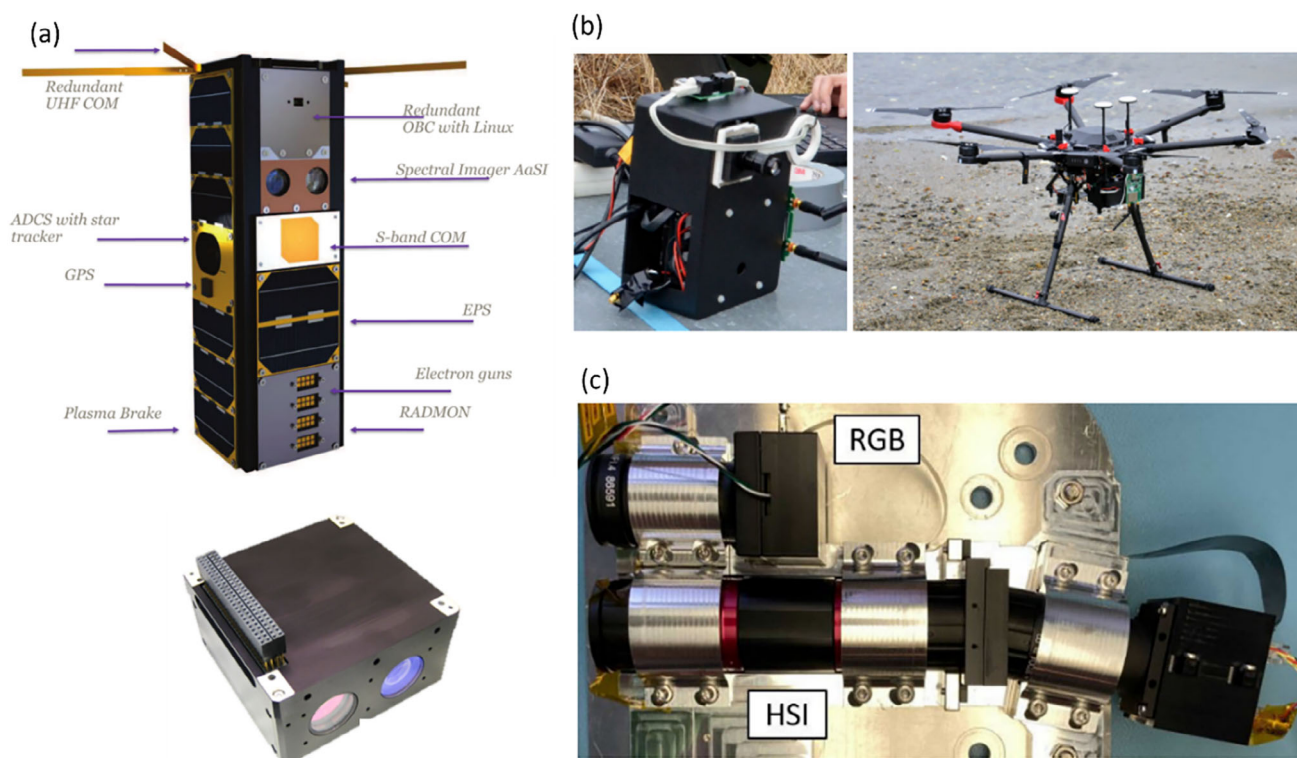


Figure 2. Illustrations of three different integrations of spectral imaging systems: a) The Aalto-1 3U CubeSat nanosatellite equipped with HSI payload and the photograph (below) of the PF filter-based spectral imager weighing less than 600 g. Reproduced with permission.^[39] Copyright 2018, IEEE. b) The modular payload of HSI systems for UAVs, including the spectral Imager and the imager mounted on a hexacopter. Reproduced with permission.^[51] Copyright 2018, IEEE. c) The optical payload unit of the HYPSON-1 CubeSat mission comprises a compact RGB camera positioned at the top and a hyperspectral imager located at the bottom. Reproduced with permission.^[52] Copyright 2018, IEEE.

alignment and tuning. Moreover, the spatial resolution of 480×750 pixels, while adequate for many applications, may fall short for advanced scenarios requiring high-resolution imagery.

In a similar vein, Mouroulis et al. developed an imaging spectrometer and telescope system integrated into a 6U CubeSat, featuring rapid acquisition capabilities and low polarization sensitivity.^[30] The system operates at an $f/1.8$ aperture with a 100 mm focal length, maximizing resolution within the CubeSat's limited volume without the need for deployable mirrors. Utilizing a Teledyne CHROMA detector, their system spans a spectral range of 350–1700 nm with a fine spectral sampling interval of 5.7 nm, a 10° field of view, and 600 cross-track pixels. This combination allows for wide-area coverage in a single pass, particularly advantageous for Earth observation and environmental monitoring. Their miniaturized design does not compromise performance; however, the absence of deployable mirrors requires highly precise pre-launch alignment, as any misalignment or damage during launch could significantly impair functionality. Additionally, the fast $f/1.8$ configuration demands meticulous focusing and stable alignment—factors that may be affected by environmental variability or CubeSat dynamics, potentially degrading image quality. Alternatively, Beasley et al. introduced the Compact Holographic Aberration-Corrected Platform (CHAP), developed by Planetary Resources Development Corporation.^[31] Designed for high-performance spectral imaging within CubeSat constraints, CHAP incorporates a disper-

sive element into the telescope's secondary mirror to achieve an extended focal length with moderate dispersion at the focal plane. Their system employs only two optical elements, enabling a simplified design capable of producing both white-light and diffracted images. A replicated grating further supports a cost-effective, spaceflight-compatible SIS. Avoiding transmissive optics and minimizing component count, the CHAP operates over the 400–900 nm spectral range—making it particularly well-suited for planetary missions such as asteroid surveys. These advancements are part of a broader trend, with numerous research efforts confirming the feasibility and effectiveness of integrating miniaturized spectral imaging systems into CubeSat platforms.^[32–39] Collectively, they highlight a transformative shift toward compact, capable, and mission-adaptable SIS technologies for next-generation space and aerial sensing applications.

Similarly, the integration of compact SIS into UAVs has significantly advanced the capabilities of remote sensing. By equipping UAVs with lightweight, miniaturized SIS, researchers and practitioners can now capture spatially and spectrally resolved data at high resolutions—enabling detailed analysis that often surpasses the limitations of conventional remote sensing platforms.^[40,41] These portable systems have made UAVs highly effective tools for a wide range of applications, including environmental monitoring, precision agriculture, and military surveillance. In environmental monitoring, UAV-mounted SIS can evaluate vegetation health, identify plant species, and

monitor water quality. The rich spectral signatures captured by these sensors allow for precise detection of vegetation stress, nutrient deficiencies, and invasive species, providing actionable data for effective ecological management and conservation strategies.^[42–44] In the domain of precision agriculture, compact SIS enables UAVs to efficiently monitor crop conditions, soil variability, and pest infestations.^[45–49] This enables data-driven decision-making that improves yield, optimizes input usage, and supports sustainable farming practices. The ability to rapidly and repeatedly survey large fields with high spectral and spatial resolution gives farmers real-time insights that were previously unattainable using traditional techniques. Beyond civilian applications, the military sector has also benefited from UAV-based SIS. Their compact size and low weight are especially advantageous for platforms with strict payload constraints, allowing longer flight durations and greater operational flexibility without compromising data quality. For instance, compact SIS operating across the visible, near-infrared, and thermal infrared bands enable UAVs to perform a range of surveillance tasks—such as detecting camouflaged targets, evaluating terrain conditions, distinguishing decoys from genuine threats, and conducting persistent wide-area monitoring.^[50] Together, these advancements illustrate how SIS are redefining the role of UAVs in remote sensing. By merging high-resolution spectral data with the mobility and accessibility of UAV platforms, these systems provide a powerful means of addressing complex challenges across agriculture, environmental science, and defense. As the technology continues to evolve, future systems are expected to become even more capable—smaller, more intelligent, and increasingly integrated with onboard processing and artificial intelligence. This ongoing evolution is poised to unlock new frontiers in remote sensing, making spectral imaging more accessible, and indispensable tool across a wide range of disciplines.

2.2. Medical Applications

Compact SIS have rapidly gained traction in medical diagnostics, offering a non-invasive means of enhancing the visualization and characterization of biological tissues. Driven by advances in miniaturized optics and sensor integration, these systems now support real-time, high-resolution imaging at the point of care. Notably, their deployment spans clinical settings ranging from operating rooms to mobile diagnostics, enabling clinicians to extract spectral information previously inaccessible with standard RGB imaging modalities.^[53–55] This miniaturization has enabled SIS platforms to be embedded in handheld devices, endoscopes, and laparoscopic tools, addressing a critical need for mobility in environments where space and equipment access are limited.^[56] The clinical impact of compact SIS is particularly evident in dermatology and wound care, where handheld spectral imagers are increasingly used to assess tissue oxygenation, melanin distribution, and early indicators of malignancy.^[57–61] The ability to perform these assessments *in situ* and in real time accelerates decision-making and improves patient management outcomes. Moreover, the integration of SIS into endoscopic and intraoperative tools has led to a new generation of minimally invasive surgical guidance systems. These devices can visualize subsurface tissue properties and delineate tumor margins with high

specificity, enabling more precise resections. For instance, Hu et al. demonstrated a compact SIS capable of operating across the visible, NIR-I (700–900 nm), and NIR-II (1000–1700 nm) ranges.^[62] Their intraoperative application in liver cancer surgeries showed that NIR-II imaging significantly enhanced tumor-to-background ratios, as illustrated in **Figure 3I**, enabling improved tumor localization at greater tissue depths. This capability is particularly critical in clinical procedures such as gastrointestinal endoscopy, where the ability to visualize subtle mucosal abnormalities, vascular structures, and lesion margins with high contrast is essential. Enhanced visualization provided by NIR-II imaging can improve the early detection and accurate delineation of neoplastic lesions, which is paramount in colorectal cancer diagnosis and staging. By enabling better discrimination between malignant and healthy tissues at greater depths, this technology supports more precise biopsy targeting, reduces the likelihood of missed lesions, and ultimately contributes to improved patient prognosis and treatment planning.

In a related development, Ebner et al. introduced an intraoperative hyperspectral imaging (iHSI) system designed for open surgical workflows.^[63] Their system, shown in **Figure 3II**, enables wide-field, real-time HSI acquisition and integrates seamlessly with existing surgical protocols. The authors provide a detailed comparison between snapshot and linescan HSI architectures, offering insight into trade-offs relevant for clinical implementation. Such systems exemplify the emerging field of surgical spectral imaging systems (S-SIS), which combines MSI or HSI modalities with real-time data analytics to support intraoperative decision-making. Strikingly, the synergy between optical instrumentation and computational interpretation is further explored in the review by Clancy et al., who emphasize the growing role of artificial intelligence (AI) in spectral data processing.^[65] By leveraging machine learning and deep learning algorithms, these systems can rapidly interpret high-dimensional datasets, providing actionable insights to surgeons without interrupting clinical workflows.

Recent work by Xiong et al. presents a significant advancement in SIS integration and performance. Their design features a spectral imaging chip composed of reconfigurable metasurfaces coupled with a CMOS image sensor (CIS). This chip comprises over 155000 adaptive microspectrometers and employs compressed sensing for real-time spectral reconstruction. The system's efficacy was validated *in vivo* during cortical hemodynamic monitoring in a rodent model, as shown in **Figure 3III**. This demonstration highlights the chip's ability to resolve dynamic changes in oxyhemoglobin and deoxyhemoglobin levels—underscoring its potential for functional brain imaging and other real-time physiological monitoring applications.^[64]

In a complementary effort, Leon et al. explored the use of HSI as an intraoperative diagnostic tool for brain tumor detection. By integrating spectral data with machine learning models, they developed a framework for *in vivo* brain tumor classification that demonstrated high accuracy and clinical relevance. As shown in **Figure 4I**, their system enhances tumor visualization during neurosurgery, offering more precise localization and tissue discrimination than conventional imaging. However, the study also underscores the need for further clinical validation, improvements in data acquisition speed, and robustness to fully enable the routine adoption of such technologies in surgical environments.^[66]

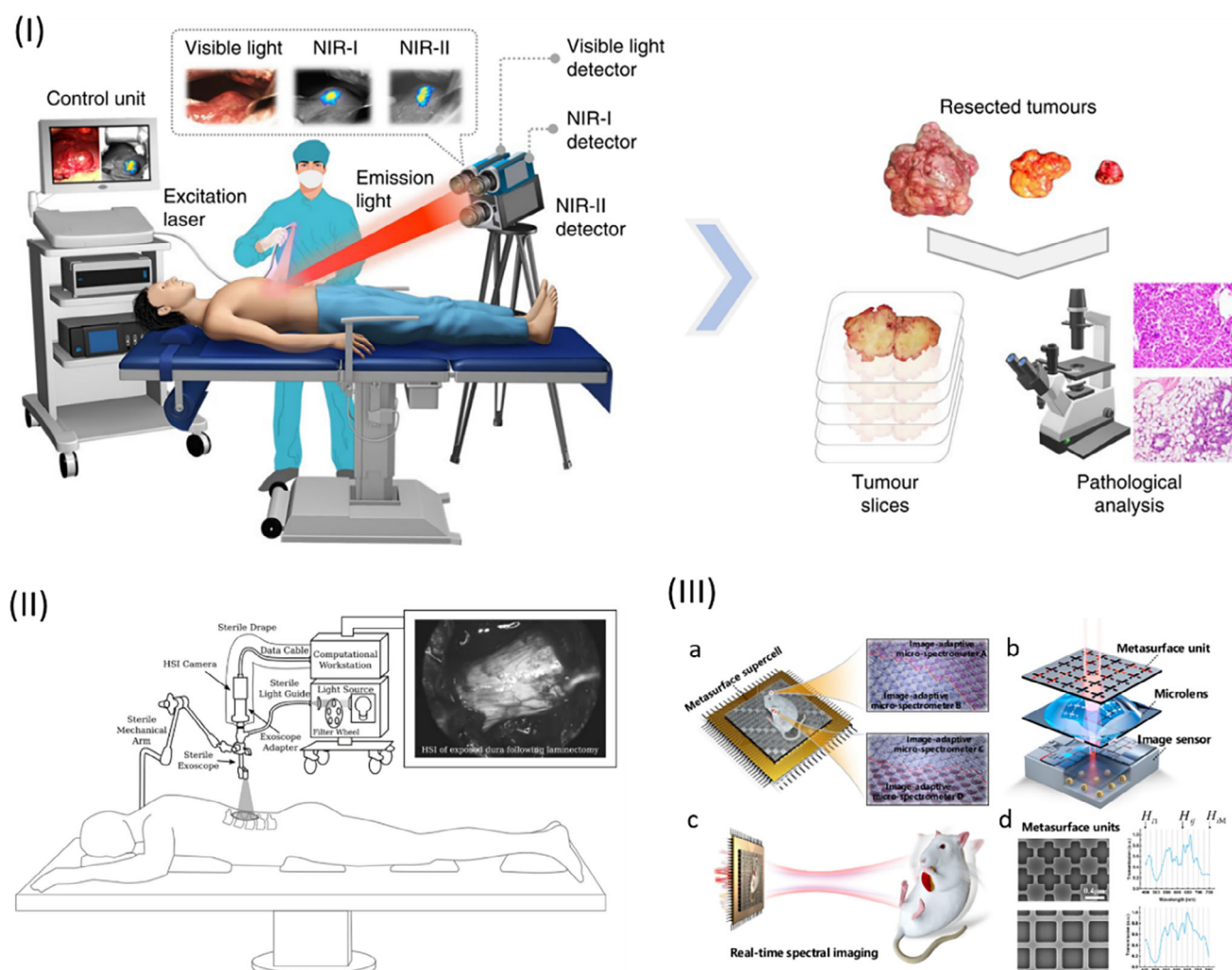


Figure 3. I) The liver surface examination of a patient with a liver cancer using integrated NIR-I/II and visible spectral instrument, and the acquired resected images during the first liver-tumor surgery guided by MS fluorescence imaging. Reproduced with permission.^[62] Copyright 2020, Springer Nature. II) Schematic of the proposed intraoperative HSI system for in-patient clinical feasibility study during spinal fusion procedure, where the system operates at video rate. Reproduced under the terms of the CC-BY 4.0 license.^[63] Copyright 2021, Copyright IOP Science. III) The proposed metasurface based spectral imaging system for the capture of dynamic brain spectrum, including a) The proposed device, b) the schematic of the illustrating the metasurface unit, the micro-lens array, and the CMOS camera, c) the snapshot spectral imaging, and d) the SEM image of the metasurfaces units and their transmission spectra. Reproduced under the terms of the CC-BY 4.0 license.^[64] Copyright 2022, Optica Publishing Group.

Their findings reinforce the potential of SIS to transform intraoperative imaging, especially in delicate and high-stakes applications like neurosurgery. Moreover, a particularly promising area of application is in minimally invasive surgery. Köhler et al. introduced a laparoscopic HSI system that enables simultaneous high-resolution video and hyperspectral acquisition across the visible and NIR ranges. Unlike many push-broom systems that suffer from motion artifacts, their design integrates a motorized spectrograph within the laparoscope, enabling scanning without requiring manual movement of the device or tissue (Figure 4). This innovation overcomes a major limitation of traditional SIS in minimally invasive surgery—namely, the trade-off between spatial resolution, acquisition speed, and mechanical complexity. The study showed strong consistency between their laparoscopic device and previously validated open-surgery HSI systems, with

a signal-to-noise ratio of 30–43 dB and acquisition times as low as 4.6 seconds.^[67]

To expand SIS technologies for point-of-care diagnostics, Cavalcanti et al. developed a smartphone-based multispectral otoscope for middle ear assessment. The system integrates nine LEDs spanning 405–700 nm and employs a standard smartphone camera for spectral data acquisition, enabling non-invasive imaging of the tympanic membrane in a highly compact and cost-effective form factor. Designed with telemedicine in mind, this device represents a compelling solution for resource-limited or remote environments, where traditional otoscopic imaging may be unavailable. In a preliminary human trial involving four subjects, the system achieved a classification accuracy of 82.8% in differentiating between healthy and pathological eardrums. Although these early results are promising, the limited sample size

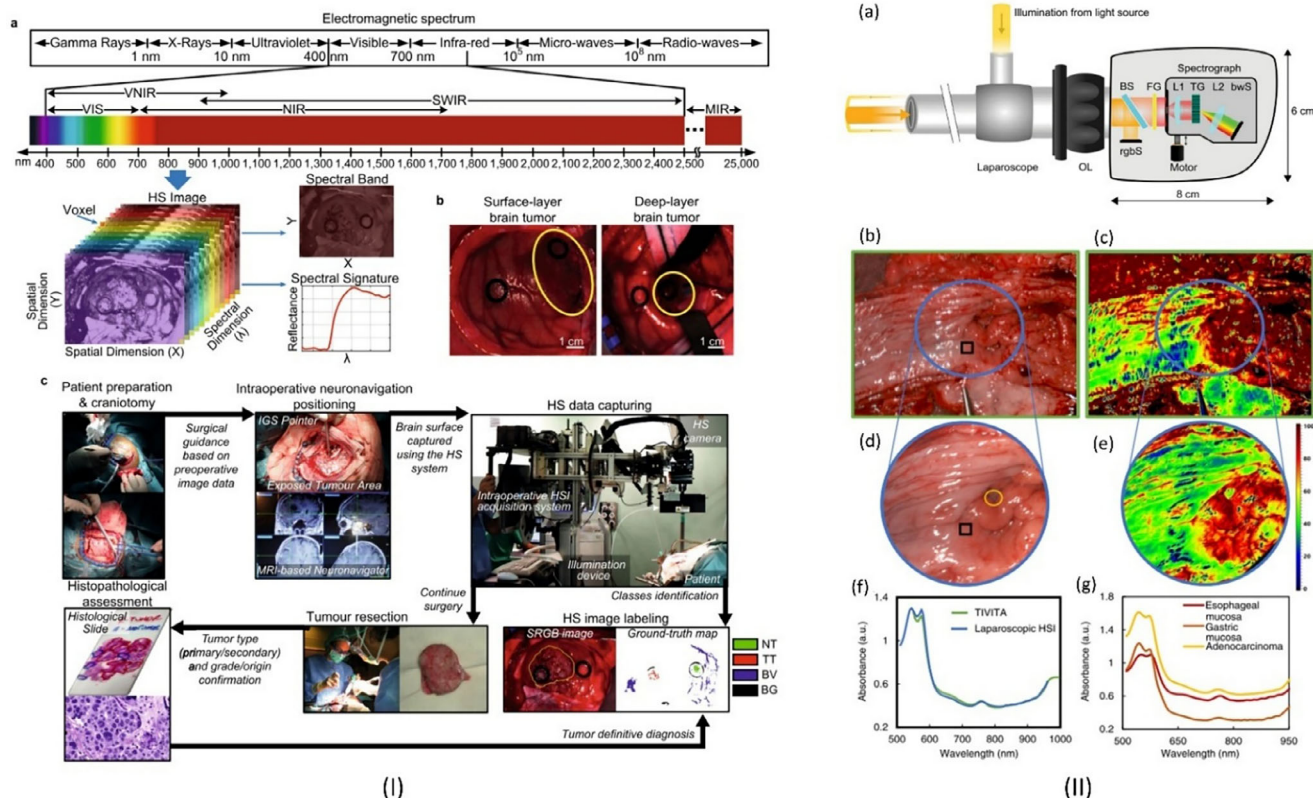


Figure 4. I) Illustration of SIS Principles and Surgical Application Workflow, a) common electromagnetic spectral regions relevant to hyperspectral imaging, spanning from the visible (VIS) to the short-wave infrared (SWIR), and mid-infrared (MIR) regions. A hyperspectral data cube is composed of two spatial dimensions (X and Y) and one spectral dimension (λ), where each pixel holds a unique spectral profile. These spectral signatures can be used to distinguish different tissue types based on their reflectance characteristics. b) Synthetic RGB representations of intraoperative brain images shown for two tumor cases: one located near the surface (left), and another situated deeper in the tissue (right). Yellow outlines mark tumor regions as annotated by clinical experts, and c) The workflow for capturing and annotating hyperspectral data during surgery. The process begins with patient preparation and skull opening, followed by using image-guided navigation tools to identify the tumor site using preoperative imaging. Hyperspectral data is then collected using a dedicated imaging setup composed of a spectral camera and illumination source. After image acquisition, pixels in the hyperspectral images are categorized into tissue types: tumor tissue (TT, red), healthy tissue (NT, green), blood vessels (BV, blue), background (BG, black), and unclassified areas (white). Resected tissue samples are analyzed under a microscope to verify tumor type and origin, informing surgical decisions and contributing to the final diagnosis. Reproduced under the terms of the CC-BY 4.0 license.^[66] Copyright 2023, Springer Nature. II) Overview of the laparoscopic HSI and the Intraluminal HSI of Resected Esophagogastric Adenocarcinoma: a) the system integrates an external light source that delivers illumination through the laparoscope. An optical lens adapter connects the laparoscope to the camera and adjusts for varying working distances. A beam splitter directs part of the visible light to an RGB sensor, while the remaining visible and near-infrared light passes through a high-pass filter into the spectrograph. The spectrograph, consisting of two achromatic lenses, a transmission grating, and a monochrome sensor, is translated by a stepper motor to enable push-broom HSI scanning. Panels (b) and (c) show representative intraluminal HSI data of a resected esophagogastric adenocarcinoma, including both color and oxygenation heatmap (OHI) images obtained using the TIVITA system. Panels (d) and (e) present corresponding data acquired using their proposed laparoscopic HSI system. The common field of view (FOV) is indicated by a blue circle. Panel (f) displays the average absorbance spectra within the black square region for both systems. Panel (g) shows mean absorbance spectra acquired using the laparoscopic HSI system from areas identified as adenocarcinoma (yellow circle) and adjacent esophageal and gastric mucosa (outside the FOV), annotated by a surgeon based on visual and tactile assessment. Reproduced under the terms of the CC-BY 4.0 license.^[67] Copyright 2020, Society of Photo-Optical Instrumentation Engineers (SPIE).

and restricted spectral range may constrain the system's ability to generalize across diverse patient populations or subtle disease phenotypes (Figure 5I).^[68]

Furthermore, SIS is also advancing histopathological diagnostics. A new study by Sravankumar et al. proposed a 3D U-Net architecture for the classification of cholangiocarcinoma using hyperspectral microscopic images. Their approach incorporates an extensive preprocessing pipeline—including flat-field correction, principal component analysis, median filtering, and normalization—which led to a 4.29% improvement in classifica-

tion accuracy compared to state-of-the-art models like IANet and SegNet. Their encoder-decoder structure with skip connections allows effective learning of spatial and spectral correlations, improving classification robustness at the pixel level.^[69]

Concurrently, SIS is becoming increasingly vital in histopathology and oncological diagnostics. One noteworthy study by Li Sun et al. introduces a benchmark dataset for microscopic HSI in cholangiocarcinoma diagnosis, comprising 880 annotated scenes. Their proposed CNN architecture includes a spectral interval convolution module, which preserves

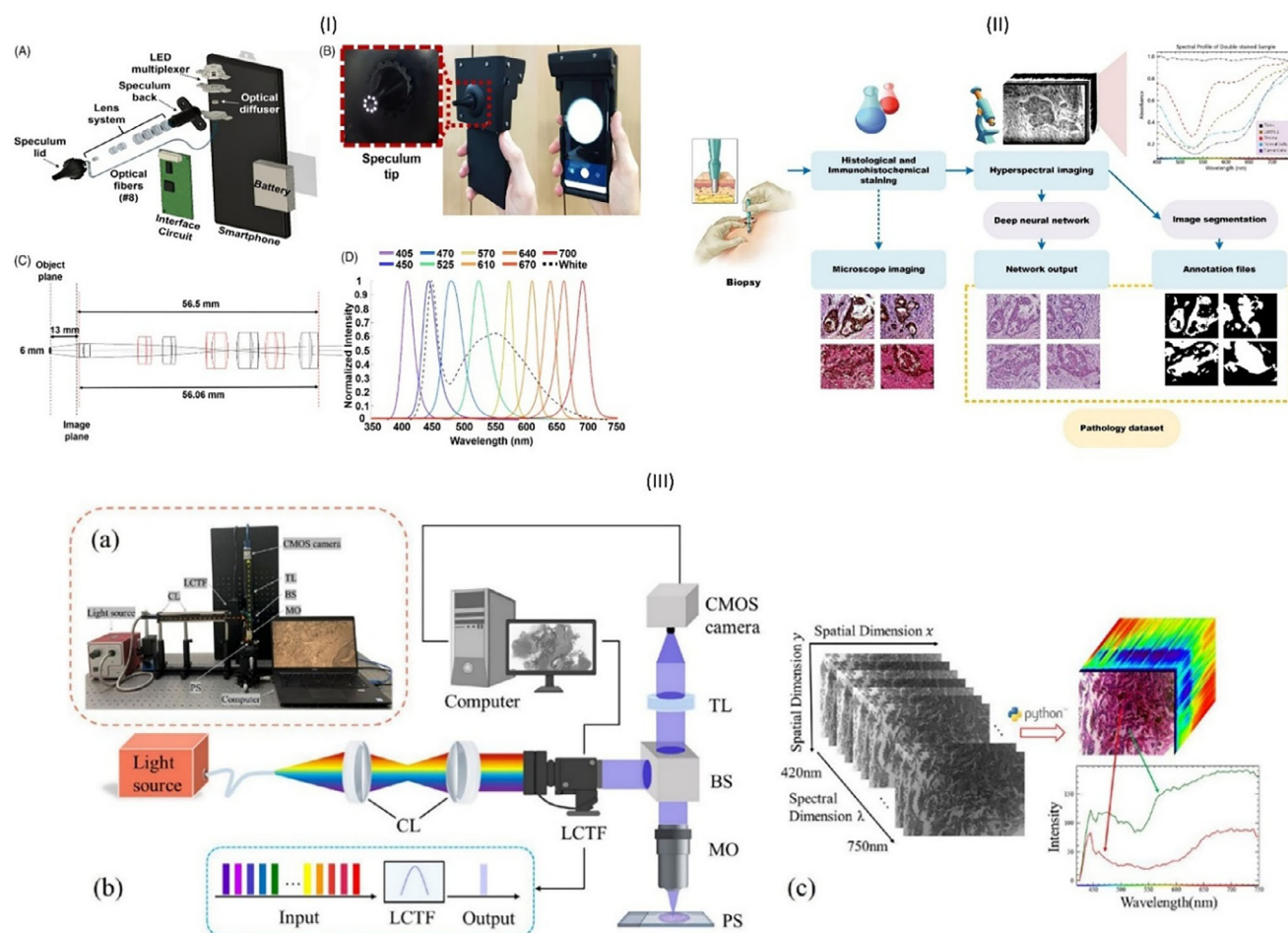


Figure 5. I) Smartphone-Compatible Spectral Oscope Design and Optical Configuration: Panel (A) illustrates the system design, featuring (from front to back): a speculum lid, seven achromatic doublet lenses, the speculum body, a LED multiplexer, optical fibers, control circuitry, a battery, and a smartphone interface, Panel (B) shows the assembled device, with a magnified view highlighting eight optical fibers emitting white light around the speculum, C) The imaging system—comprising seven achromatic lenses—is optimized for tympanic membrane visualization, offering a 6 mm field of view, 13 mm working distance, and 56.5 mm image distance, within a 56.06 mm overall length, and panel (D) presents the spectral profiles of the nine illumination LEDs (solid lines) alongside a standard white LED (dashed line), showcasing the system's multispectral capability. Reproduced with permission.^[68] Copyright 2020, John Wiley and Sons. II) Workflow for Histological Stain Translation and Automated Annotation Using SSHGAN: This process begins with double-stained tissue sections acquired from two different hospitals, captured using a microscope under consistent staining protocols. Hyperspectral imaging produces a 3D spectral data cube representing the tissue. A deep neural network (SSHGAN) then reconstructs corresponding H&E-like stained images. These synthesized images are used to train an image classifier, which subsequently generates annotation files automatically. Together, the generated H&E-style images and annotations form a comprehensive benchmark dataset for pathological analysis. Reproduced under the terms of the CC-BY 4.0 license.^[71] Copyright 2023, Elsevier. III) Hyperspectral Imaging Apparatus and Data Structure: Panel (A) shows the physical setup of the hyperspectral microscopy imaging (HMI) system, panel (b) provides a schematic of the optical configuration, and the system includes a collimating lens (CL), a liquid crystal tunable filter (LCTF), a beam splitter (BS), a microscopic objective (MO), pathological slides (PS), a tube lens (TL), and a CMOS sensor for image capture, and panel (c) illustrates the structure of the acquired hyperspectral data cube, comprising 67 spectral channels and a spatial resolution of 3088 × 2064 pixels. Each pixel encodes spectral information across all bands, with visual examples shown using red and green channel projections. Reproduced under the terms of the CC-BY 4.0 license.^[77] Copyright 2025, Springer Nature.

spectral integrity by processing bands in grouped intervals. This architecture enables accurate tumor classification from spatial-spectral patches and outperforms RGB-based methods in slide-level diagnostics. Additionally, their method aggregates patch-level predictions using a random forest classifier to obtain slide-level diagnostic results, achieving robust performance in a clinical-like workflow.^[70]

Wang et al. further tackled the data bottleneck in AI pathology by introducing a pipeline for the automatic generation of anno-

tated pathological datasets using hyperspectral images of double-stained tissues. Their pipeline integrates a Spatial-Spectral Hyperspectral GAN (SSHGAN) for synthesizing hematoxylin-eosin (H&E)-equivalent images from CAM5.2 double-stained inputs and uses gradient-boosted decision trees with spectral priors for segmentation. The approach demonstrated high fidelity in lesion identification—even for small or diffuse tumors often missed by manual annotation.^[71] Figure 5II illustrates the complete workflow of their approach, from spectral image acquisition to

automated dataset generation, marking a significant step toward scalable and accurate dataset creation for AI-based pathology.^[72]

Furthermore, expanding the capabilities of SIS in histopathology, Zhang and Dong recently introduced SAHIS-Net, a deep learning framework tailored for the segmentation of cholangiocarcinoma (CCA) in hyperspectral microscopy. The model operates on 60 spectral bands ranging from 550 to 1000 nm and integrates a Spectral Attention (SA) module, which uses 1×1 depth-wise convolution and a cosine annealing learning rate scheduler to dynamically prioritize the most informative spectral features.^[73] This design improves spectral discriminability and contributes to the model's strong quantitative performance, achieving a mean Intersection over Union (mIoU) of 0.6163 and an Average Symmetric Surface Distance (ASSD) of 8.01, outperforming other architectures such as U-Net and HLCA-UNet.^[74,75]

Addressing the challenge of cellular segmentation in complex histological images, Qing Zhang et al. propose TAJ-Net, a two-stage network that first segments clustered cells using a few-shot GAN-based model and then classifies them using an ensemble of 3D and 1D CNNs. The inclusion of spectral information into both segmentation and classification stages enhances robustness to annotation noise and improves classification of visually ambiguous cell types, such as hyperplasia. Their results demonstrate the clear spectral distinction among tumor, hyperplastic, and normal pulmonary cells—information crucial for accurate pathological interpretation.^[76]

One of the key bottlenecks in spectral pathology is the labor-intensive nature of dataset annotation. A possible solution by Yan et al. introduced a semi-automated refinement method for generating accurate segmentation masks in hyperspectral lung cancer imaging. Utilizing a dataset of 65 lung squamous cell carcinoma (LSCC) samples acquired across 67 spectral bands (420–750 nm), their study combines K-means clustering with guided expert selection to produce high-quality pixel-level annotations. This hybrid strategy reduces annotation time from over 30 minutes to approximately 5 minutes per whole-slide image, while improving segmentation accuracy from 77.33% to 92.52%, thereby significantly increasing both efficiency and diagnostic reliability. A carefully designed preprocessing pipeline—including Savitzky-Golay smoothing, Standard Normal Variate (SNV) normalization, and Principal Component Analysis (PCA)—was employed to mitigate spectral noise and enhance feature discrimination (Figure 5III).^[77]

While this section highlights a selection of notable recent advances in medical SIS, it is important to recognize that the field encompasses a much broader and rapidly evolving landscape. Numerous other studies and technologies—ranging from algorithmic innovations and clinical validation trials to hardware miniaturization and multimodal system integration—continue to shape the trajectory of SIS in both research and clinical domains. Given the scope and complexity of ongoing developments, a full account lies beyond the aims of this review. For readers seeking a more in-depth exploration of the field, including comprehensive comparative analyses of imaging architectures, clinical performance benchmarks, and engineering trade-offs in device deployment and integration, we encourage consultation of specialized review articles and foundational studies that focus specifically on medical SIS technologies and their applications.^[15,78–82]

In summary, the rapid evolution of compact SIS technologies has significantly expanded their reach across domains such as environmental monitoring, precision agriculture, and medical diagnostics. Through innovations in optical design, sensor integration, and computational reconstruction, these systems are now capable of achieving remarkable performance within stringent size, weight, and power consumption (SWaP) constraints. Compact SIS are redefining technical possibilities beyond the laboratory, whether they are deployed on CubeSats orbiting Earth, UAVs scanning agricultural landscapes, or handheld devices assisting surgeons in real time. Notably, the integration of metasurfaces, compressive sensing, and adaptive optics—as explored by Xiong et al.^[64] for instance—is paving the way toward highly reconfigurable and application-specific imaging solutions, capable of operating in challenging, dynamic environments. Simultaneously, advancements in machine learning and real-time data processing are unlocking new possibilities for interpreting complex spectral datasets—transforming raw spectral cubes into actionable insights faster than before. However, while these strides are impressive, challenges remain. Issues such as calibration complexity, data acquisition bottlenecks, and algorithmic robustness under real-world conditions must be addressed to ensure reliable deployment across different platforms. Additionally, continued cross-disciplinary collaboration—spanning optics, microelectronics, machine learning, and domain-specific expertise—will be critical in maturing these systems from promising prototypes into fully realized tools for global-scale impact. Thus, compact SIS stands at a compelling intersection of engineering ingenuity and real-world application. As the technology continues to mature, its influence will likely become increasingly ubiquitous—quietly embedded in the systems that observe, diagnose, and navigate the world around us. The recent advancements in image processing techniques, combined with the rapid progress in AI, have the potential to significantly elevate the capabilities of SIS. These technologies enable SIS to non-invasively extract highly detailed and accurate biological information, facilitating deeper insights into complex biological processes without the need for invasive procedures. In particular, the integration of SIS with precision agriculture and remote sensing applications—especially through the deployment of CubeSats—opens new frontiers for large-scale, cost-effective monitoring of crop health, soil conditions, and environmental factors. By leveraging AI-enhanced spectral data captured via CubeSats, farmers and researchers can gain real-time, high-resolution biological and ecological insights that support sustainable farming practices and optimized resource management. This convergence of advanced image processing, AI, and satellite-based remote sensing promises to revolutionize the precision, efficiency, and scope of spectral data acquisition, with broad implications for agriculture, environmental monitoring, medicine, and beyond.

3. Progress in Compact SIS design

3.1. DIY Approaches to Compact SIS Design

Significant strides have been made in the development of DIY spectral imaging systems, driven by advances in 3D printing technology and the growing accessibility of commercially available optical components. Today, researchers and enthusiasts alike

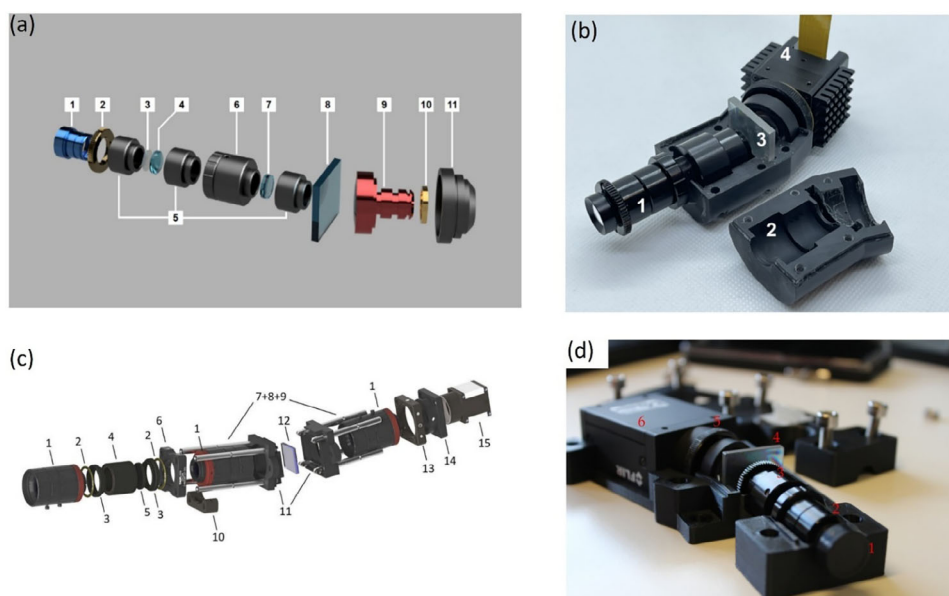


Figure 6. Illustration of four different designs of DIY spectral imaging systems: a) The schematic diagram provides an exploded view of the components in the DIY spectral imager: The parts include: 1) the front lens, 2) a lock nut, 3) an air slit, 4) a field lens, 5) three thin lens mounts, 6) a focus tube, 7) a collimator lens, 8) a transmission grating, 9) a detector lens, 10) focus spacers, and 11) a C-mount lens adapter. Reproduced under the terms of the CC-BY 4.0 license.^[84] Copyright 2018, Optica Publishing Group. b) The Open-source HSI system solution for all: The parts include: 1) The S-mount collimator tube 2) The Collimator mount, 3) The dispersive component, and 4) The Camera. Reproduced under the terms of the CC-BY 4.0 license.^[85] Copyright 2022, MDPI. c) The schematic diagram presents an exploded view of the components for a variant of a DIY spectral imager: the components includes: 1) the objective lens, 2) a spacer ring, 3) an adapter ring, 4) a lens tube, 5) the slit, 6) a cage plate for the lens tube, 7–9) steel rods and swivels, 10) a mounting bracket, 11) a grating holder, 12) the grating, 13) a cage plate for the detector mount, 14) the detector mount, and 15) the detector. Reproduced under the terms of the CC-BY 4.0 license.^[86] Copyright 2022, Optica Publishing Group. d) The photograph illustrates a DIY-SIS design: light first passes through the front lens (1) and continues through a series of S-mounts that contain the slit and field lens (2). It then moves through the S-mount focus tube and collimator lens (3). The diffraction grating (4) disperses the light into its constituent wavelengths. This dispersed light is then focused by the detector lens (5) and captured by the machine vision camera (6). Reproduced under the terms of the CC-BY 4.0 license.^[87] Copyright 2021, MDPI.

can design and construct custom SIS tailored to specific applications and experimental needs. The advent of 3D printing has transformed the DIY landscape by enabling the rapid prototyping and precise fabrication of complex optical mounts, housings, and other structural elements—at a fraction of the cost associated with traditional manufacturing techniques. This democratization of technology has made it possible to customize optical layouts and seamlessly integrate a variety of readily available commercial off-the-shelf (COTS) components such as lenses, mirrors, filters, and sensors, leading to the creation of portable, flexible, and cost-effective SIS. Moreover, DIY spectral imaging platforms serve as valuable educational tools, offering students and hobbyists a hands-on opportunity to explore the principles of spectral imaging and gain practical experience with cutting-edge optical systems.

Furthermore, the community-driven nature of DIY projects fosters collaboration and rapid innovation by encouraging the open exchange of designs, enhancements, and application strategies through open-source platforms. This collective ecosystem accelerates progress and empowers a wide range of users—from researchers to hobbyists—to contribute meaningfully to the field. A notable example is the work of Seoung et al., who in 2016 demonstrated an innovative framework for reconstructing hyperspectral images using multiple consumer-level digital cameras within “DIY” paradigm. Their approach leverages the vary-

ing spectral sensitivities of different camera sensors to create hyperspectral images—providing a cost-effective alternative to traditional HSI systems. Their proposed algorithm models both the spectral reflectance of scene points and the illumination spectrum simultaneously, enabling high-quality reconstructions for both indoor and outdoor environments. This approach promotes further development and customization within the community, paving the way for continued innovation in DIY-SIS design.^[83]

Sigernes et al. developed a low-cost, lightweight pushbroom HSI system for handheld and airborne uses. They combined 3D-printed structures with commercial optics to make it affordable, portable, and easy to assemble. The design features S-mount components, off-the-shelf camera heads, and a transmission grating, all within a 3D-printed thermoplastic frame (Figure 6a).^[84]

Mao et al. introduce an OpenHSI project, a lightweight, open-source pushbroom HSI designed for accessibility and affordability (Figure 6b). Constructed using COTS components and housed in a compact 3D-printed enclosure, the system operates over the 400–830 nm spectral range and delivers ≈ 4 nm spectral resolution (binned) across 108 bands. With a mass of just 119 g and physical dimensions of $116 \times 46 \times 35$ mm³, the system achieves a ground sample distance (GSD) of 5 cm when flown at 120 m altitude, making it well-suited for drone-based remote sensing. The optical design improves upon the Sigernes et al. model by

incorporating an achromatic collimator lens, significantly reducing chromatic aberration and enhancing image quality across the spectrum, as validated by Zemax simulations. Key optical components include a 600 lines/mm transmission grating blazed at 28.7° , a $25\text{ }\mu\text{m}$ entrance slit, and a XIMEA MX031MG-SY camera featuring a Sony IMX252 CMOS sensor (2064×1544 pixels, $3.45\text{ }\mu\text{m}$ pixel size), capable of capturing data at 75 FPS with a 10 ms exposure time. The accompanying open-source Python library, *openhsi*, supports a comprehensive processing pipeline that includes spectral calibration (via HgAr lamps, with $\pm 0.3\text{ nm}$ accuracy using cubic interpolation), efficient smile correction using 1D convolution, radiance conversion based on integrating sphere data, and atmospheric correction through either the 6SV radiative transfer model or empirical line calibration (ELC). Integrated ancillary sensors—IMU, GPS, and BME280—enable georectification and atmospheric parameterization, extending the system's field readiness. Limitations of the system include negligible keystone correction, dust-induced imaging artifacts, and sensitivity to atmospheric variability in the 6SV model. Future improvements may involve the addition of order-sorting filters to suppress second-order diffraction, wider slits to enhance signal-to-noise ratio (SNR), and more advanced georectification to mitigate motion artifacts during flight. All design files and software are openly available on GitHub, supporting reproducibility and lowering technical and financial barriers for a range of applications, including precision agriculture, environmental monitoring, and education.^[85]

Henriksen et al. introduce the HSI V6, an optimized DIY pushbroom HSI designed for VIS/NIR applications (400–800 nm). Built using COTS components, the system offers a cost-effective and replicable platform for compact HSI, particularly in resource-limited settings (Figure 6c). Their system measures $220 \times 65 \times 65\text{ mm}^3$, weighs 650 g, and includes three 50 mm focal length objectives, a 300 grooves/mm transmission grating blazed at 17.5° , and a $50\text{ }\mu\text{m}$ entrance slit. The configuration yields a theoretical full width at half maximum (FWHM) spectral resolution of 3.33 nm, with experimental verification showing 3.69 nm at 546.1 nm. A Sony IMX174 CMOS sensor provides high quantum efficiency in the visible range, while alternative sensors (e.g., IMX249, IMX252) are explored for cost reduction or enhanced NIR sensitivity. Optical distortions such as smile and keystone are notably reduced compared and further correction is supported via referenced software methods. To address second-order diffraction—particularly problematic above 780 nm—their system employs a novel correction approach using a 575 nm long-pass filter. While this method improves spectral purity, its effectiveness is dependent on the spectral composition of the incident light. System calibration was performed using argon and mercury-argon lamps alongside an integrating sphere, confirming a reliable spectral operating range. Field tests at the Longyearbyen harbor validated both spatial and spectral performance, capturing environmental signatures including vegetation and sky reflectance with high fidelity. With an estimated build cost of $\sim \$2600$ and open access to CAD files for all 3D-printed components, their system balances performance, portability, and affordability. However, limitations remain, particularly in the precision required for manual assembly and sensitivity to spectral variability in second-order correction. Future improvements may include automated assembly processes and refined

spectral correction models to enhance robustness and usability across a broader range of applications.^[86]

Riihiho et al. introduce a low-cost, open-source HSI platform based DIY push-broom architecture (Figure 6d). Building upon the Sigernes design, the system employs commercially available optical components mounted within a custom 3D-printed enclosure, allowing for rapid prototyping without the need for optical laboratory infrastructure. The authors address two major sources of distortion—spectral smile, a common curvature artifact in HSI systems, and tilt, a misalignment-induced aberration specific to manual assembly. Their Python-based correction software estimates and compensates for both distortions by fitting circular arcs to detected emission lines and applying pixel-wise shift corrections, eliminating the requirement for monochromators or spectral calibration lamps. The full pipeline includes tools for real-time camera preview, acquisition control, image cube construction, and post-processing visualization, with user interaction managed via a lightweight IPython-based interface. To evaluate the system, the authors conducted experiments on synthetic datasets and a physical color checker chart imaged under standard office lighting. The correction algorithm effectively reduced tilt from approximately 0.9° to 0.005° , and minimized curvature metrics by over 95%, confirming the robustness of the correction strategy. The system's total cost, under $\text{€}2000$ —with the camera accounting for roughly half—can be further reduced by substituting lower-end components. All hardware design files, correction datasets, and software (available via GitHub and OSF) are shared under permissive licenses, supporting reproducibility and community adaptation. This work represents a significant advancement in democratizing DIY SIS, particularly for educational institutions, low-resource labs, and field-based applications where compactness, affordability, and flexibility are critical.^[87]

Similarly, Morales et al. present a custom-designed laboratory HSI tailored for pushbroom cameras compliant with the GenI-Cam standard, leveraging a gigabit Ethernet interface (Figure 7a). The setup incorporates a motorized linear stage (Zaber A-LST1000B, $0.4961\text{ }\mu\text{m}$ microstep resolution), a 150 W Quartz Tungsten-Halogen illumination source (400–2500 nm), and two Specim cameras: FX10 (400–1000 nm) and FX17 (900–1700 nm). A Python-based graphical user interface (GUI), developed with DearPyGui, automates camera and stage control, calibration, and real-time visualization, streamlining the acquisition pipeline. They performed callibration using stored white and dark references, minimizing redundancy and enabling high repeatability, with mean standard deviations of 0.11% for FX10 and 0.05% for FX17. Their system undergoes validation through keystone and smile aberration assessments, achieving up to 98.07% keystone correction using Specim's Automatic Image Enhancement. Spectral calibration against a Zenith Polymer standard yields errors below 10% for central wavelengths. Ground sampling distance (GSD) is refined using a morphological analysis based on a chessboard pattern, resulting in an ellipse axis ratio of 0.986 for FX10, confirming geometric precision. Besides, a case study involving plastic classification (HDPE and LDPE) demonstrates the system's spectral fidelity, with spectral angle deviations of 5.378° and 6.846° compared to USGS spectral library references. While the system's modular architecture and open-source software enhance its utility for research, its dependence on controlled

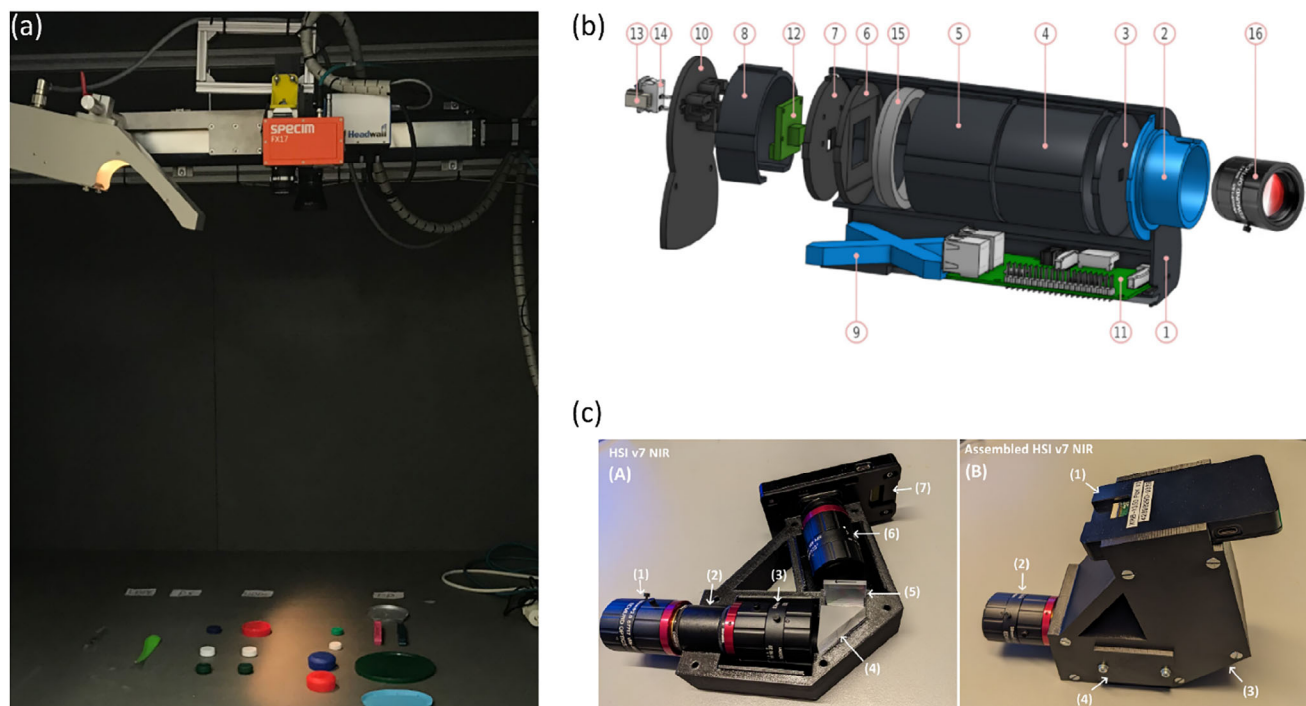


Figure 7. a) Layout of plastic samples within the proposed hyperspectral imaging system. Reproduced under the terms of the CC-BY 4.0 license.^[88] Copyright 2022, MDPI. b) Components of the hyperspectral imaging (HSI) system: 1) Main housing case, 2) Front lens element, 3) Square aperture, 4–5) Dual-stage optical extension tubes, 6) Diffraction grating, 7) Camera mounting base, 8) Sealing extension, 9) Lateral extension module, 10) Top cover lid, 11) Raspberry Pi 3 Model B+, 12) Raspberry Pi NoIR Camera V2 (8 MP), 13) Power supply connector, 14) Power on/off switch, 15) +10 macro lens (52 mm), and 16) 35 mm C-Mount lens (Edmund Optics, model 59 872). Reproduced under the terms of the CC-BY 4.0 license.^[89] Copyright 2020, Elsevier. c) The photograph of the black silicon Pushbroom HSI system. The enclosure includes. A 3D printed press fit base (Left panel (A)), where 1) front lens, 2) slit housing (tube), 3) collimator lens, 4) right-angle prism, 5) transmitting grating, 6) detector lens, and 7) SiOnyx development board. The assembled instrument (Right panel (B)), where 1) is camera head, 2) front lens, 3) press fitted 3D housing, and 4) inspection lid. Reproduced under the terms of the CC-BY 4.0 license.^[90] Copyright 2025, Optica Publishing Group.

environments and the absence of quantitative smile aberration metrics limit broader applicability. Future improvements could include advanced aberration quantification and the development of an online spectral reference database to promote accessibility and reproducibility.^[88]

Salazar-Vazquez and Mendez-Vazquez present a low-cost, single-shot HSI system based on Computed Tomographic Imaging Spectrometer (CTIS) technology, designed for accessibility, portability, and ease of assembly (Figure 7b). With an estimated cost of $\approx \$550$, the system weighs 300 g, measures $18 \times 9 \times 7 \text{ cm}^3$, and is constructed entirely from commercial off-the-shelf (COTS) components. Core elements include a Raspberry Pi 3 B+, a Raspicam NoIR v2.1 (8 MP), and a double-axis diffraction grating ($13\,500 \text{ lines inch}^{-1}$). The system features no moving optical parts, enhancing durability and field-deployability. Their CTIS system captures hyperspectral datacubes with a spatial resolution of 116×110 pixels and supports up to 315 wavebands across the 400–1052 nm spectral range. The effective spectral resolution is $\sim 2.07 \text{ nm}$, with optimal performance between 415–920 nm (64–105 bands) for reflectance applications. The optical configuration, housed in custom 3D-printed PLA parts, includes a 35 mm Edmund Optics lens, a +10 macro lens, and a square aperture to condition incoming light. The compact design enables rapid, single-shot acquisition suitable for mobile and low-resource environments. A Linux-compatible open-source software suite fa-

cilitates image acquisition, system calibration, and hyperspectral reconstruction via an Expectation-Maximization (EM) algorithm, significantly reducing computational time. Calibration is performed using spectral photographs (SPs) taken under fluorescent (Sylvania CFL 2700 K, with emission peaks at 405, 440, 545, 613, 710 nm) and incandescent lighting. These SPs define key image features such as the region of interest (ROI), zero-order mode (ZM), and pixel-wavelength mappings, resulting in high spectral accuracy. Validation experiments included controlled imaging of a Rubik's cube under halogen lighting, demonstrating accurate spectral reconstruction across distinct color regions. An outdoor scene was also captured and processed through image stitching, revealing material-specific spectral peaks (e.g., a yellow flower showing distinct reflectance at 570, 600, and 645 nm), affirming the system's applicability to real-world spectral mapping. Despite its promising performance, the system exhibits limitations: relatively low spatial resolution, diffraction grating texture artifacts, and sensitivity to illumination balance in spectral photographs, which necessitates dual exposure times (e.g., 2000 μs and 50 μs) to mitigate saturation and noise. Nonetheless, the comprehensive open-source framework—including 3D models, calibration guidelines, and software hosted on OSF.io under a GPL license—enables rapid assembly and encourages adoption for applications in precision agriculture, remote sensing, material classification, and educational environments.^[89]

Stuart et al. present a low-cost, laboratory-based HSI system ($\approx$ £6000), designed to improve accessibility for environmental research and reduce reliance on high-cost commercial systems (£30 000–£150 000). The system employs a push-broom scanning architecture using commercially available components, including a Thorlabs plano-convex lens (LA1401-A), a Hamamatsu C12880MA miniature spectrometer (spectral range: 340–850 nm, spectral resolution: $\approx$ 15 nm), and a rotary mirror system (New Scale Technologies DK-M3-RS-U-2M-20-L) for precise beam steering. The system achieves a theoretical instantaneous field of view (IFOV) of 2.50×4.90 mm and a total field of view (TFOV) of up to 36.4 cm, depending on scanning parameters. A bi-directional raster scanning pattern enables user-defined control over image resolution and exposure time. Custom LabVIEW-based control software converts 1D spectral data into hyperspectral cubes in real-time, facilitating immediate visualization and analysis. Radiometric calibration is traceable to the UK's National Measurement Institute. The system uses a custom white reference “orb” (coated with Edmund Optics white reflectance material, $>97\%$ reflectivity across 350–850 nm) and a silicon photodiode-based radiometer to quantify radiance, resulting in a per-pixel power estimate of ≈ 1.10 nW. Uniform scene illumination is provided by a 30 cm integrating sphere coated with the same high-reflectance material, effectively minimizing shadows and stray light. Despite its strengths, the system has notable limitations: relatively low spatial resolution due to a basic lens design, elevated noise in the blue region of the spectrum, and prolonged scan times for high-resolution imaging. However, its compact form factor and open-source software platform provide a solid foundation for further development—potential directions include extending spectral range into the infrared, improving spatial resolution, and adapting the system for field-deployable applications. Overall, the system offers a cost-effective, adaptable HSI platform for research in agriculture, geology, environmental science, and dental diagnostics, with promising potential for further innovation.^[91]

Recently, Sigernes et al. introduced the NIR HSI v7—a 3D-printed near-infrared pushbroom HSI developed for CubeSat and drone applications (Figure 7c). This compact system builds upon the HYPISO-1 satellite's HSI v6 and incorporates a black silicon CMOS sensor (SiOnyx XQE-1350) to enhance sensitivity across the 600–1000 nm range. Their optical design includes a 50 μ m entrance slit, a 300 lines/mm blazed grating, and 50 mm focal length lenses, enabling a spectral resolution of 3.33 nm and a field of view of 11.4° , while operating at 90 frames per second. Their calibration results confirmed a usable spectral range of 600–1000 nm, constrained by the declining quantum efficiency of their sensor and lens transmission beyond 1000 nm. Within this range, their new system achieved a signal-to-noise ratio of 100–200. Field tests conducted in both winter and summer conditions in Longyearbyen demonstrated the imager's effectiveness in capturing hyperspectral data cubes, successfully identifying vegetation, asphalt, and atmospheric absorption features. However, the use of a rolling shutter introduces potential distortions when imaging fast-moving targets. With a total component cost of approximately \$3145 USD, the NIR HSI v7 adheres to the OpenHSI framework, offering a low-cost, high-speed solution for remote sensing. Despite its strong performance, further optimization may be

needed to extend its spectral range and address shutter-related artifacts.^[90]

Table 1 offers a comparative overview of various HSI systems, evaluating key parameters such as weight, size, cost, number of spectral channels, spectral range, spectral and spatial resolution, acquisition technique, and the availability of open-source software. The analysis also relates these systems to previously discussed instruments, providing insight into trade-offs and application contexts. As seen from the table, the system weights range from as little as 0.12 kg ideal for UAV deployment, to an estimated 1.5–2.7 kg, better suited for satellite or ground-based use. The HSI v6 developed by Henriksen et al. weighs 0.65 kg, striking a balance for CubeSat and drone compatibility,^[86] while the lab-based system by Morales et al. is significantly heavier (~ 1.2 –1.5 kg), due to its more extensive setup.^[88]

Size varies from compact, pocket-sized units to larger systems such as the $150 \times 85 \times 71$ mm³ variants,^[88] with HSI v6 ($220 \times 65 \times 65$ mm³) offering a moderate footprint. Cost spans a wide range—from \$626 and \$2589 (HSI v6, using commercial off-the-shelf components), to high-end systems estimated at \$50000. Lower-cost solutions trade off some capabilities in favor of accessibility, while more expensive systems reflect proprietary or specialized hardware. The number of spectral channels ranges from 31 to 224. Spectral resolution varies from 4 nm to 15 nm, with HSI v6 achieving 3.69 nm—finer than most, though slightly coarser than the 2.7 nm resolution of the FX10 used by Morales et al.^[88] Spatial resolution also varies significantly, from 2064×1544 pixels to 3376×2704 pixels. HSI v6 defines spatial resolution through an IFOV metric, yielding a ground sampling distance of ~ 60 m from 500 km altitude, while Morales et al. achieves 0.641–1.014 mm/pixel, highlighting a much higher spatial precision. Most systems adopt the pushbroom acquisition method, consistent with HSI v6, while snapshot imaging is employed by Refs [83] and [89]. Open-source software support is also variable: Refs [85] and [87] offer accessible tools (Open HSI Calibration, Documentation, and the De-smiler Project Repo, respectively).

In Summary, DIY approach of SIS design offer a compelling blend of cost-effectiveness, customization, educational value, rapid prototyping, adaptability, and community-driven innovation. The systems leverage affordable, commercially available components and 3D printing technologies to significantly reduce overall costs. This affordability democratizes access to SIS design, enabling a wider range of users to engage in research, experimentation, and application development that might otherwise be financially out of reach. Additionally, the hands-on nature of constructing and calibrating such systems offers tremendous educational value. It provides students, educators, and enthusiasts with a deeper understanding of the underlying principles of optics, spectroscopy, system integration, and data analysis. Furthermore, advances in open-source software for spectral calibration, image registration, and data analysis have significantly reduced the barrier to entry, enabling even non-experts to achieve reliable results. As 3D printing resolution continues to improve and as low-cost sensors become increasingly sophisticated, the performance gap between DIY and commercial systems continue to narrow. This convergence of affordability, accessibility, and performance opens new frontiers for real-time, field-deployable SIS in resource-limited settings, potentially transforming how data

Table 1. Comparative Overview of the Advances in DIY based SIS.

Ref	Weight [kg]	Size	Cost [\$]	No. of spectral channels	Spectral range [μm]	Spectral resolution [nm]	Spatial resolution	Acquisition technique	Open-source software
[83]	2.7 (Est.)	0.5 m ³ (Est.)	3000 (Est.)	31	0.4–0.7	10	–	Snapshot	–
[84]	0.2	Pocket size	626	~214	0.4–0.7	1.4–5	–	Pushbroom	–
[85]	0.12	116 × 46 × 35 mm ³	–	108	0.4–0.83	4	2064 × 1544	Pushbroom	Open HSI Calibration Documentation
[89]	0.3	18 × 9 × 7 cm ³	550	64–105	0.42–0.95	2.07	116 × 110	Snapshot	Hypercube Import Lib Tutorial
[91]	1–2 (Est.)	30 × 15 × 15 cm ³ (Est.)	7600	34	0.340–0.850	15	1.4–2.8 mm	Pushbroom	–
[87]	1–2 (Est.)	20 × 10 × 10 cm ³ (Est.)	2280	~230	0.4–1	~2.6	3376 × 2704	Pushbroom	De-smiler Project Repo.
[88]	1.26 1.56	150 × 85 × 71 mm ³ 150 × 75 × 85 mm ³	50000 (Est.)	~224 ~224	0.4–1 0.9–1.7	2.7 3.5	0.641– 1.014 mm/pixel	Pushbroom	–
[86]	0.65	220 × 65 × 65 mm ³	2600	~120–160 (Est.)	0.4–0.8	3.69	–	Pushbroom	–
[90]	0.782	12.3 × 9.9 × 10 mm ³	3145	93	0.6–1	3.33	–	Pushbroom	–

is collected and analyzed across science and industry. However, despite these advantages, DIY SIS design face several technical challenges. For instance, 3D-printed components may lack the precision required for high-accuracy optical alignment, which can degrade system performance. Calibration processes can be complex and time-consuming, and errors in wavelength mapping or radiometric calibration may compromise data accuracy. Moreover, while DIY systems can perform well under controlled conditions, they may struggle in demanding operational environments where robustness, stability, and high signal-to-noise ratios are critical. Nevertheless, these limitations do not overshadow the broader value of DIY spectral imaging systems—particularly for educational use, proof-of-concept development, and niche applications where commercial systems are impractical. As 3D printing resolution improves, sensor technology advances, and the open-source ecosystem continues to evolve, many of these challenges may be progressively addressed. In this context, DIY spectral imaging stands not only as a practical alternative but also as a catalyst for innovation, collaboration, and the broader dissemination of optical imaging technologies across diverse disciplines.

3.2. Freeform Optics Based Designs

Freeform optics represent a rapidly evolving class of optical design, enabling unprecedented control over light propagation through surfaces that lack traditional rotational symmetry. These non-conventional optical elements are increasingly being integrated into SIS to achieve compact, lightweight, and high-performance designs that would be difficult—or even impossible—to realize with standard spherical or aspherical components. Unlike traditional optics that rely on symmetrical surfaces, freeform optics utilize complex, non-rotationally symmetric shapes, allowing for unprecedented control over light

propagation. This flexibility enables the correction of aberrations, improvement of image quality, and reduction in the size and weight of optical components [92]. Recent advancements in fabrication technologies—such as computer-aided design (CAD), ultra-precision diamond turning, and additive manufacturing—have enabled the production of freeform surfaces with high accuracy. These innovations have propelled progress across multiple fields, including consumer electronics, where freeform optics contribute to the development of lightweight, high-performance imaging systems. Their integration into optical design is redefining system capabilities, enabling more efficient, compact solutions with superior optical performance.^[5]

At the core of freeform optical design is the general surface description, which departs from rotational symmetry. A freeform surface can be mathematically described using a series expansion, such as the XY-polynomial or Zernike polynomial expansion. A commonly used representation is the XY-polynomial form:

$$z(x, y) = \sum_{i=0}^N \sum_{j=0}^N a_{ij} x^i y^j \quad (1)$$

Here, $z(x, y)$ is the sag (surface height) as a function of the Cartesian coordinates x and y , while a_{ij} are the polynomial coefficients that define the surface shape. This flexible mathematical representation allows for the modeling of arbitrarily complex surface geometries, enabling the correction of higher-order and off-axis aberrations that conventional optical surfaces cannot efficiently address.

In SIS design, the adoption of freeform optics unlocks several substantial advantages. By allowing multiple optical functions—such as focusing, collimation, and aberration correction—to be combined into a single surface, freeform elements reduce the

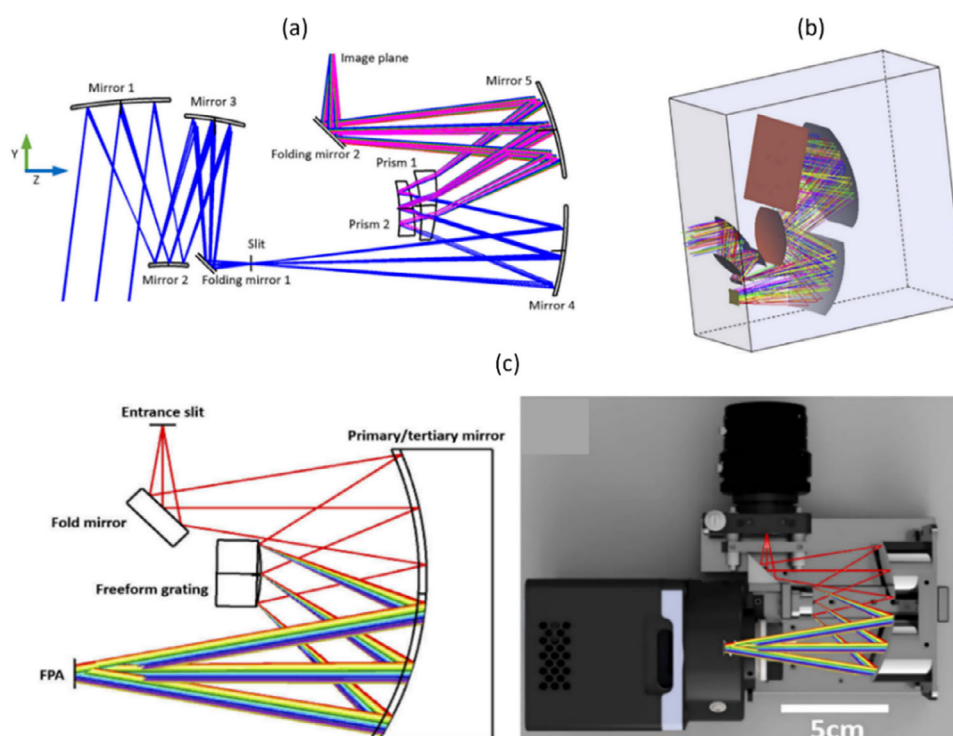


Figure 8. Illustration of three different freeform optics based spectral imaging systems: a) The schematic illustration of the freeform optics based spectral imager. Reproduced with permission.^[99] Copyright 2020, Optica Publishing Group. b) Schematic overview of the three-dimensional structural model of the HSI system based on freeform optics, highlighting the key optical components and their spatial configuration within the instrument. Reproduced under the terms of the CC-BY 4.0 license.^[100] Copyright 2021, Optica Publishing Group. c) The FYMOS hyperspectral imaging system includes a schematic representation of the optical layout, illustrating the configuration of key optical elements (left panel), as well as a photograph of the assembled mechanical design showcasing the system's compact integration (right panel). Reproduced under the terms of the CC-BY 4.0 license.^[101] Copyright 2021, Optica Publishing Group.

number of components required in the optical path. This integration not only minimizes alignment challenges and system volume but also enhances optical throughput and robustness. The result is higher spatial and spectral resolution, greater system compactness, and improved imaging fidelity.^[93–98]

The growing viability of freeform optics has been driven by advances in both design software and precision manufacturing. Modern optical design tools now support freeform surface modeling with optimization routines that account for wavefront errors, tolerances, and stray light analysis. On the fabrication side, technologies such as ultra-precision diamond turning, 5-axis CNC machining, and additive manufacturing enable the realization of these complex surfaces with nanometer-level accuracy and repeatability. Several recent studies demonstrate the power of freeform optics in SIS. For instance, Zhang et al. (2020) developed a compact hyperspectral spectrometer based on an Offner relay configuration enhanced with a freeform mirror. The system uses two curved prisms for spectral dispersion and incorporates a freeform surface to mitigate off-axis astigmatism, resulting in improved image quality across a wide field of view. This design significantly reduces the system's footprint, making it well-suited for remote sensing applications in space-constrained environments (Figure 8a).^[99]

In a related effort, Liu et al. (2021) introduced a HSI system combining a freeform reflective triplet imager with a freeform

triplet spectrometer in a double-pass configuration. Operating at $f/2$ with a 30 mm entrance pupil, and a 15° cross-track field of view, the system achieves high performance within a volume of less than 2 liters, making it ideal for 3U CubeSat deployment (Figure 8b).^[100]

Extending the use of freeform optics into system-level innovation, Graham et al. (2021) developed a Freeform-based Hyperspectral Imager for Moisture Sensing (FYMOS). This design incorporates a custom-engineered freeform blazed diffraction grating within an Offner spectrograph configuration. The grating is specifically optimized to correct beam path asymmetries, resulting in a 30% reduction in root-mean-square (RMS) wavefront error compared to conventional implementations. FYMOS operates over the $0.7\text{--}1.7\ \mu\text{m}$ near-infrared spectral range and is tailored for use with InGaAs detectors, highlighting the growing role of freeform optics in specialized spectral sensing applications (Figure 8c).^[101]

Furthermore, detailed stray light analysis demonstrates the robustness of freeform-optics-based SIS designs in suppressing parasitic reflections, which is critical for maintaining imaging quality in compact form factors.^[100] However, the integration of freeform surfaces introduces notable manufacturing complexities, potentially increasing production costs and lead times. The highly application-specific nature of these designs can also limit their adaptability to broader use cases.^[93] Additionally, achieving

and maintaining precise optical alignment is crucial to realizing the intended system performance, yet this remains a significant challenge under real-world operational conditions.^[102] Despite the challenges associated with fabrication complexity, alignment sensitivity, and mission-specific designs, the integration of freeform optics into SIS designs represents a transformative advancement in the field. As manufacturing technologies such as ultra-precision machining, freeform metrology, and hybrid additive-subtractive techniques continue to mature, the barriers to implementing freeform elements will likely diminish, making them more accessible for both research and commercial deployment. In conclusion, freeform optics not only enhance the performance of SISs—they fundamentally redefine their design potential. Their continued advancement holds the promise of enabling a new generation of compact, high-precision, and application-tailored imaging platforms that can address challenges across remote sensing, biomedical diagnostics, environmental monitoring, and beyond.

3.3. Integrated Spectral Filters-on-Chip based Designs

Advancements in on-chip spectral imagers have revolutionized the field by enabling the integration of advanced spectral imaging capabilities directly onto microchips, resulting in highly compact and efficient devices. This miniaturization is largely driven by the development of sophisticated fabrication techniques that embed spectral filters directly onto imaging sensors.^[103] These on-chip systems can capture multiple wavelengths of light simultaneously, eliminating the need for bulky external optics and significantly reducing the system size, weight, and complexity. A notable example is the work by Zhang et al., who proposed a miniature computational spectrometer that integrates silver nanoparticles (AgNPs) embedded in Fabry-Pérot (FP) microcavities with AI-driven spectral reconstruction. The design exploits the strong coupling between the localized surface plasmon resonance of AgNPs and the FP cavity modes, enhancing the spectral discrimination capability of the microfilters (**Figure 9a**).^[104]

The system uses a CMOS image sensor to record the optical responses of these microfilters, which are then processed through a neural network to reconstruct the incident light spectrum. Their approach achieves a high average spectral resolution of 0.65 nm across the visible range and offers a scalable architecture suitable for portable and in situ spectral analysis and imaging. While the design holds promise for widespread deployment, and because all fabrication steps are CMOS-compatible and maskless, the microfilter array (< 200 μm thick) can be mounted on existing image sensors or even smartphone cameras, making the system highly compact and scalable. However, the design entails significant manufacturing complexity—requiring precise nanoparticle sizing, cavity uniformity, and UV exposure control—which can reduce yield. Densely correlated filter responses make reconstruction computationally intensive and sensitive to noise and fabrication nonuniformity, demanding extensive calibration and potentially hardware acceleration. Extending beyond the 395–725 nm range adds further material and process challenges.

Wang et al., demonstrated an on-chip spectrometer based on photonic crystal (PC) slabs that generate complex spectral responses, enabling high spectral resolution while maintain-

ing compatibility with CMOS technology for low-cost, scalable manufacturing. The proposed spectrometer utilizes micrometer-thin dielectric layers with periodic nanostructures to extend the effective optical path length, thereby enhancing spectral resolution. Within these PC slabs, light undergoes multiple internal reflections, producing diverse transmission spectra through guided-mode resonances, Fabry-Pérot resonances, and Fano interference—mechanisms that collectively support efficient single-shot spectral acquisition. As shown in **Figure 9b**, arrays of PC slabs with varying periodicities, lattice constants, and hole sizes were fabricated directly on a CMOS sensor. This configuration enabled each photodetector pixel to capture distinct spectral features. To demonstrate HSI capability, a 10 × 10 array of PC slab spectrometers was integrated onto the CMOS detector. Their system successfully captured a hyperspectral image of numerals illuminated at different wavelengths, showcasing the potential for detailed spatial-spectral analysis.^[105]

Building upon similar principles, Zhu et al. introduced a compact CMOS-compatible spectral sensor utilizing two-dimensional photonic crystal (PC) slabs integrated directly onto photodetector arrays (**Figure 9c**). These PC slabs, fabricated from silicon nitride (Si₃N₄), enable the generation of diverse transmission spectra through guided-mode resonances. They employed advanced signal processing algorithms to reconstruct the incident spectra with high resolution from these distinct optical signatures. Their design offers a low-cost, scalable platform for HSI, making it suitable for use in consumer electronics, industrial inspection, and compact analytical devices. However, the design is limited to visible band, effectiveness hinges on photonic-crystal quality and uniformity, and optical losses in Si₃N₄ may reduce throughput.^[106]

In another approach, Vasireddy and Meldrum developed a spectroscopic system based on linear variable optical filters (LVOFs) deposited directly onto the protective glass of OmniVision monochrome and color image sensors. Specifically, the filter structure consists of a variable-thickness SiO₂ dielectric spacer layer, sandwiched between two semi-reflective silver (Ag) mirrors, forming a Fabry-Pérot interferometer with a spatially varying resonant wavelength along the length of the filter. This gradient in dielectric thickness enables the filter to transmit different wavelengths at different spatial locations, effectively mapping spectral information onto the sensor array during image acquisition. By depositing the LVOFs directly on the cover glass—rather than as a separate component in the optical path—the system avoids alignment issues and reduces optical losses typically associated with free-space filter assemblies. Additionally, the design supports both monochrome and Bayer-patterned color sensors, offering flexibility for integration into existing imaging architectures. As illustrated in **Figure 9d**, the result is a highly compact, solid-state spectral imager with no moving parts, well-suited for applications requiring low power, small volume, and real-time multispectral capture. Their method shows strong potential for use in medical diagnostics, environmental sensing, and consumer electronics, where cost, size, and system simplicity are critical constraints.^[107]

Similarly, Yako et al. introduced a CMOS-compatible spectral imager utilizing an array of Fabry-Pérot filters with spatially varying cavity thicknesses integrated on a monochrome sensor (**Figure 9e**). These FP filters are fabricated with sub-wavelength

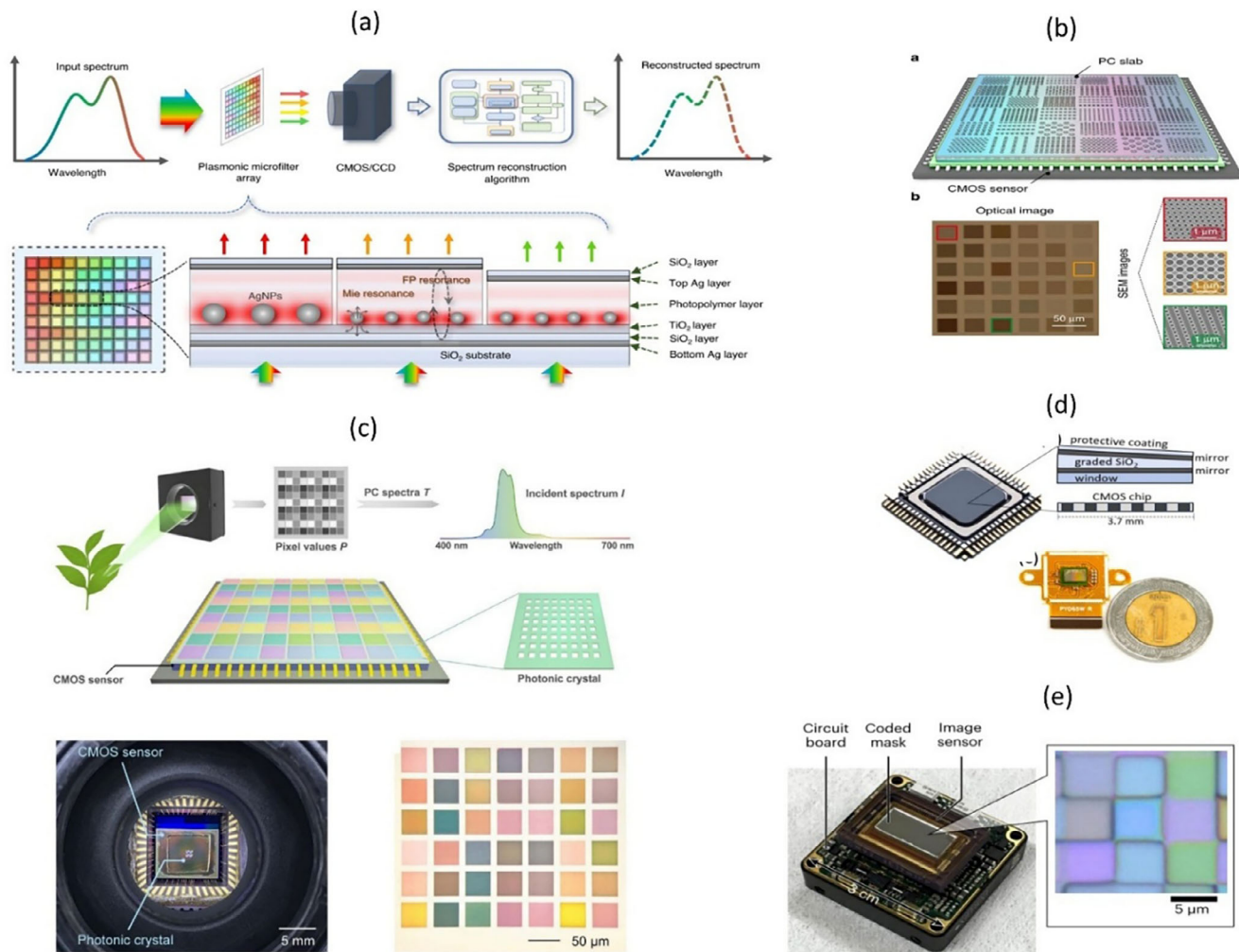


Figure 9. Illustration of five distinctive designs of compact on chip spectral imaging systems: a) The schematics illustrating the working fundamentals of the computational spectral imager based plasmonic nanoparticles in a cavity microfilter. Reproduced under the terms of the CC-BY 4.0 license.^[104] Copyright 2024, Springer Nature. b) The schematics demonstrating the miniaturized spectral imager based on photonic crystal, including the integrated PC slab on to the CMOS sensor chip and the optical and SEM images of the fabricated photonic crystal slabs. Reproduced under the terms of the CC-BY 4.0 license.^[105] Copyright 2024, Springer Nature. c) A photonic crystal (PC) based on chip spectral imager, including the operating principle and Schematics: The sensor's functionality relies on an array of PC slabs situated on a CMOS image sensor. Each PC slab exhibits a unique transmission spectrum, forming a sampling basis T . When light hits the sensor, the pixel values and the sampling basis T work together to determine the unknown spectrum. At bottom is the device Photograph depicting an array of PC slabs fabricated on a silicon nitride (SiN_x)/quartz substrate, integrated with a CMOS image sensor, and the optical micrograph displaying the array of PC slabs. Reproduced under the terms of the CC-BY 4.0 license.^[106] Copyright 2019, Optica Publishing Group. d) The schematic of an on chip spectral imager, including the coated CMOS camera, schematic cross section of the filter and the camera, photographic illustration of the fabricated device. Reproduced with permission.^[107] Copyright 2023, Elsevier. e) Photograph of the spectral imager based on coded mask with the mask consisting of square cells array shown in the inset. Reproduced under the terms of the CC-BY 4.0 license.^[108] Copyright 2023, Springer Nature.

precision to create a set of miniaturized optical resonators, each designed to transmit a narrow spectral band based on its specific optical cavity length. The resulting filter array covers a wide spectral range and produces spatially and spectrally randomized transmittance patterns across the image sensor surface. This spectral randomness is not a byproduct but a fundamental feature of the system design—optimized for compressive sensing (CS). Unlike traditional spectral imagers that rely on sequential scanning or fixed spatial dispersion mechanisms, this architecture captures a compressed representation of the hyperspectral data cube in a single snapshot. Each pixel essentially re-

ceives a unique, encoded mixture of spectral information, governed by the local FP filter transmission profile. To reconstruct the original spectral data cube from these compressed measurements, the authors employ an iterative reconstruction algorithm based on the Two-step Iterative Shrinkage/Thresholding (TwIST) method. TwIST is a well-established algorithm in sparse signal recovery and compressed sensing, particularly effective for solving large-scale, ill-posed inverse problems. In this context, it enables the recovery of high-dimensional spectral information by exploiting the sparsity and low-rank structure of natural hyperspectral scenes. Their approach delivers high sensitivity and

spectral resolution while significantly improving temporal performance, achieving frame rates far beyond those of conventional scanning-based HSI systems. The fully solid-state design, enabled by CMOS-compatible fabrication, not only ensures seamless integration into modern imaging pipelines but also offers low power consumption, compact form factor, and robustness against mechanical vibration—qualities that are particularly beneficial for real-time applications in fields such as robotic vision, biomedical imaging, remote sensing, and industrial inspection. However, the design is limited to 20 bands (450–650 nm), still relies on iterative reconstruction, fixed mask design, and filter weight/cost unspecified.^[108]

Additionally, Zhang et al. presented the development of a handheld snapshot multispectral camera, termed THETA, capable of real-time video-rate multispectral imaging at tens-of-megapixel resolution (Figure 10a). This system represents a significant advancement in the field of portable spectral imaging, uniquely combining low-cost materials, computational imaging, and high-resolution sensing to enable practical field deployment without sacrificing performance. At the core of THETA is a thin-film spectral encoding mask, fabricated using conventional film camera techniques and commercially available color-positive film. This thin mask encodes a unique spectral transmission pattern across its surface, acting as a spatially varying filter that modulates incoming light with spectral content before it reaches the sensor. The use of color-positive film offers a cost-effective, easily manufacturable solution that can be tailored to encode specific spectral bases. The mask is mounted directly in front of a high-resolution CMOS image sensor, and critically, a fiber optic plate (FOP) is used to relay the encoded optical signal from the mask onto the sensor array with high spatial fidelity.

The FOP ensures precise alignment and minimizes light loss, distortion, and parallax, enabling sharp transfer of the spectral modulation pattern to each pixel. This configuration eliminates the need for traditional bulky relay optics or lens assemblies, dramatically reducing the size and complexity of the optical system—making it truly handheld and field-deployable. To decode the complex spatio-spectral information embedded in each captured frame, the system leverages a deep neural network (DNN), trained to reconstruct the underlying spectral image cube from the compressed 2D projection. The DNN is capable of mapping the encoded patterns to their corresponding spectral signatures, allowing for real-time spectral video reconstruction at high spatial resolutions and full-frame rates. The THETA system exemplifies how emerging trends in computational imaging and machine learning can be harnessed to bypass traditional limitations in optics, offering a pathway toward scalable, cost-effective, and high-performance multispectral imaging solutions that are both accessible and field-ready. However, the cons include delicate mask alignment, reliance on trained neural network (GPU), and overall system cost.^[109]

Tang et al. reported the development of a high-performance HSI system that utilizes mercury telluride (HgTe) colloidal quantum dots (CQDs) integrated with a distributed Bragg reflector (DBR) filter array to achieve a multi-channel, narrowband photoresponse (Figure 10b). Their design leverages the broad spectral tunability and high infrared absorption efficiency of HgTe CQDs, which are solution-processable and compatible with large-area, low-cost fabrication methods. Each pixel in the sensor ar-

ray is spectrally selective, thanks to the integration of a Fabry-Pérot (FP) optical cavity, formed by placing a dielectric optical spacer layer between two highly reflective DBRs. The DBRs, composed of alternating high- and low-refractive-index materials, act as broadband mirrors, creating well-defined optical resonances within the cavity. By adjusting the cavity thickness—i.e., the spacer layer dimensions—across the array, the system enables control over the spectral transmission windows, effectively segmenting the array into multiple, narrowly defined spectral channels. The resulting sensor array exhibits high responsivity and detectivity, with values exceeding 10^{10} Jones across a wide spectral and thermal operating range—from room temperature (295 K) down to cryogenic levels (75 K). This performance under varying temperature conditions makes the system especially suitable for infrared imaging in harsh environments or thermally sensitive applications. Furthermore, the sensor exhibits a fast response time of approximately 120 nanoseconds, which is crucial for applications requiring high frame rates or dynamic scene capture, such as defense surveillance, optical communications, or industrial inspection. The combination of narrow spectral response, fast operation, and compact architecture positions this technology as a promising candidate for next-generation, high-speed infrared hyperspectral imagers. However, challenges remain in CQD stability and uniformity, precise alignment of filters is critical, the number of spectral channels per pixel is limited, and accurate cross-band reconstruction requires meticulous calibration.^[110]

In contrast, August et al. developed a miniature ultra-spectral imaging system based on a single liquid crystal phase retarder (LCPR), designed to perform compressive hyperspectral imaging in a highly compact form factor (Figure 10c). The LCPR, a voltage-controlled birefringent device, is used to modulate the phase of incident light in a wavelength-dependent manner. This phase modulation encodes spectral information into the intensity domain when combined with polarization optics and a conventional monochrome image sensor. The use of compressive sensing techniques allows the system to acquire hyperspectral data cubes from a limited set of measurements, overcoming the conventional Nyquist sampling limit. This is particularly beneficial in reducing acquisition time, data storage requirements, and computational load, while still enabling the recovery of high-resolution spectral content. Their approach utilizes dynamic spectral tunability, facilitated by the electrically addressable LCPR, and fine spectral resolution within a mechanically simple, lens-free architecture. The absence of mechanical scanning components increases system reliability, reduces power consumption, and allows for miniaturization. However, reconstruction is iterative and computationally intensive, often requiring seconds to minutes per frame. Mask fabrication demands precise liquid crystal layering, and spatial resolution is ultimately limited by the sensor's pixel size.^[111]

However, Sun et al. take a slightly different yet complementary route by emphasizing spectral-response modulation mechanisms using solution-processable semiconductors and 2D materials. They highlight the role of bound states in the continuum (BIC) and perovskite materials to realize low-cost, high-resolution on-chip spectrometers with customizable wavelength extraction characteristics. Such materials enable the construction of reconfigurable and wavelength-selective components that are

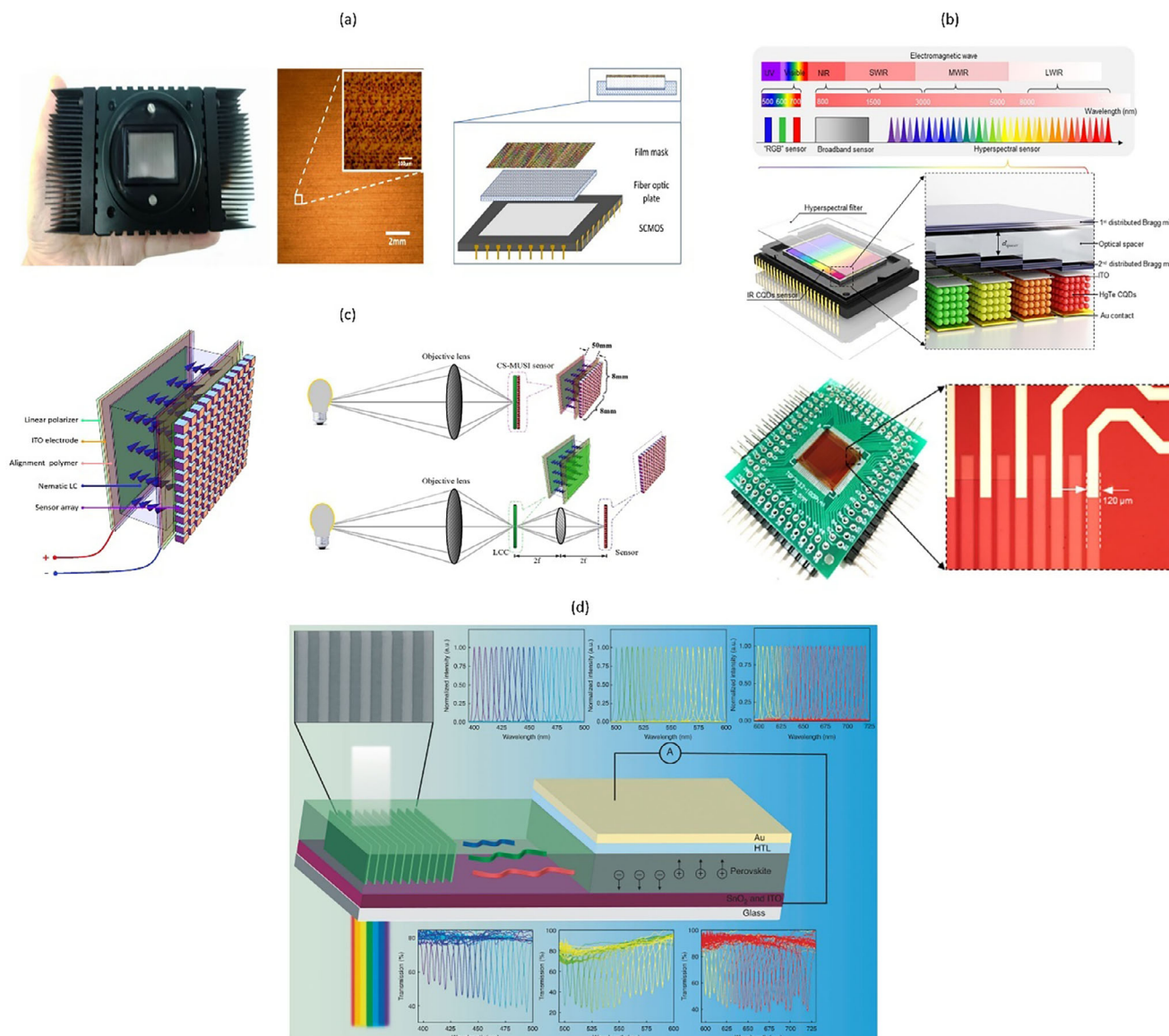


Figure 10. Illustration of three assorted designs of compact on chip spectral imaging systems: a) the tens-of-megapixels handheld multi-spectral videography approach (THETA) including the photograph of the device, the image of the film mask proposed, and the schematic of the proposed system. Reproduced under the terms of the CC-BY 4.0 license.^[109] Copyright 2023, Springer Nature. b) The schematics illustrating the working fundamentals of colloidal quantum dot IR HSI imager, including (top) the schematic highlighting the differences in spectral sensing ranges among “RGB” sensors, broadband sensors, and hyperspectral sensors. The primary infrared imaging bands encompass near-infrared (NIR, 0.7–1.5 μm), short-wave infrared (SWIR, 1.5–3 μm), mid-wave infrared (MWIR, 3–5 μm), and long-wave infrared (LWIR, 8–12 μm), all aligning with the transparent infrared atmospheric window. A detailed schematic representation of a CQD HS imager (middle), which includes a CQD sensor and a HS filter. This setup features an optical spacer positioned between two distributed Bragg mirrors, with the CQD sensor array directly fabricated onto the filter array of the proposed system. The photograph of the fabricated device (bottom). Reproduced under the terms of the CC-BY 4.0 license.^[110] Copyright 2019, John Wiley, and Sons c) The spectral imaging system featuring a grayscale standard imaging array sensor (depicted in purple) mounted on a thick liquid crystal (LC) phase retarder, which measures tens of micrometers in thickness with a clear aperture of around one centimeter. The phase retarder consists of an LC layer sandwiched between two flat glass plates, which are coated with Indium Tin Oxide (ITO) (shown in orange), and two linear polarizers (depicted in green). Additionally, the glass plates are covered with a polymer alignment layer (shown in pink). The LC cavity is filled with the commercially available LC material E44 from Merck, selected for its high birefringence and rapid response. The birefringence is adjustable through an electric field applied to the ITO electrodes. The figure illustrates how the anti-parallel alignment of the LC molecules (indicated in blue) is altered in response to the applied voltage, thereby modifying the effective retardance. On the right, is the schematic diagram of the experimental setup showing an objective lens projecting the image onto the LCC. Reproduced under the terms of the CC-BY 4.0 license.^[111] Copyright 2016, Springer Nature. d) Schematic illustrating the operating principle of the conjugated-BIC photonic spectrometer, which combines conjugated-BIC photonic structures with integrated waveguides and perovskite photodiodes. Ultra-narrowband spectral responses are achieved by tailoring the grating design within the perovskite-based system. Reproduced under the terms of the CC-BY 4.0 license.^[112] Copyright 2023, Springer Nature.

not only scalable and cost-efficient but also compatible with flexible substrates, thereby broadening the scope of integration beyond rigid silicon chips (Figure 10d). Their approach holds significant promise in terms of adaptability and cost-effectiveness. It leverages low-cost materials and printing techniques to deliver spectral imaging platforms that could be used in disposable diagnostics or flexible electronics. However, the use of non-standard materials like perovskites may raise concerns about long-term stability, environmental robustness, and integration with mature semiconductor processes.^[112]

In addition, one compelling direction in this domain involves the integration of pyroelectric materials with plasmonic metasurfaces, as demonstrated by Stewart et al. (Figure 11i). They present an ultrafast pyroelectric photodetector design that incorporates an aluminum nitride (AlN) thin film layered beneath a resonant plasmonic metasurface.^[113] This configuration allows the detector to exhibit a response time in the sub-nanosecond regime—700 picoseconds rise time and 1.7 nanoseconds full width at half maximum (FWHM)—while maintaining wavelength selectivity across the visible to near-infrared range (660–2000 nm). Notably, the device bypasses the conventional limitations of pyroelectric sensors such as sluggish response times and low responsivity.

By tailoring the metasurface resonance rather than relying on intrinsic material absorption, this approach separates thermal generation from wavelength selection, enabling the development of ultrafast hyperspectral detection platforms that operate at room temperature. This design combines speed and selectivity without the need for cryogenic cooling, making it suitable for high-speed optical sensing. However, its reliance on thermal mechanisms means that responsivity can be lower than that of semiconductor-based photodiodes under certain conditions. Additionally, fabricating uniform plasmonic arrays on a chip scale may require precise lithographic alignment, which could affect scalability and cost.

Parallel to thermal detector advancements, the field of diffractive optics has made significant strides in simplifying and miniaturizing hyperspectral imaging. Jeon et al.^[114] propose a DOE-based snapshot hyperspectral camera that eliminates traditional prisms, coded masks, and relay optics. Instead, a single diffractive optical element affixed directly to a CMOS sensor encodes spectral information into a spatially varying point spread function (PSF). The innovation lies in the spectral rotation of the PSF, which remains spatially invariant in size but rotates anisotropically with wavelength. This property, coupled with a custom-trained unrolled neural network, facilitates accurate reconstruction of hyperspectral data cubes. The design notably achieves snapshot imaging in a minimalist form factor, pushing forward the feasibility of hyperspectral imaging in consumer-grade and mobile platforms (Figure 11ii). This architecture replaces an entire HSI assembly with a single patterned layer, reducing weight and power consumption. Furthermore, its reliance on machine learning for reconstruction introduces flexibility. On the downside, spectral fidelity is highly dependent on algorithmic recovery, which can be sensitive to noise and may struggle with scenes outside the training domain. The rotational PSF approach also limits spatial resolution and places demands on calibration and deblurring accuracy.

Another strategy to achieve compact spectral imaging involves mechanical filtering elements, as explored by Wang et al. in

their miniaturized filtering-wheel-based multispectral imaging system. Although not purely planar or metasurface-based, their system exemplifies how thoughtful integration of micro-scale electromechanical components, such as a custom PCB-based motor and tightly packed filter array, can yield high-resolution MSI in a compact ($\Phi 38 \text{ mm} \times 23 \text{ mm}$) and lightweight ($\approx 98 \text{ g}$) form factor. The device supports modularity, allowing it to be tuned for various detection scenarios. It also demonstrates the viability of spectral imaging in real-time operational environments like camouflage detection, where minimal hardware complexity is desired.^[117]

Danz et al. presents the design, fabrication, and demonstration of a highly compact, wafer-level integrated array of microspectrometers aimed at enabling parallel, snapshot spectral sensing. Each spectrometer channel employs a compact 6.6 mm optical path composed of a prism-grating combination and two microlenses to spectrally resolve light from a $10 \times 100 \mu\text{m}$ slit. Arranged in a tiled 2D array and directly mounted on a $4/3''$ CCD sensor, the design enables over 2000 parallel spectrometer channels covering the 450–900 nm range. The system is effective for microscale spectral imaging and fluorescence detection under microscopy, with high spatial resolution, low stray light, and scalable wafer-level fabrication. Their proposed design offers several notable advantages. Its compact, wafer-level design allows for large-scale integration and parallel spectral acquisition, enabling real-time snapshot MSI without mechanical scanning. The use of planar micro-optics and prism-grating combinations supports efficient dispersion over a small footprint, making the system well-suited for integration into microscopes or portable imaging platforms. Additionally, the high fill-factor array enables over 2000 spectrometer channels on a single CCD, providing good spectral resolution ($\approx 11.5 \text{ nm/pixel}$) and acceptable spatial sampling within each slit.^[116]

However, these benefits come with trade-offs. The system's fixed geometry and slit-based sampling constrain its spatial resolution across the field of view and reduce flexibility for various applications. Miniaturization presents fabrication and alignment challenges, especially in ensuring uniformity and minimizing stray light across channels. Additionally, the narrow slits and small apertures limit light throughput, making the design less optimal for low-light environments or high-dynamic-range imaging scenarios.

In extending beyond spectral analysis to spectropolarimetry, Li et al. propose a compact snapshot spectropolarimeter (MINI-ORRISp) that employs a passive polarization modulator with a single multi-order retarder and linear polarizer (Figure 11iii). The encoded spectral-polarimetric data is decoded using a self-calibrating deep prior-based reconstruction algorithm. The entire system weighs only 410 g and fits in a $60 \times 66 \times 53 \text{ mm}^3$ envelope. It is capable of simultaneously capturing the full Stokes parameters over 72 spectral bands in the 400–850 nm range. The removal of mechanical scanning mechanisms enhances its applicability to dynamic and real-time sensing tasks, especially in field-deployable instruments. Their design expands spectral imaging capabilities to include polarization, enhancing information richness in scientific and defense applications. The passive configuration reduces energy consumption and simplifies alignment. Nonetheless, the system's accuracy can depend heavily on the quality of reconstruction algorithms, and it may face

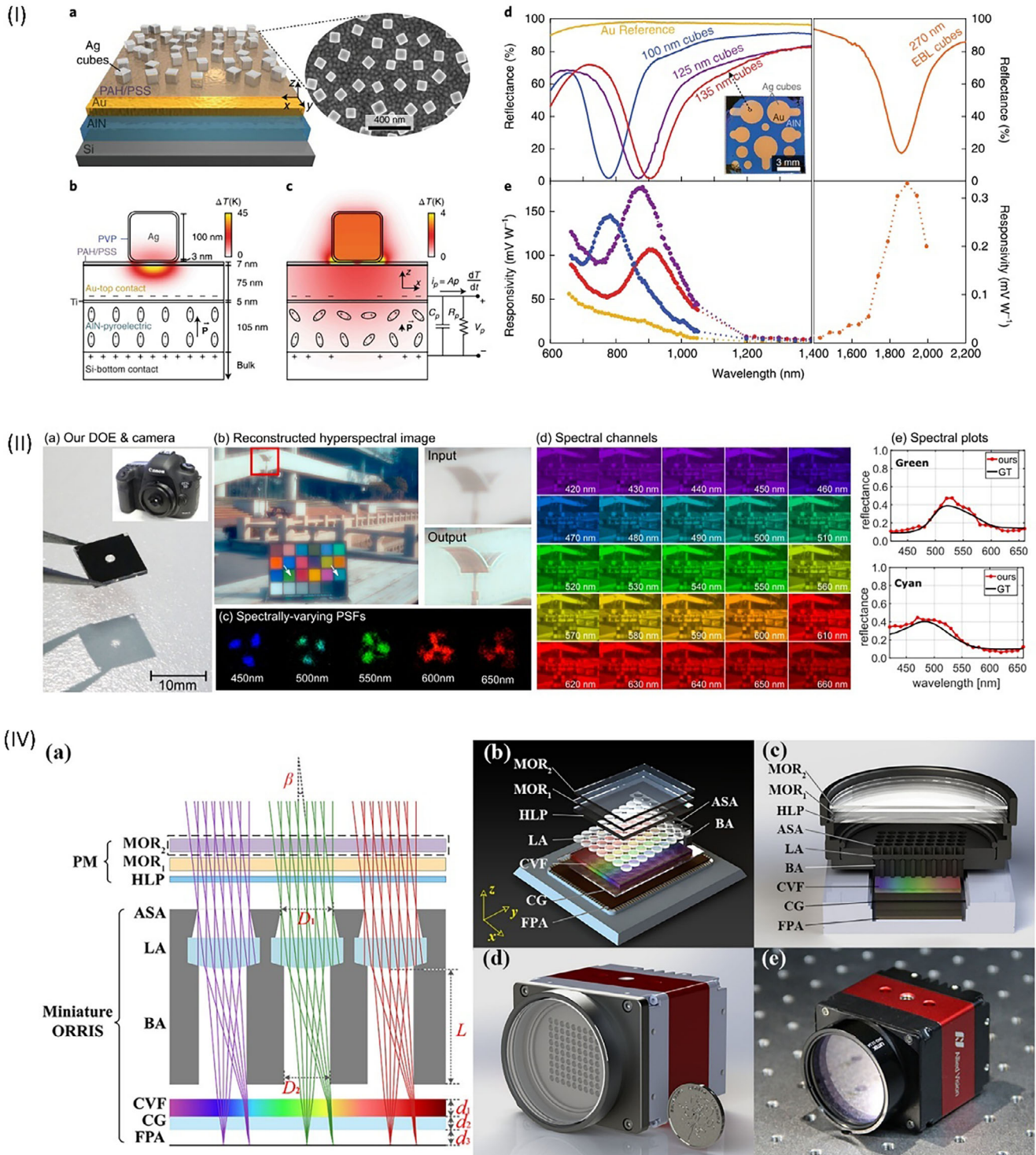


Figure 11. I) a) Schematic of a vertical detector integrating a plasmonic metasurface of 100 nm Ag cubes on a polycrystalline AlN pyroelectric layer, alongside an SEM image of the fabricated structure, b,c) Simulated thermal response of a metamaterial element at 1 ps and 40 ps post-excitation, illustrating temperature-driven changes in spontaneous polarization (P) and the corresponding pyroelectric signal. An equivalent circuit model shows C_p (capacitance), R_p (resistance), and V_p (voltage), d) Reflection spectra from three colloidal metasurface-pyroelectric detectors, a gold-based reference device, and a metasurface fabricated via electron beam lithography (EBL). The inset shows detectors with 135 nm Ag cube metasurfaces of varying diameters (3.0, 2.0, 1.5, and 1.0 μm). e) Normalized photovoltage spectra from all five detectors, highlighting strong agreement with their respective absorption profiles. PAH: polyallylamine hydrochloride; PSS: polystyrene sulfonate; PVP: polyvinylpyrrolidone. Reproduced with permission.^[113] Copyright 2019, Springer Nature. II) A diffraction-based HSI approach demonstrated using a custom-designed diffractive optical element (DOE) mounted directly on a conventional image sensor, including (a) The fabricated DOE (inset) mounted on a DSLR camera, (b) Hyperspectral image reconstructed from

Table 2. Comparative Overview of the Advances in Filters-on-Chip SIS design.

Ref	Weight [kg]	Size	Cost	No. of Spectral Channels	Spectral Range [μm]	Spectral Resolution [nm]	Spatial Resolution	Acquisition Technique
[114]	< 0.5	Compact (lens + DOE on DSLR)	low	30–40	0.4–0.7	10–20	–	Snapshot with DOE, data-driven reconstruction
[115]	Not specified	~mm scale chip size	Low to medium	>100	0.4–1	<10	–	On chip spectral modulation and metasurface Optics
[116]	Array Module only	6 mm per channel	Low for mass production	2000	0.45–0.9	~10	–	Snapshot via parallel micro-spectrometer array
[112]	Chip-level, (< 0.1)	~ micron scale chip	Very low	10–100	0.4–1.7	~10	116 × 110 pixels	Single detector with spectral modulation using BIC/perovskites
[113]	< 0.001	0.0044–7.0 mm ²	Low (scalable)	5	0.66–2	~30–60	–	–
[110]	–	Microscale sensor + chip	Very low	64	SWIR - LWIR	~30–40 cm ⁻¹	–	Snapshot via CQD with Bragg filter
[103]	<0.05	~mm scale (chip scale)	Very low	–	0.4–1	0.3	–	–
[103]	–	–	Medium	–	0.38–1	0.3	–	–
[107]	–	Compact Module	\$100	–	0.33–0.715	10	–	–
[108]	–	Compact Module	Medium	20	0.45–0.65	10	~16.5 μm per 3 pixels	Single-shot compressive measurement
[109]	<0.3 (Est.)	Handheld form factor	High	12	0.4–0.7	~25	7000 × 9000 pixels	Single-shot compressive snapshot

difficulty capturing subtle polarization shifts in highly dynamic or low-light scenes.^[115]

Table 2 illustrates how on-chip spectral imagers have evolved from multi-component DSLR setups to wafer-scale microdevices, shifting the emphasis from mere miniaturization to optimized trade-offs between spectral fidelity, spatial resolution, and system complexity. Early designs such as the DOE-based snapshot imager (< 0.5 kg, 30–40 bands, 10–20 nm resolution) retain DSLR optics for spatial detail but rely on data-driven inversion to recover a 0.4–0.7 μm spectral cube^[114]. In stark contrast, chip-scale gratings and metasurface modulators compress hundreds to thousands of channels into sub-millimeter footprints: the chirped-grating spectrometer on silicon hosts more than 150 channels at 0.3 nm resolution over 0.4–1 μm ,^[115] while photonic-crystal slab arrays offer ~1 nm resolution across 0.38–1 μm with no post-capture deconvolution. Parallel micro-spectrometer modules push this further, placing 2000 micro-spectrometers in a 6 mm array to achieve ~10 nm resolution from 0.45–0.9 μm in a single snapshot. These high-channel-count systems inher-

ently demand precision lithography and yield-sensitive fabrication, yet they unlock spectral discrimination at a level previously reserved for bench-top instruments. Cost-efficiency is another central theme, with several systems emphasizing affordability either in raw cost (e.g., \$100 in^[107]) or through scalability for mass production (e.g.,^[116]). Systems such as^[112] and^[103] achieve very low costs while retaining moderate spectral resolution, making them attractive for widespread deployment in resource-limited settings. In contrast, devices like^[109] are relatively expensive but provide significant trade-offs in performance, including very high spatial resolution (7000 × 9000 pixels), making them suitable for high-end medical or scientific imaging. The spectral capabilities of these systems vary significantly. Most devices operate in the visible to near-infrared (VNIR) range, with a few extending into short-wave (SWIR) and long-wave infrared (LWIR) regions. The number of spectral channels spans a wide range, from as few as 5^[113] to as many as 2000,^[116] depending on the sensor architecture and intended application. Spectral resolution values tend to hover around 10 nm for many systems, with some offering sub-10 nm

real-world input, c) Measured PSFs across wavelengths, d) Captured spectral channels, and e) Spectral profiles from two Color Checker patches compared to ground truth. Reproduced with permission.^[114] Copyright 2019, Association for Computing Machinery (ACM). III) MINI-ORRISp hyperspectral polarimetric imager, including: a) optical schematic, b) internal architecture of the mimic module, c) mimic assembly layout, d) mimic prototype, and e) practical working prototype. Key components: AQWP – achromatic quarter-wave retarder (detachable); MOR – multi-order retarder; HLP – horizontal linear polarizer; ASA – aperture stop array; LA – lens array; BA – baffle array; CVF – continuously variable filter; CG – cover glass; FPA – focal plane array. Reproduced with permission.^[115] Copyright 2023, Elsevier.

resolution—sufficient for identifying fine material or tissue differences. However, certain systems prioritize broader wavelength coverage over fine spectral precision. In terms of spatial resolution, many systems omit specifications, likely due to design constraints or the use of single-pixel detectors. However, some notable exceptions include (116×110 pixels),^[112] ($\sim 16.5 \mu\text{m}$ per $3 \mu\text{m}$ pixels),^[108] and (7000×9000 pixels),^[109] showing that high-resolution HSI is possible even in compact form factors, though often at a higher cost. These figures reflect the trade-off between spatial and spectral fidelity, where systems must balance the number of spectral bands with the ability to resolve fine spatial details. The acquisition techniques employed are diverse, revealing the underlying engineering innovations that make these designs possible. Several systems use snapshot imaging, which enables real-time or near-instantaneous data capture—ideal for applications requiring rapid scene analysis or where motion artifacts are problematic. Others leverage compressive sensing to reconstruct hyperspectral data from a reduced number of measurements, significantly reducing the sensor complexity and data throughput while maintaining acceptable performance. Innovative designs also include spectral modulation using metasurfaces or perovskite-based BIC structures, pointing toward a growing integration of nanophotonics in spectral sensing. In summary, Table 2 highlights a rapidly evolving landscape of on-chip SIS that are pushing the boundaries of portability, affordability, and performance. The trade-offs between spectral range, resolution, and spatial coverage are managed through diverse sensor architectures and computational imaging techniques. The choice of system ultimately depends on the application—ranging from remote sensing and agriculture (requiring wide spectral bands) to biomedical diagnostics (demanding high spatial detail and compact design). The move toward affordable, scalable, real-time SIS shows the field is ready for wider commercial and industrial use.

In summary, in the evolution of chip-scale SIS is underpinned by shared goals: miniaturization, cost-efficiency, and enhanced functionality. By employing advanced materials and fabrication techniques—such as photonic crystal structures, Fabry–Pérot resonators, and plasmonic components—these systems deliver high spectral resolution and sensitivity within compact, integrated platforms. These innovations are further amplified by the incorporation of sophisticated signal processing approaches, often powered by machine learning algorithms. This fusion enables real-time spectral reconstruction and analysis, paving the way for detailed material identification and biological tissue assessment with unprecedented accuracy. Importantly, advances in sensor materials and readout electronics have improved performance in low-light conditions, while supporting high frame rates. This makes on-chip spectral imagers well-suited for dynamic applications requiring both speed and sensitivity, such as live biomedical diagnostics, high-throughput screening, and rapid field-based monitoring.^[108]

In parallel, the adoption of CMOS-compatible fabrication has dramatically lowered manufacturing costs, enabling scalable production and seamless integration into existing electronic ecosystems. As a result, SIS is becoming more accessible across diverse sectors—from consumer electronics and precision agriculture to industrial automation and environmental monitoring.^[108]

Looking forward, the trajectory of miniature spectral imaging will be defined by continued breakthroughs in functional materials, scalable microfabrication, and AI-integrated processing. These advances are expected to drive the development of next-generation systems that are smaller, faster, and more intelligent—transforming spectral imaging into a widely deployable tool for both everyday use and high-end scientific exploration. However, despite these impressive gains, on-chip spectral imagers still face several challenges. One key limitation is the restricted spectral range and resolution resulting from the integration of fixed spectral filters directly onto the chip. While this compact integration reduces system complexity, it can limit the specificity of detectable wavelengths—potentially excluding critical spectral features required for certain applications. Additionally, these devices often demand precise calibration to maintain accuracy, which can be difficult to achieve and sustain in varying operational environments. Minor misalignments or filter degradation can introduce significant errors in the captured spectral data. Manufacturing complexity also remains a major hurdle. Integrating optical components at the microchip level requires nanoscale precision and is susceptible to defects, which can affect consistency across units and drive up production costs. As a result, while CMOS compatibility enables mass production, achieving uniform performance across large batches is still a challenge. Nonetheless, the continued evolution of cost-effective fabrication techniques and AI-driven reconstruction algorithms is steadily mitigating these issues. As materials science, nanofabrication, and computational imaging advance in tandem, the potential applications of on-chip spectral imagers are rapidly expanding—from handheld consumer diagnostics and wearable health monitors to industrial inspection and autonomous sensing platforms. This synergy is likely to propel the technology into widespread use, transforming how we capture and interpret spectral information across disciplines.^[104]

3.4. Designs Based on Metasurfaces

Classic optical phenomena—refraction, reflection, absorption, and diffraction—are utilized to manipulate light in conventional optical devices such as lenses, wave plates, mirrors, and light modulators. The gradual buildup of phase during propagation through these components, influenced by their structures, refractive indices, and dielectric constants, shapes the wavefronts. However, achieving high integration features such as on chip integration, ultra-thin structures, wavelength, and polarization control remains challenging with conventional optical components due to their limited manipulation mechanisms and material choices. Thus, the wave properties of light can be harnessed to control conventional optics through boundary conditions, a fundamental concept in electromagnetic theory that governs the establishment of magnetic and electric fields at the interfaces of optical media.^[118] By varying the topology or structure of these interfaces, the characteristics of light propagation can be finely tuned. Similarly, certain boundary conditions, such as gradient and nonlinear, would cause the classical law of refraction and reflection to be generalized.^[119–121]

Likewise, the revised Fresnel equations may be produced, by varying boundary conditions for orthogonal polarization states,

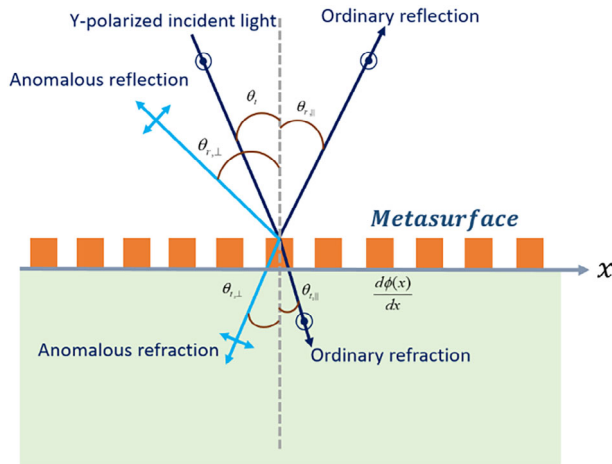


Figure 12. Illustration of the generalized Snell's law of refraction and reflection, depicting the anomalous paths of refracted and reflected beams for cross-polarized scattered light.

which implies that both the light's transmission/reflection coefficients and polarization states are controllable.^[122–124] The general relation between the incident and scattered waves is given as:^[119]

$$\begin{aligned} n_i \sin(\theta_i) - n_t \sin(\theta_t) &= \frac{\lambda_0}{2\pi} \frac{d\phi(x)}{dx} \\ n_i \sin(\theta_r) - n_i \sin(\theta_i) &= \frac{\lambda_0}{2\pi} \frac{d\phi(x)}{dx} \end{aligned} \quad (2)$$

In this context, n_i and n_t represent the refractive indices on either side of the interface. The term λ_0 denotes the wavelength in free space. Angles θ_i , θ_r , and θ_t correspond to the angles of incidence, reflection, and transmission, respectively. The term indicates $d\phi/dx$ the gradient of the phase discontinuity along the interface, which is introduced by the meta-atoms of the metasurface as illustrated in **Figure 12**. The refracted and reflected beams can be directed in arbitrary directions by introducing an appropriate constant gradient of phase discontinuity along the interface $\frac{d\phi}{dx}$. (By designing this phase discontinuity, it is possible to achieve complete control over the light's path, enabling unprecedented manipulation of anomalous reflection and refraction as governed by the generalized Snell's law (Equation 2).

The prevailing term for artificial boundaries is metasurfaces. Unlike metamaterials, metasurfaces primarily modify electromagnetic waves by leveraging boundary conditions rather than intrinsic properties in three-dimensional space.^[125] They are composed of two-dimensional arrays of sub-wavelength scatterers that are engineered to control the polarization, phase, intensity, or spectral composition of light. By employing these planar structures, metasurfaces facilitate the development of "Planar Optics".^[126] Remarkably, research interests in metasurfaces have recently experienced a tremendous growth, as they provide a comparable and unconventional phenomenon due to their remarkable characteristics in light manipulation, and flexibility in ultrathin optical applications.^[127] Unlike metamaterials, metasurfaces benefit tremendously from compatibility with the streamlined micro and nano fabrication process, and their pla-

nar profiles make them a good candidate for integration with on-chip nanophotonic devices.^[128] The incident light on optical metasurfaces is captured and re-radiated by the subwavelength nano-resonators known as meta-atoms, and depending on how these meta-atoms are constructed, the scattered light may differ from the incoming incident light in certain ways. This innovative approach has enabled the development of ultra-thin, lightweight, and highly efficient optical components that can perform complex functions such as focusing, filtering, and dispersing light with remarkable precision and flexibility.

3.4.1. Micro/Nano Optical Filters Based on Metasurfaces

The micro/nano optical filters are essential components in modern spectroscopic and imaging systems. These filters can generally be classified into two main categories: resonant narrow-band filters and random broadband filters. Resonant narrow-band filters are designed to selectively transmit a specific range of wavelengths while effectively blocking others. They rely on resonant optical structures—such as Fabry–Pérot cavities, ring resonators, or photonic crystal slabs—precisely tuned to specific wavelengths. These filters are characterized by their high spectral selectivity and sharp cutoffs, making them ideal for applications that require precise wavelength discrimination. However, one limitation of these filters is their constrained spectral resolution, often dictated by the number of discrete filters available, which in turn limits the number of usable spectral channels. In contrast, random broadband filters are designed to transmit a broad range of wavelengths without relying on narrow resonances. These filters often utilize structures that scatter or modulate light in a spectrally dependent but spatially random manner. Technologies employed in their implementation include quantum dot arrays, photonic crystal plate arrays, multilayer thin-film coatings, bandgap-tunable nanowires, and metasurface arrays. Instead of isolating individual wavelengths, these filters encode spectral information into spatially varying responses across a detector array. In these architectures, incident light interacts with a filtering structure—such as a metasurface or disordered nanostructure—that alters the light's properties based on its spectral content. The resulting spatially encoded patterns serve as fingerprints of the incoming spectra. Advanced computational methods, such as compressive sensing and deep learning algorithms, are then applied to reconstruct the full spectrum from these encoded measurements with high accuracy. This section of the review provides a comprehensive overview of micro/nano optical filter technologies, with a particular emphasis on recent advancements in metasurface-based broadband filters, which are at the forefront of compact, high-performance spectral imaging and sensing systems.

The advancements over the past few decades on the metasurface-based broadband filters, primarily driven by the development of engineered surfaces, enable unprecedented control over electromagnetic waves, leading to the creation of innovative optical devices. The metasurface-based spectral filters operate on the principle of controlling light-matter interactions at the sub-wavelength scale. These interactions are facilitated by the precise engineering of the meta-atoms that dictate the phase, amplitude, and polarization of the incident light, and allow for the creation

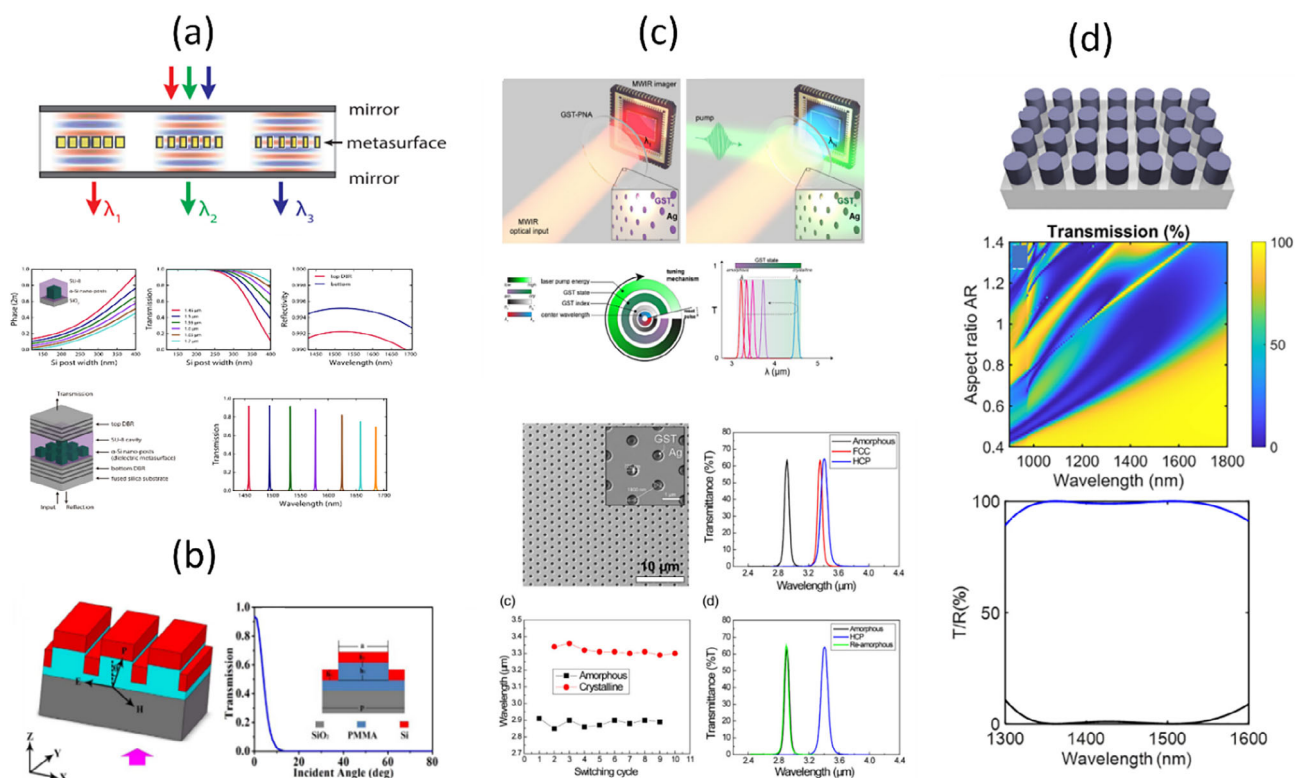


Figure 13. The proposed vertical DBR-based micro-cavities bandpass filter array, including, the schematic illustrating the transmissive dielectric metasurface layers integrated within the micro-cavities to serve as phase-shifting layers to tune their resonance wavelengths across a broad bandwidth, and the transmission characterization and the schematic diagram of the of the filter. Reproduced under the terms of the CC-BY 4.0 license.^[129] Copyright 2016, Optica Publishing Group. b) The schematic illustration of the grit grating metasurface filter, and the transmission spectra at different angle of incident. Reproduced with permission.^[130] Copyright 2019, Optica Publishing Group. c) The conceptual illustration of the Tunable GST based plasmonic metasurface filter, showing the initial stage (left) and spectral shift (right) after phase change, and the SEM image of the fabricated filter and the transmission characterization of the device. Reproduced under the terms of the CC-BY 4.0 license.^[131] Copyright 2020, Optica Publishing Group. d) The schematic representation of the silicon nano-disk metasurface filter proposed in, the computed transmission spectra depicting the relationship between the aspect ratio (AR) and the wavelength, and the computed transmission and reflection spectra for 1.05 AR at different wavelength. Reproduced with permission.^[132] Copyright 2021, AIP Publishing.

of ultra-compact broadband filters. These filters can be as thin as a few nanometers, which is beneficial for space-constrained applications in portable and miniaturized SIS, such as smart-phones and small satellites. Furthermore, the planar nature of metasurfaces allows for easy integration of these filters into existing optical systems without significantly increasing their size or weight. One of the early works in this domain, is the work of Horie *et al.* (2016),^[129] where they present a novel design for a planar array of optical bandpass filters that can be integrated with photodetector arrays to create high-resolution on-chip spectrometers. These filters are composed of low-loss dielectric metasurface layers placed between two DBRs, forming a Fabry-Pérot resonator. The key innovation is the use of the metasurface layer as a phase-shifting element, which allows for independent control of the center wavelengths of each filter by altering the in-plane geometry of the metasurfaces. Their configuration enables scalable, wavelength-selective filtering across a broad spectral range spanning approximately 250 to 1550 nm, with experimentally demonstrated quality factors exceeding 700. The metasurfaces consist of amorphous silicon nano-posts arranged in a square lattice and embedded in a low-index polymer matrix, providing high transmission efficiency and precise phase

control—characteristics critical for achieving narrowband spectral response and high spectral resolution. The DBRs are composed of alternating layers of silicon and silicon dioxide, delivering high reflectivity to enhance the finesse of the FP resonator (Figure 13a).

In 2019, Qian and colleagues presented an all-dielectric ultra-thin metasurface angular filter employing a double-layer grid grating to achieve high-efficiency directional filtering for linearly polarized light at the wavelength of 633 nm. The structure consists of a low refractive index layer of polymethacrylate (PMMA) on a silica (SiO₂) substrate, with a silicon high refractive index layer deposited on top (Figure 13b). This layered configuration enables the filter to exhibit strong angular selectivity, characterized by high transmission at normal incidence and a sharp suppression of transmission at oblique angles. Specifically, the device achieves a transmission efficiency of 93.6% at normal incidence, which drops to nearly zero for incident angles greater than 12°, effectively blocking off-axis light. Additionally, the transmitted beam maintains a narrow divergence angle of only 7.76°, indicating precise control over the output direction. These properties make the proposed metasurface filter a promising candidate for compact optical systems re-

quiring directional selectivity and spatial light filtering such as SIS.^[130]

In 2020, Julian et al. presented the development of a tunable, reversible optical filter based on phase-change metasurfaces utilizing GeSbTe (GST). Specifically, the filter employs GST-225, a phase-change material known for its substantial refractive index contrast between its amorphous (a-GST) and crystalline (c-GST) states. The GST is embedded within a plasmonic nanohole array (PNA), which enhances the local electromagnetic field and enables dynamic tuning of the spectral response (Figure 13c). The proposed design demonstrates a high transmission efficiency (~70%) and a narrow bandwidth (~74 nm) within the mid-wave infrared (MWIR) range (3–5 μm). This approach offers considerable potential for real-time reconfigurable optical filtering in applications such as SIS and thermal sensing. However, long-term stability of GST materials and achieving reliable, reversible phase transitions remain key challenges for practical deployment.^[131]

In 2021, Zheng et al. introduced a planar narrowband filter based on silicon resonant metasurfaces (Figure 13d). This design harnesses the multipolar resonance characteristics of dielectric metasurfaces to achieve high spectral selectivity in the near-infrared region. The filter combines electric dipole (ED) and magnetic dipole (MD) resonances to establish a broad suppression region, while a sharp transmission peak is generated by breaking a BIC, thus forming a high-quality leaky mode resonance. Their proposed architecture provides strong angular stability and high filtering performance. Nonetheless, the fabrication of such structures demands nanometer-scale precision, and the realized efficiency is still somewhat below theoretical limits.^[132]

In the same year, an inverse design methodology for metasurface optical filters was proposed by Han et al. utilizing DNNs to dramatically expand design freedom. Their two-stage framework consists of a simulator network, which predicts the spectral response of a metasurface design, and a generator network, which produces structural parameters to meet specific optical targets. This DNN-based strategy significantly reduces design iteration time and supports the exploration of non-traditional geometries and complex unit cell patterns, thereby enhancing the capability to tailor desired spectral responses. However, the effectiveness of this approach is highly contingent on the quality and diversity of the training dataset, and translating these AI-generated designs into real-world devices introduces challenges in nanofabrication precision and material constraints.^[133]

In 2022, Kim et al. presented the development of an ultra-wideband transmission filter based on guided-mode resonance (GMR) in terahertz (THz) metasurfaces. The filter exploits GMR—a phenomenon where specific wavelengths couple into leaky guided modes supported by a dielectric slab, leading to resonantly enhanced transmission (Figure 14a). The metasurfaces are composed of periodically arranged gold disks, which serve as the resonant elements. By symmetrically placing identical metasurfaces on both the upper and lower surfaces of a thick quartz substrate, the structure enables the simultaneous excitation of even and odd GMR modes, thereby broadening the transmission bandwidth. This dual-resonance interaction significantly enhances both the spectral width and peak efficiency of the filter. However, the need for precise nanoscale patterning of gold disks introduces fabrication complexity, and scaling such high-

resolution metasurface architectures for large-area commercial applications remains a technical challenge.^[134]

Similarly, De Proft et al. proposed a highly selective color filter design that combines both dielectric and plasmonic resonances to achieve narrowband filtering with excellent out-of-band rejection. The structure integrates a hybrid configuration that combines Fabry–Pérot and guided-mode resonances across two spatially separated dielectric slabs, while also coupling to surface plasmon polaritons via dual parallel aluminum gratings. These gratings are laterally offset to enhance resonance coupling and are embedded within a silicon nitride slab, which supports both dielectric and plasmonic interactions. This hybrid resonance approach significantly improves full width at half maximum (FWHM) characteristics and peak transmittance. Notably, the design emphasizes scalability and cost-effectiveness, showing resilience against minor fabrication imperfections. Nonetheless, achieving the optimal balance between plasmonic confinement and dielectric resonance, as well as maintaining precise grating alignment, are key challenges for consistent large-scale manufacturing.^[135]

In the same vein, Wu et al. introduced a method for compact SIS using randomly distributed color filters based on an all-dielectric metasurface architecture. The metasurface is composed of dielectric gratings, an anti-reflection (AR) coating, and an integrated PN-junction for photodetection. The filters are arranged in a spatially random pattern, resulting in distinct, uncorrelated spectral responses at each pixel location. When combined with a CS algorithm, the system can reconstruct high-resolution spectral data-cubes from a reduced number of spatial-spectral measurements. The CS algorithm leverages the low spectral correlation induced by the random filters to recover fine spectral detail efficiently. While this approach significantly reduces hardware complexity and enables miniaturization, the precise fabrication of spatially random metasurface patterns remains nontrivial. Moreover, the fidelity of spectral reconstruction is highly contingent on the robustness and accuracy of the computational reconstruction algorithms, making algorithmic design and optimization a critical component of system performance.^[136]

In 2023, Wang et al. introduced a compact snapshot compressive spectral imaging system (SCSI) that combines dual-channel switchable metasurface filters with a deep-learning-based reconstruction algorithm. The system incorporates anisotropic titanium dioxide (TiO_2) meta-atoms integrated with twisted nematic liquid crystals (TNLCs), allowing for rapid switching between x - and y -polarized light states through voltage control (Figure 14b). This dynamic polarization control enables the capture of two distinct spectral measurements in a single shot. For reconstruction, they employed an end-to-end residual neural network (ResNet), replacing traditional iterative compressive sensing algorithms. Trained on a synthetic dataset using two-frame snapshot data from orthogonal polarization channels, ResNet significantly enhances both reconstruction speed and spectral fidelity. Their study highlights the potential of integrating metasurface technology with deep learning to overcome the limitations of conventional HSI systems. While the system achieves high spectral resolution and a compact form factor, challenges remain in scaling the architecture and ensuring robust integration between metasurface filters and learning algorithms for broader applications.^[137]

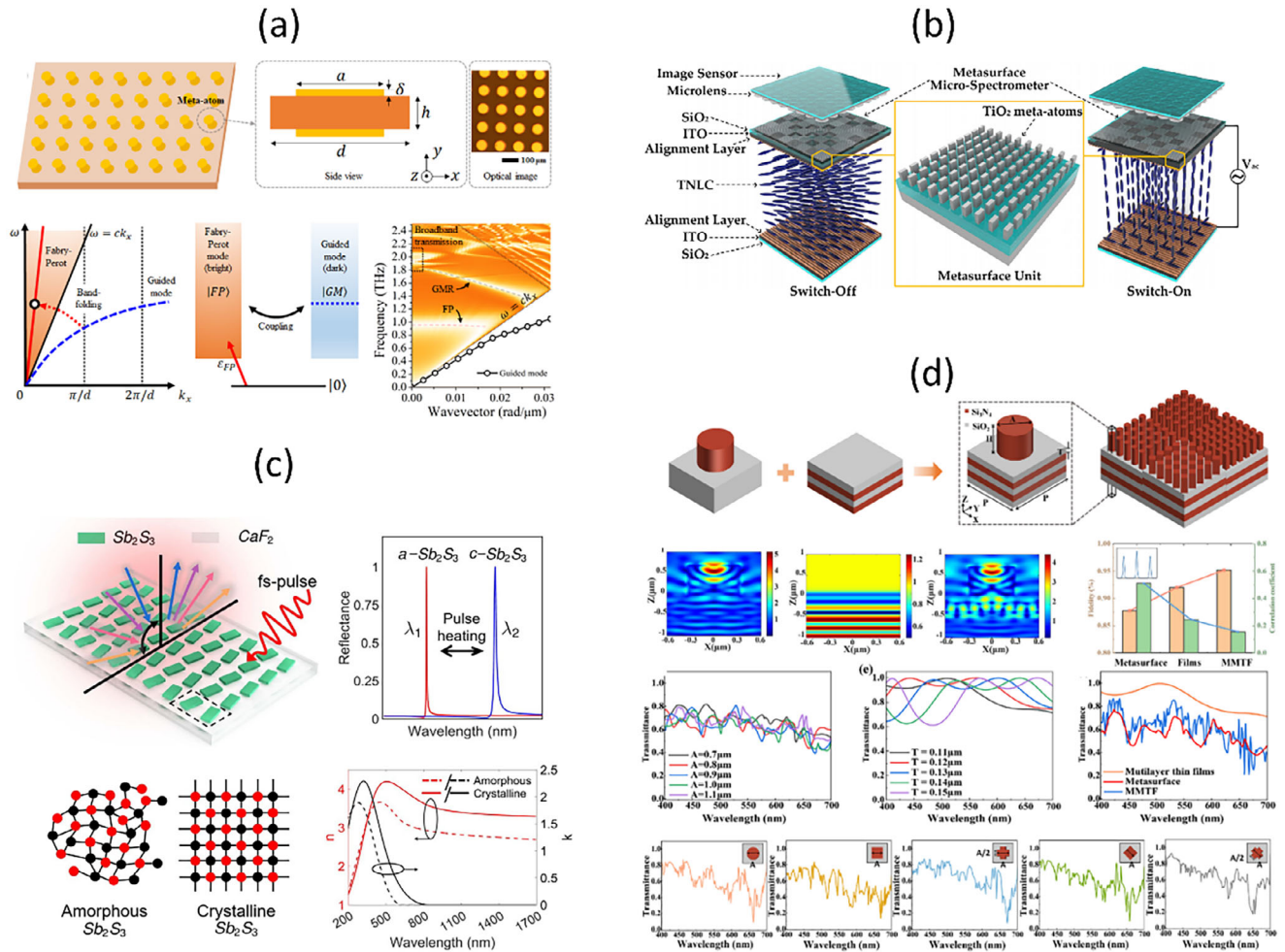


Figure 14. a) The proposed GMR in two THz metasurface-based filters includes the schematic, geometry, and an image of the fabricated device shown in the inset. It also depicts the band structure, coupling between FB and GMR within the device, and contour region. Reproduced under the terms of the CC-BY 4.0 license.^[134] Copyright 2022, Optica Publishing Group. b) Schematic illustrating the switchable metasurface based filter. Reproduced under the terms of the CC-BY 4.0 license.^[137] Copyright 2023, MDPI. c) The conceptual schematic of the BIC-based metasurface filter, including the proposed design, the resonance tuning after modeling the PCM with FS laser, the amorphous phase transition, and the crystalline phase transition. Reproduced with permission.^[138] Copyright 2023, IOP Science. d) The schematic of the broadband filter based on the combination of metasurface and multilayer thin film, including, field intensity, fidelity of the reconstructed spectrum, the transmission spectra of the metasurface at different values of A (diameter of the meta-atom), the transmission spectra of the multilayer thin film at different values of T (layer thickness), the transmission spectra of the filter at fixed T and H (Meta-atom height), and the transmission spectra of different designs of the meta-atom. Reproduced with permission.^[139] Copyright 2024, IOP Science.

In the same year, Tin et al. proposed a tunable all-dielectric metasurface filter that leverages a BIC mechanism in combination with an optical phase-change material (OPCM) for dynamic spectral tuning. The filter is based on a two-dimensional metasurface consisting of antimony (III) sulfide (Sb_2S_3) bars arranged on a calcium fluoride (CaF_2) substrate, supporting sharp resonances with minimal energy leakage due to BIC confinement (Figure 14c). These resonances are further modulated by tuning the refractive index of the OPCM via pulsed laser heating, which enables transitions between amorphous and crystalline states. Additional tuning is achieved by varying the incident angle, providing precise control over resonance wavelengths. The device demonstrates extremely sharp resonances with a high Q-factor of ~ 1100 , indicating excellent spectral resolution and minimal

loss. However, concerns remain regarding the long-term stability and reliability of the OPCM under repeated switching and environmental fluctuations. Although both simulations and initial experiments show promise, extensive real-world testing is required to fully validate the design for practical applications such as bio-sensing, thermal imaging, and spectral analysis.^[138]

Recently, a proposed design of random broadband filters using a combination of metasurfaces and multilayer thin films (MMTF) was demonstrated by Guo et al. The design seeks to enhance the degrees of freedom in optical design, enabling more intricate and efficient light modulation. The architecture features silicon nitride nanopillars atop alternating layers of Si_3N_4 and silicon dioxide, forming a highly customizable filtering platform (Figure 14d). Their structure produces low-correlation

transmission spectra, which are critical for accurate spectral reconstruction using CS. The filter responses are optimized via a particle swarm optimization algorithm coupled with CS techniques to minimize reconstruction error and maximize spectral resolution. The proposed system demonstrates a strong capability for compact, high-performance spectral imaging, showing potential for integration into miniature spectrometers. Nevertheless, the complex fabrication process and potential sensitivity to noise present challenges that need to be addressed through further refinement and development.^[139]

Similarly, Xiang et al. introduce a novel Fabry–Pérot-based multispectral filter array (MSFA) that exploits high-order resonances and subwavelength patterning to achieve high transmission, narrow bandwidths (13–31 nm), and angular insensitivity. The filters span 455–960 nm, with average transmissions exceeding 60%. The distinguishing feature is the use of ultrathin metallic layers and refractive index modulation within a fixed cavity geometry, enabling high spectral resolution without increasing fabrication complexity. These filters are CMOS-compatible and can be downscaled to 1 μm pixel sizes, aligning with the demands of modern digital image sensors. The approach circumvents common limitations such as spectral overlap, angular sensitivity, and low efficiency seen in conventional diffractive filter designs. The clear advantage of their design is its compatibility with CMOS sensors, allowing monolithic integration of spectral filters on existing imaging platforms. Its high spectral fidelity and robustness to angle and polarization variations make it attractive for demanding imaging tasks. On the downside, fabrication involves multiple layers and subwavelength features, which may complicate large-scale production. Furthermore, tuning beyond the designed passbands would require substantial redesign or re-fabrication.^[140]

In summary, metasurface-based broadband filters are compact and lightweight, utilizing nanoscale structures instead of bulky optical components. They offer great design flexibility, enabling control of the optical response across various wavelengths. This allows for custom spectral filters with tailored bandwidths and resolutions. Moreover, these filters achieve high spectral resolution and efficiency within the desired wavelength range. The potential for dynamic tunability by incorporating materials responsive to external stimuli such as electric fields or temperature changes, as demonstrated in the work of Tin et al.^[138] further enhances their functionality.

Despite these strengths, performance trade-offs are inherent in the design of metasurface-based filters. A common compromise exists between bandwidth and spectral resolution, where achieving high spectral resolution may limit the bandwidth and vice versa, depending on specific design parameters and material properties. Achieving high efficiency across a broad wavelength range also remains challenging, with filters frequently performing optimally only within a narrow range. Moreover, while dynamic tunability is advantageous, it introduces challenges regarding response time. The response to external stimuli may not be fast enough for certain applications, and implementing tunability often requires complex control systems and additional power sources, increasing the overall complexity and cost. Likewise, the performance of these filters depends critically on the dimensions and arrangements of the subwavelength structures which need to be controlled at a nanometer scale. Current fab-

rication techniques, such as electron-beam lithography and focused ion beam milling, offer high precision but are often time-consuming and costly. Scaling up these methods for mass production without compromising accuracy remains a significant hurdle.

Another challenge is the choice of materials for the metasurfaces, which is crucial for their optical performance. Metals are commonly used in plasmonic metasurfaces but suffer from high losses at optical frequencies, which degrades filter performance. Dielectric materials, while offering lower losses, are challenging to pattern with the necessary precision. Identifying materials that offer minimal optical losses while maintaining ease of fabrication is crucial for the progression of metasurface-based spectral filters. Future research should focus on developing advanced fabrication techniques that offer high precision and scalability.^[141] Techniques such as nanoimprinting and self-assembly offer potential for producing large area metasurfaces at reduced costs. Improving these methods can help overcome the current limitations in fabrication precision and scalability.^[142–145] In addition, exploring new materials with desirable optical properties and ease of fabrication is a key direction for future research. For instance, emerging materials such as transition metal dichalcogenides (TMDs), two-dimensional (2D) materials, and phase-change materials offer opportunities for designing metasurfaces with enhanced performance and tunability.

Developing metasurfaces that can dynamically adjust their optical properties in response to external stimuli, such as electrical, thermal, or optical signals, is a promising direction. Tunable metasurfaces can enable reconfigurable spectral filters that adapt to changing operational requirements, expanding their applicability. Enhancing the integration of metasurface-based spectral filters with photonic platforms, such as silicon photonics and integrated optics, can open new possibilities for compact and efficient optical systems. This integration requires designing metasurfaces that are compatible with standard photonic fabrication processes and can be seamlessly incorporated into existing devices. Finally, exploring new applications of metasurface-based spectral filters not only in SIS but also in emerging fields such as biosensing can drive innovation and development.

3.4.2. Metasurface-Based Spectrometers

Recent advancements in optical spectroscopy, driven by metasurfaces, have enabled the design of compact, lightweight, and high-performance spectrometers. These new spectrometers are now used in environmental monitoring, medical diagnostics, consumer electronics, and industrial inspection. Traditionally, optical spectrometers have relied on bulk components such as prisms and diffraction gratings to disperse light into its constituent wavelengths. While effective, these systems are often large, heavy, and mechanically complex, limiting their use in portable or on-chip platforms, where miniaturization and integration are crucial. To address these limitations, significant efforts between the early 1990s and 2000s led to the development of a variety of miniaturized spectrometers,^[146] which can be broadly classified into four categories based on their underlying spectral analysis mechanisms:^[14]

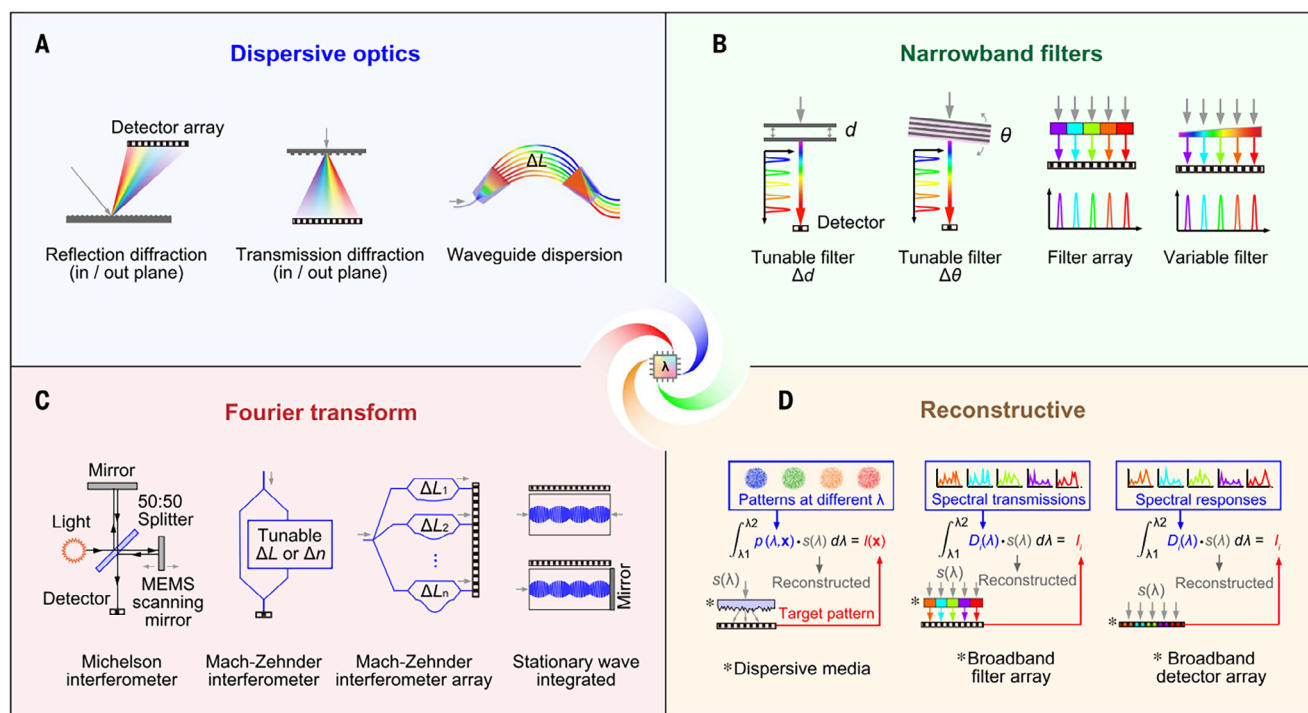


Figure 15. Summary of the advancements made in compact and miniaturized spectrometers over the past three decades. These advancements include dispersive optics, narrowband filters, Fourier transform interferometers, and computational spectral reconstruction techniques. Reproduced with permission.^[14] Copyright 2021, American Association for the Advancement of Science.

- Miniaturized Dispersive Systems:** These spectrometers retain the principle of spatially separating light into different spectral components using compact gratings or prisms, directing them to an array of photodetectors. While they maintain good spectral resolution, the reduction in size can limit throughput and alignment tolerance.
- Narrowband Filter-Based Systems:** This category includes spectrometers that use arrays of fixed or tunable narrowband optical filters (such as Fabry–Pérot cavities, thin-film filters, or metasurface-based filters) to selectively transmit specific wavelengths. These are well-suited for snapshot imaging or multispectral detection.
- Fourier Transform Spectrometers (FTS):** These devices generate interferograms using integrated interferometers, often realized with microelectromechanical systems (MEMS). The recorded temporal or spatial interferograms are converted to spectral data using Fourier transformation algorithms. FTS offer high spectral resolution and are particularly useful for broadband applications.
- Computational Spectrometers:** These systems reconstruct the spectral content of incident light using precalibrated spectral responses from a set of broadband filters or detectors. Advanced computational algorithms, including compressive sensing and machine learning, are used to decode the spectral information, significantly reducing hardware complexity.

These diverse approaches are summarized in **Figure 15**, illustrating how the strategies for achieving compact and miniaturized spectrometers have evolved over the past three decades. The extensive body of literature reflects continued innovation

in this space, incorporating integrated photonic circuits, CMOS-compatible fabrication techniques, metasurface engineering, and powerful computational methods to realize spectrometers that are not only small in size but also cost-effective, scalable, and high-performing. Recent developments have also emphasized the synergistic integration of optics and electronics, particularly through on-chip platforms that allow for direct coupling of the spectrometer to detectors, signal processors, and computational algorithms. These compact devices are paving the way for a new generation of portable, wearable, and even implantable spectral sensors, enabling real-time and in situ measurements in previously inaccessible settings. For a more comprehensive examination of recent progress and state-of-the-art methodologies in this domain, readers are encouraged to consult key reviews and research articles that delve deeper into each category and technology trend.^[10,147–158]

The transformative potential of metasurfaces in spectroscopy arises from their ability to engineer optical responses through the arrangement of nanoscale structures. These metasurfaces can function as highly efficient dispersive elements, capable of splitting light into its constituent wavelengths with exceptional resolution and compactness. Over the past decade, significant research has been devoted to exploring and expanding the capabilities of metasurfaces for spectroscopic applications.^[159] Plasmonic metasurfaces, for instance, leverage surface plasmon resonances to enhance spectral sensitivity at specific wavelengths. These are particularly useful for applications requiring narrowband filtering and high field enhancement. However, their intrinsic losses can limit overall efficiency. In contrast, dielectric metasurfaces, typically made from low-loss materials such as Si_3N_4

or TiO₂, offer broader spectral coverage and higher transmission efficiency. This makes them well-suited for applications demanding high spectral fidelity across the visible and near-infrared regions. This section of the review aims to provide a comprehensive overview of recent advancements in metasurface-based spectrometers, highlighting major innovations, design strategies, and practical implementations. By examining landmark studies, we will elucidate the working principles of these devices, assess their performance metrics, and discuss current challenges and future research directions in this rapidly evolving field.

One of the earliest significant demonstrations was presented by Xu et al., who introduced an innovative method for high-resolution color filtering and spectral imaging using plasmonic nano-resonators. These were designed as periodic metal-insulator-metal (MIM) stacks consisting of a zinc selenide (ZnSe) dielectric layer sandwiched between aluminum (Al) layers, fabricated on a magnesium fluoride (MgF₂) substrate. Each resonator supported surface plasmon modes at visible wavelengths, enabling efficient light coupling and precise color filtering. Different periodicities (360 nm, 270 nm, and 230 nm) were tailored to produce red, green, and blue filters, respectively. This work showcased the potential of plasmonic metasurfaces for compact and high-efficiency spectrometers, particularly for spectral imaging applications.^[160]

Further progress was marked by Ni et al., showcasing an ultrathin metalens that efficiently focuses visible light with minimal aberration. Their approach utilized Babinet-inverted (complementary) plasmonic nano-antennas—specifically, concentrically arranged nanovoids milled in a 30-nm-thick gold film—to manipulate the phase of cross-polarized scattered light. These nano-antennas enabled discrete phase shifts spanning the full 0 to 2 π range, allowing full control of the outgoing wavefront in an ultracompact geometry. Unlike conventional dielectric lenses that rely on bulk refractive phase accumulation, this planar design achieved focusing through surface-confined phase modulation, opening new opportunities for miniaturized, integrable, and chromatically tunable optics. This early demonstration of metasurface-based flat optics highlighted the potential of engineered subwavelength structures to enable lightweight, integrable, and multifunctional optical components for future photonic systems.^[161]

Building on this foundation, researchers began exploring metasurfaces for spectral manipulation, aiming to replace bulky dispersive elements in conventional spectrometers with compact, flat alternatives. Early metasurface-based spectrometers primarily utilized plasmonic structures,^[162] leveraging resonant nanostructures to achieve wavelength-selective behavior. However, these designs often suffered from significant ohmic losses, limiting their overall transmission efficiency and thermal stability—particularly in the visible and near-infrared ranges. A pivotal advancement in metasurface technology occurred with the transition to dielectric metasurfaces, constructed from low-loss, high-refractive-index materials.^[163,164] These structures support Mie-type resonances that allow efficient control of phase, amplitude, and polarization across broad wavelength ranges, while avoiding the absorptive drawbacks of metals. This transition enabled the development of high-efficiency, broadband metasurface devices, including dispersive optics and planar spec-

trometers that could be readily integrated with CMOS sensors and photonic platforms.

A key milestone in this trajectory was the work of Khorasaninejad et al. in 2016, who introduced an innovative class of super-dispersive off-axis dielectric meta-lenses for compact, high-resolution spectroscopy.^[165] Their design employed silicon nanofins acting as half-wave plates, where the phase profile was engineered using the Pancharatnam–Berry (PB) geometric phase principle.^[166] As shown in Figure 16a, by precisely rotating the nanofins across the metasurface, the metalens imparts a spatially varying geometric phase to the incident light, enabling it to focus at off-axis positions and achieve enhanced spectral dispersion. The working principle of the off-axis metalens can be summarized as follows: the nanofins are strategically distributed across the lens surface such that the phase shifts introduced at each location (x, y) redirect incoming light to a focal point located at an angle α , off the optical axis (Figure 16a). The phase delay due to the optical path difference between a nanofin located at position (x, y) and a reference point at the center of the metalens ($x = 0, y = 0$) is given by:

$$\phi_d(x, y) = \frac{2\pi}{\lambda_d} \left(\sqrt{(x - x_f)^2 + (y - y_f)^2 + z_f^2} - f \right) \quad (3)$$

where $\phi_d(x, y)$ is the phase delay due to the path difference, λ_d is the design wavelength, (x, y), and the focal length f is calculated as $\sqrt{x_f^2 + y_f^2 + z_f^2}$, and (x_f, y_f, z_f), are the coordinates of the focal spot in the focal plane in 3D space. To ensure constructive interference at the focal point, the phase delay introduced by optical path differences must be exactly canceled by the phase shift induced by the nanofins at (x, y). Thus, the required offset phase is:

$$\phi_{\text{offset}}(x, y) = -\phi_d(x, y).$$

Following the Pancharatnam–Berry phase principle^[166], the rotation of these nanofins by an angle $\phi_{\text{offset}}(x, y)$ imparts a phase change $\phi_{\text{offset}}(x, y) = 2\theta_{\text{offset}}(x, y)$ for a right-handed circularly polarized light, and $\phi_{\text{offset}}(x, y)$ is given by:

$$\phi_{\text{offset}}(x, y) = \frac{\pi}{\lambda_d} \left(f - \sqrt{(x - x_f)^2 + (y - y_f)^2 + z_f^2} \right) \quad (4)$$

This principle allows the metasurface to implement complex, wavelength-selective phase profiles using only variations in nanofin orientation, rather than changing material thickness or refractive index. As a result, off-axis metalenses can achieve strong angular dispersion and sub-nanometer spectral resolution in a planar and highly compact format. In cases where the metalens functions as a cylindrical lens, the phase variation depends primarily on the y -coordinate, simplifying the offset phase function $\phi_{\text{offset}}(x, y)$ to:

$$\phi_{\text{offset}}(x, y) = \frac{2\pi}{\lambda_d} \left(f - \sqrt{(x - x_f)^2 + z_f^2} \right) \quad (5)$$

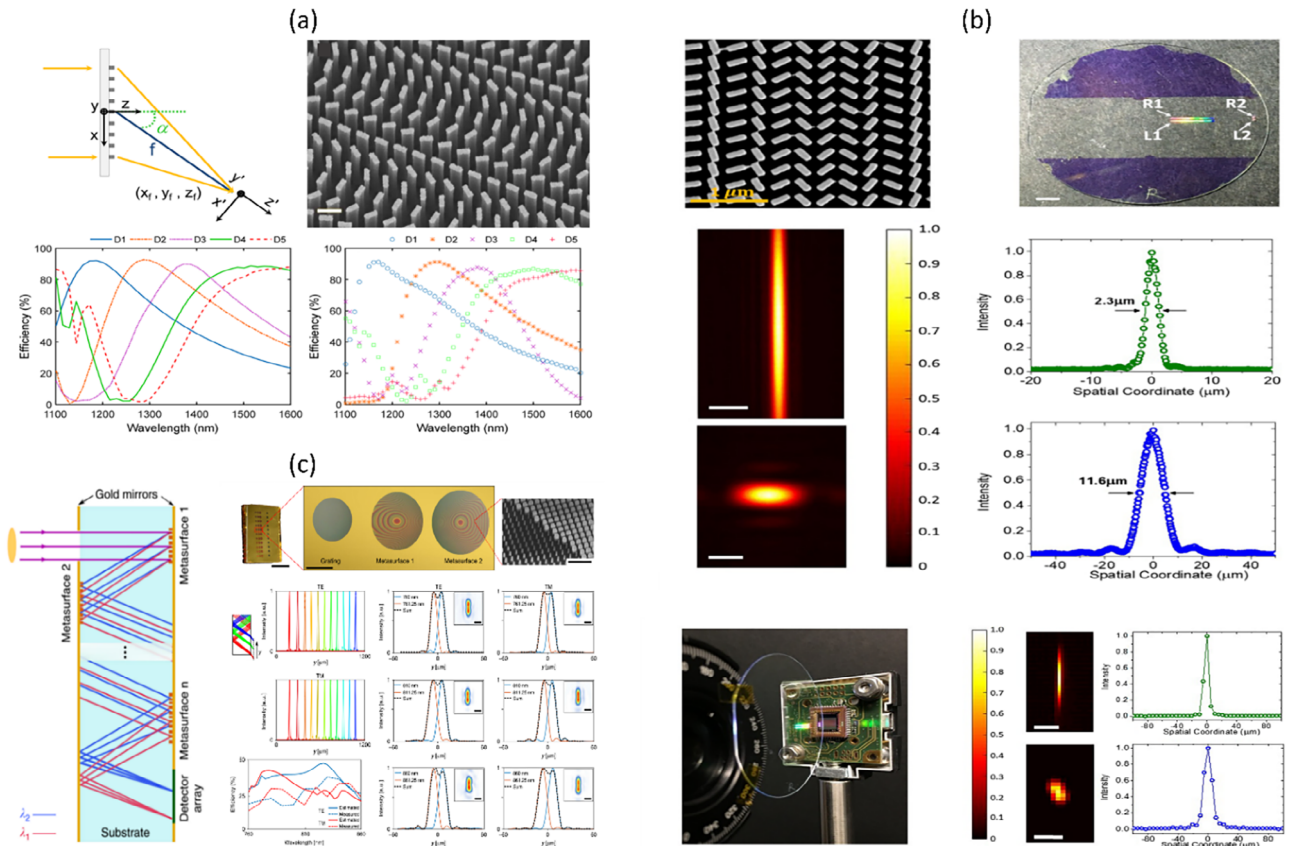


Figure 16. a) Off-axis metalens focusing: the schematic of the lens' coordinates, SEM image of the fabricated metalens, simulated conversion efficiency under circularly polarized light, and measured efficiency matching the simulation. Reproduced with permission.^[165] Copyright 2016, American Chemical Society b) The proposed off-axis metalens for chiral spectrometer, including, the scanning Electron Microscope Image, photograph of the metalens, and the intensity profiles as well as the photograph of the fabricated metalens placed in front of the camera and the intensity profiles. Reproduced under the terms of the CC-BY 4.0 license.^[167] Copyright 2017, AIP Publishing. c) The concept of a folded metasurface-based spectrometer, including the schematic design, and the experimental characterization results of the proposed scheme. Reproduced under the terms of the CC-BY 4.0 license.^[168] Copyright 2018, Springer Nature.

In general, the metalens is designed to direct the light into a line at an angle α and a focal length f . The focusing angle α , defined as the angle between the focal spot and vertical axis, and the focal length f are expressed as:

$$\alpha = \sin^{-1} \left(\frac{x_f}{f} \right) \quad (6)$$

$$f = \sqrt{x_f^2 + y_f^2} \quad (7)$$

These equations describe how the off-axis metalens directs light and focuses it at specific angles and distances, essential for high-resolution imaging and spectral applications. Moreover, the dispersive behavior of the metalens is derived using the generalized Snell's law, where the deflection angle $\gamma(x)$ of normally incident light is proportional to the local phased gradient and the incident wavelength:

$$\gamma(x) \propto \frac{d\phi_{\text{offset}}(x)}{dx} \cdot \lambda \quad (8)$$

$$\sin(\gamma(x)) = \frac{\lambda}{2\pi} \frac{d\phi_{\text{offset}}(x)}{dx} \quad (9)$$

This spatially dependent diffraction mechanism enables the metalens to distinguish closely spaced wavelengths even over short propagation distances—an essential feature for ultra-compact spectrometers. The lateral shift in the focal spot, expressed as $(f \cdot \Delta\alpha)$, occurs for a small change in the incident wavelength $\Delta\lambda$ when $\alpha > 0$.^[167] The overall displacement $|\Delta\vec{r}|$ is given as: $\sqrt{\Delta f^2 + (f \Delta\alpha)^2}$. This displacement varies with the wavelength, and the linear dispersion relation $|\frac{\Delta\lambda}{\Delta r}|$ can be computed by substituting Equation (6) into Equation (7), yielding:

$$\frac{\Delta\lambda}{\Delta r} = \frac{\Delta\lambda}{f \left\{ \sin^{-1} \left[\left(1 + \frac{\Delta\lambda}{\lambda_d} \right) \sin(\alpha) \right] - \alpha \right\}} \quad (10)$$

Finally, the spectral resolution defined as the smallest resolvable wavelength difference $\delta\lambda_{\text{min}}$, is limited by the Rayleigh criterion:

$$\delta\lambda_{\text{min}} = \left(\frac{\Delta\lambda}{\Delta r} \times \frac{0.61\lambda}{NA} \right) \quad (11)$$

This relationship ties together the physical design of the metasurface, its phase profile, and its resolving power, highlighting how nanoscale engineering of phase enables performance comparable to or exceeding traditional diffractive optics.

Zhu et al. extended this concept by integrating multiple off-axis meta-lenses onto a single planar substrate, each tailored for different spectral resolutions and wavelength bands. This enabled a multichannel meta-spectrometer capable of simultaneous spectral analysis without increasing system complexity or footprint (Figure 16b). Notably, the device could distinguish the helicity of incident light, a task that typically requires additional polarimetric components in conventional spectrometers. Their design, based on TiO₂ nanofins on a glass substrate, efficiently coupled circularly polarized light into diffracted orders with a spectral resolution as fine as 0.3 nm.^[167]

Further refinements were made by Yi et al., who conducted a comprehensive parametric analysis of off-axis dielectric metalenses. Their study systematically explored the influence of structural parameters—including nanofin dimensions, spacing, and orientation—on the spectral resolution and angular dispersion of the system. These insights offered valuable guidelines for optimizing metasurface-based spectrometers across diverse application spaces, from on-chip sensing and portable diagnostics to astronomical instrumentation.^[169]

Together, these advances illustrate a clear trajectory: from the early use of plasmonic metasurfaces for wavefront shaping, to high-efficiency dielectric metasurfaces capable of precise spectral control. With continued improvements in fabrication techniques, algorithmic reconstruction, and materials engineering, metasurface-based spectrometers are rapidly emerging as powerful, scalable alternatives to traditional dispersive optics—paving the way for miniaturized, multifunctional hyperspectral and polarimetric imaging systems.

However, off-axis metalenses often face challenges related to optical aberrations, causing defects such as blurring, distortion, and color fringing in the resulting images. To overcome this, Zhu and colleagues proposed a more compact design using dispersion-tailored metasurfaces that addresses the chromatic aberrations by engineering the group delay (GD) and group delay dispersion (GDD). This enables the metalens to maintain a consistent focal spot across a broad spectral range, and a working distance of just 4 cm between the lens and the detector. The spectrometer achieved a spectral resolution of approximately 1 nm across a bandwidth of 200 nm in the visible spectrum. The focal spots remained diffraction-limited, demonstrating the effectiveness of their proposed aberration correction scheme.^[170]

In parallel, Faraji-Dana et al., introduced a compact spectrometer using a novel folded metasurface design which leverages metasurfaces to significantly reduce the size and complexity of traditional spectrometers, while maintaining high resolution and efficiency. Fabricated on a single 1-mm-thick glass substrate, the device uses multiple reflective dielectric metasurfaces to confine and guide light through folded optical paths (Figure 16c). This architecture achieves a compact volume of just 7 mm³, while delivering a spectral resolution of ~1.2 nm across 80 spectral points in the 760–860 nm range. The metasurfaces perform wavefront shaping and focusing at each reflective interface, demonstrating that high-resolution spectroscopy can be achieved in a highly miniaturized, integrable form.^[168]

In 2020, Ding et al. demonstrated an ultra-compact, fully integrated spectrometer leveraging metasurface-decorated photonic integrated waveguides. The proposed device consists of all-dielectric meta-atoms placed on top of the waveguide, which locally modify the effective mode indices through evanescent coupling. By varying the dimensions of these meta-atoms, the propagation phase is controlled, enabling in-plane waves focusing akin to a gradient-index (GRIN) lens. The proposed integration allows for the routing of guided light waves of different wavelengths to distinguish focal positions within the waveguide, thus achieving wavelength demultiplexing.^[175] However, the trade-offs between bandwidth and resolution, along with challenges in maintaining high efficiency across a broad spectral range, present notable limitations.

In 2021, Du et al. proposed a transmission imaging spectrometer based on dielectric metasurfaces, operating within a spectral width of 700–800 nm, with a spectral resolution of approximately 1 nm, and a compact size of 13 mm³ (Figure 17a). The metasurface is composed of metal nanorods that manipulate the phase and polarization of incident circularly polarized light, enabling anomalous refraction. The design ensures high diffraction efficiency and polarization conversion, which are critical for the device's performance. Their approach combines metasurface technology with traditional optical design principles to overcome the limitations of traditional spectrometers, offering a promising solution for the future of miniaturized and high-performance spectral imaging devices. Despite these advantages, the complex fabrication process and response time limitations for dynamic applications are areas that may need further improvement.^[171]

In the same year, Kong et al. introduced the use of a single-detector spectrometer based on a superconducting nanowire. The design eliminates the need for traditional wavelength multiplexing optics, thus significantly reducing the complexity and physical footprint of the spectrometer. The spectrometer utilizes a computational spectroscopic approach to reconstruct the incident light spectrum. Thus, by measuring photon count rates at various bias currents, the system builds a response matrix that is used to reconstruct the spectrum through mathematical algorithms. The elimination of traditional optical components and the use of computational reconstruction methods offer a promising path towards miniaturized, high-resolution spectroscopic and spectral imaging devices. However, improving response time and minimizing reconstruction artifacts are areas still under research.^[176]

In 2022, Bendoy et al. demonstrated the feasibility of a metamaterial-based spectrometer for hyperspectral imaging in the mid-wave infrared region, emphasizing reduced size, weight, and power consumption (SWaP-C). The core geometrical components of their proposed design include a low-loss sub-wavelength dielectric resonator metasurface, which is integrated directly into existing focal plane array (FPA) detectors, and a Distributed Bragg Stacks filter, composed of ZnS and Ge, providing narrow passband functionality (Figure 17b). To achieve flat dispersion curves, their design manipulates the electromagnetic coupling between the resonators, where they demonstrated that the sub-wavelength dielectric resonator could be divided into multiple parallel spectral channels, each with a separately engineered central wavelength and bandwidth. This flexibility allows for the creation of hyperspectral or multi-spectral filters,

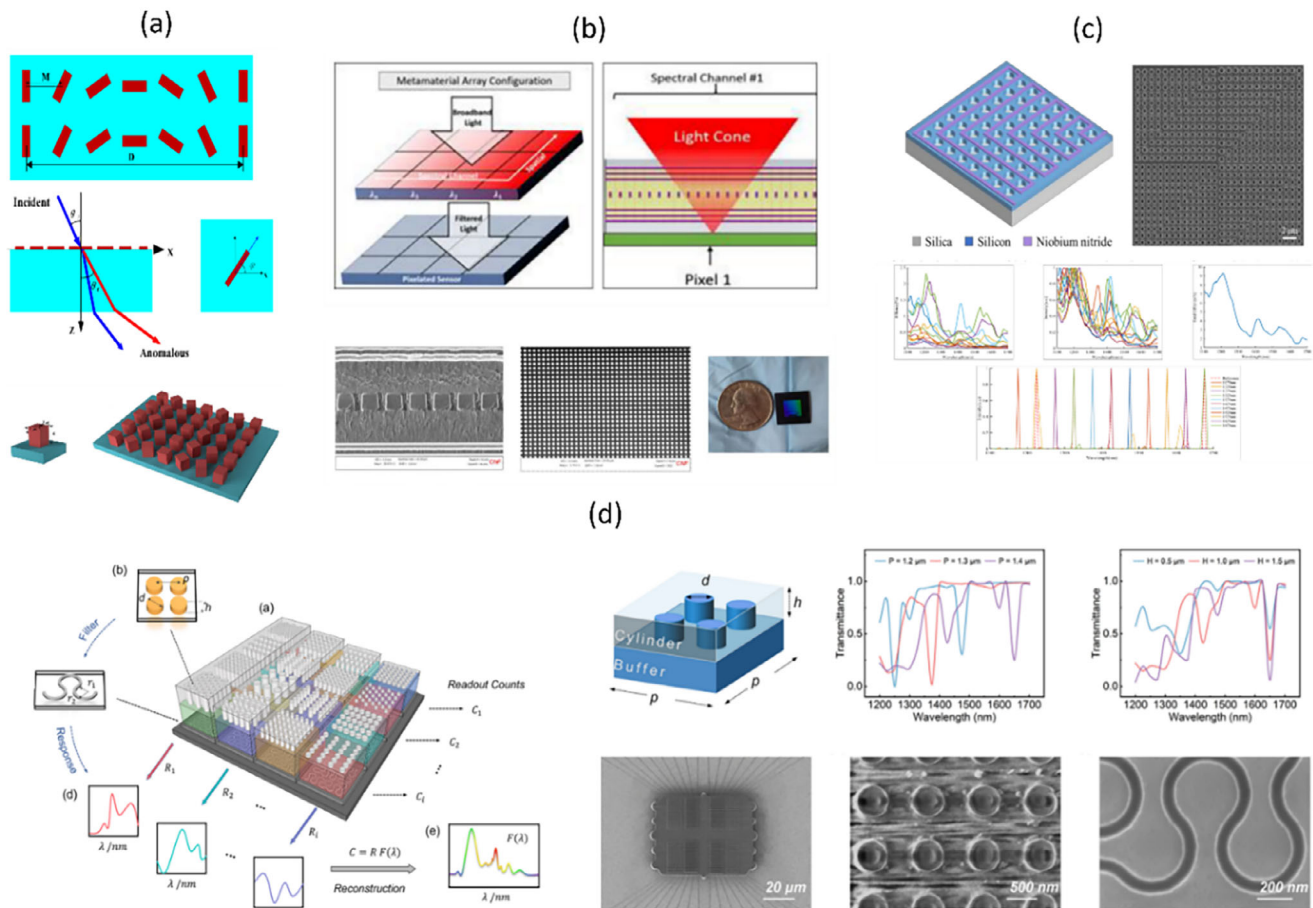


Figure 17. a) Working principle of the dispersive metasurface for a transmission spectrometer, and the simulated structure. Reproduced with permission.^[171] Copyright 2021, Society of Photo-Optical Instrumentation Engineers (SPIE). b) Low SWaP-C HS metamaterial spectrometer, including a schematic illustration of the filter integration with the FPA, along with the SEM image and the physical chip size. Reproduced with permission.^[172] Copyright 2022, Society of Photo-Optical Instrumentation Engineers (SPIE). c) Metasurface-based single-photon sensitive spectrometer, along with the SEM image of the fabricated device, and the measured filter spectra. Reproduced with permission.^[173] Copyright 2022, Optica Publishing Group. d). Schematic illustration of the proposed computational spectrometer based on SNSPF and 3D-printed array of cylinders, and the design and fabrication of the proposed device. Reproduced with permission.^[174] Copyright 2022, American Chemical Society.

significantly enhancing the spectrometer's capability for detailed spectral analysis.^[172]

In the same year, Zheng et al. demonstrated the use of metasurfaces combined with superconducting nanowire single-photon detectors (SNSPDs) to create a single-photon sensitive spectrometer. The proposed spectrometer achieved a total detection efficiency ranging from 1.5% to 9.5% within the wavelength range of 1100 nm to 1700 nm. The SNSPDs, composed of niobium nitride (NbN) on a silicon-on-insulator (SOI) substrate (Figure 17c). This combination of metasurfaces and SNSPDs allows for highly sensitive spectral measurements. Their proposed spectrometer uses computational spectral reconstruction to enhance photon utilization. By utilizing the distinct spectral responses of the metasurface structures, the system accurately reconstructs the incident spectrum, even with fewer detection units than the number of discrete wavelength points. The combination of metasurfaces and SNSPDs offers a significant advantage in photon-scarce environments. However, the complex-

ity in fabrication and calibration poses challenges for widespread adoption.^[173]

Similarly, Xiao et al. proposed the design of a spectrometer that uses a 4×4 array of SNSPDs (Figure 17d), fabricated using niobium titanium nitride (NbTiN) and feature a fractal structure to ensure polarization insensitivity and reduce current crowding effects. The proposed 3D-printed PC filters are designed to have diverse spectral responses, which encode the incident light before detection. The device achieves a system sensitivity down to -108.2 dBm at 1310 nm and a spectral resolution of 5 nm over the 1200–1700 nm range. The spectral information of the proposed design is encoded by the PC filters and decoded computationally using compressed sensing algorithms. This approach allows the spectrometer to reconstruct the incident spectrum with high accuracy using fewer detection channels than wavelength channels. While the proposed spectrometer design shows significant potential for various applications. However, challenges related to fabrication complexity, response time, reconstruction accuracy,

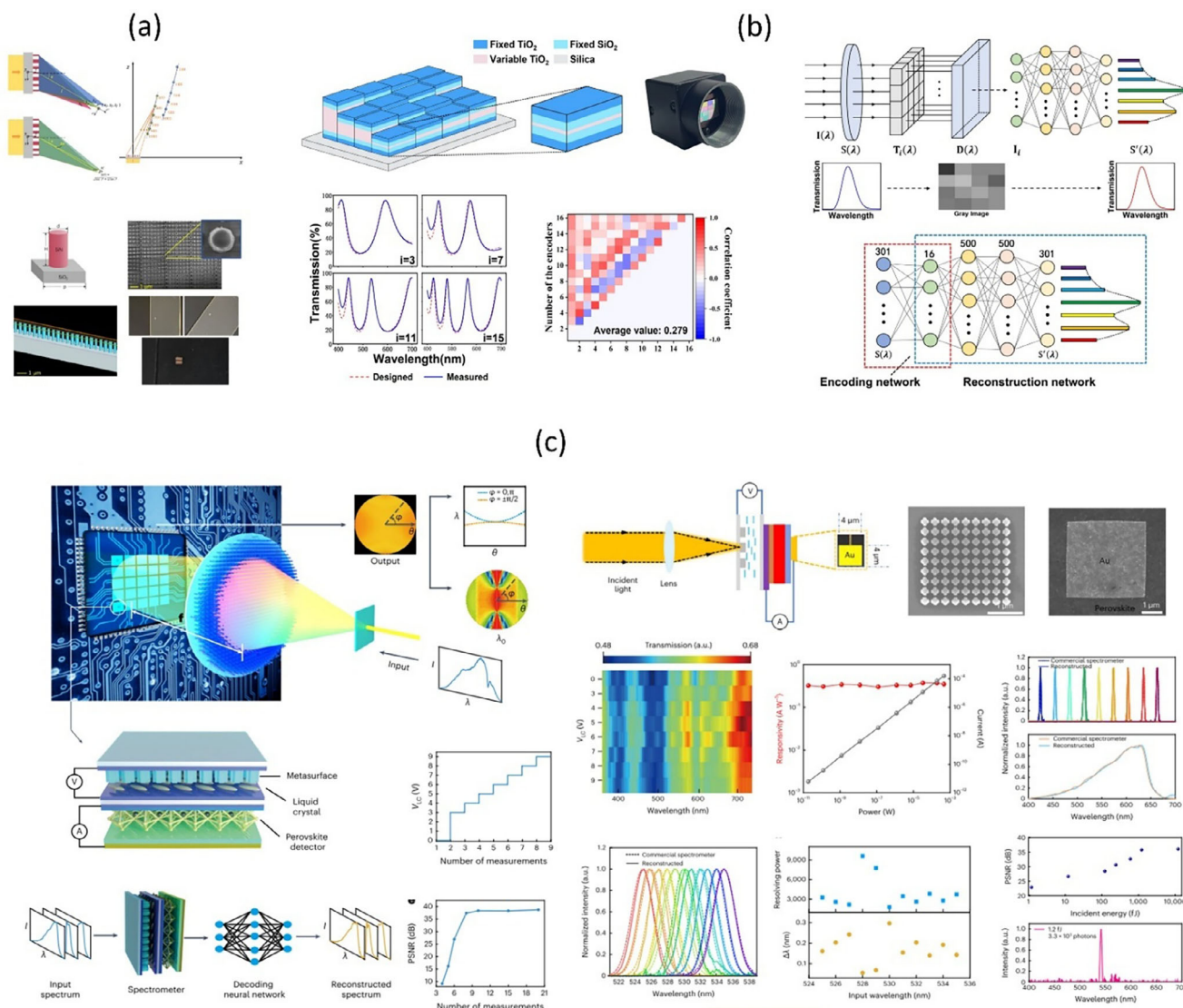


Figure 18. a) Focusing and dispersion scheme of off-axis metalens, the unit cell, and the SEM image of the fabricated device. Reproduced with permission.^[177] Copyright 2023, Optica Publishing Group. b) Miniaturized all-dielectric based spectrometer, including the schematic, system photograph, transmission spectra with their correlation coefficient, and the method for spectral reconstruction showing the workflow diagram and the schematic of the neural network. Reproduced with permission.^[178] Copyright 2022, American Chemical Society. c) Angle-resolved metasurface-based spectrometer, including the schematic illustrating the integration of the filter with the FPA, the optical characteristics, SEM images of the unit cell, and the measured performance. Reproduced with permission.^[179] Copyright 2024, Springer Nature.

and operational requirements need to be addressed for broader adoption and practical use.^[174]

Ke et al. presented a design for a compact multispectral variable spectrometer based on high-efficiency metalenses fabricated using Si₃N₄ nanopillars on a SiO₂ substrate.^[177] This approach offers an alternative to earlier TiO₂-on-SiO₂ configurations, such as the one reported by Zhu et al.,^[167] with comparable functionality and improved fabrication compatibility. As illustrated in Figure 18a, the proposed design utilizes an off-axis metalens for simultaneous focusing and dispersion. The bottom panel of the figure displays both the unit cell layout and SEM image of the fabricated structure, confirming the high-resolution pattern fidelity necessary for spectral performance. Wen et al. proposed a thin-

film-based spectrometer composed of only five dielectric layers, achieving miniaturization through a highly simplified structure. By tuning the thickness of a single layer, they generated distinct transmission spectra across just 16 spectral encoders, simplifying both design and fabrication. Light transmitted through these encoders is captured by a CMOS sensor (Figure 18b), and the recorded intensities are decoded using a deep neural network (DNN).^[178] The measured intensity at each detector element is modeled as:

$$I_i = \int_{\lambda_1}^{\lambda_2} I(\lambda) S(\lambda) T_i(\lambda) D(\lambda) d\lambda \quad (12)$$

where $I(\lambda)$, $S(\lambda)$, $T_i(\lambda)$, and $D(\lambda)$ represent the source emission spectrum, the spectrum under test, the transmission spectra, and the normalized spectral response of the detector, respectively. The reconstruction optimization objective $\arg \min \|S(\lambda) - S'(\lambda, W)\|_2^2$ is used to train the deep learning network to reconstruct the spectra from the measured intensities by minimizing the error between the true $S(\lambda)$ and reconstructed $S'(\lambda, W)$ spectra based on the learned network parameter W . The integration of deep learning with ultrathin all-dielectric optics positions this design as a notable step toward low-cost, portable spectral analysis. However, the approach remains sensitive to layer thickness tolerances and manufacturing precision, which are key areas for future optimization.

A significant advancement in miniaturization was reported by Cai et al., who developed an ultra-compact angle-resolved metasurface spectrometer with a total footprint of just $4 \times 4 \mu\text{m}^2$, representing a substantial reduction compared to previous designs. The system integrates a metalens with a metasurface spectrometer array, achieving a wavelength accuracy of 0.17 nm and a spectral resolution of 0.4 nm, along with an angular resolution of 4.88×10^{-3} rad. High-resolution spectral data is reconstructed using advanced computational algorithms, including deep neural networks, allowing efficient recovery of spectra from a reduced number of measurements (Figure 18c).^[179]

In a similar approach, Xu et al. introduced a reconstructive spectrometer using monocrystalline silicon metasurfaces, designed to offer high fidelity spectral reconstruction across the visible range. The system employs a bank of broadband filters (Figure 19a) to encode the incoming light spectrum spatially, eliminating the need for traditional dispersive optics. The encoded light is passed through a focusing lens and metasurface filters before reaching the detector. The measured intensity at each pixel is modeled as the combined effect of the lens, metasurface filter, and the sensor.^[180]

$$I_i = \int_{\lambda_1}^{\lambda_m} L(\lambda) M_i(\lambda) D(\lambda) S(\lambda) d\lambda \quad (13)$$

where, I_i is the measured intensity at the i -th detector element, λ_1 and λ_m are the lower and upper bounds of the operational wavelength range, $L(\lambda)$ is the focusing lens transmittances, $M_i(\lambda)$ is the i -th metasurface filter transmittance, $D(\lambda)$ is the sensor's quantum efficiency, and $S(\lambda)$ is the incident light spectrum. The continuous integral in Equation (13) can be simplified into a discrete form, which is easier to handle computationally, and its matrix form is given by Equation (14). This matrix A encapsulates the entire spectral response of the system, and S is the vector of spectral components of the incident light.

$$I = AS, A \in \mathbb{R}^{n,m} \quad (14)$$

where I is vector containing the measured intensities for all detector elements, A is the system transmission matrix with each element $A_{i,j} = L(\lambda_j)M_i(\lambda_j)D(\lambda_j)$ represents the response of the i -th detector element at the j -th wavelength. S is a vector representing the incident light spectrum at discrete wavelengths, n represents the number of metasurface-based filters, and m denotes the number of spectral channels. The spectrum is reconstructed by solving

this inverse problem using convex optimization via the CVX toolbox, minimizing:

$$\min_S \|I - AS\|_2^2 \quad (15)$$

To ensure aperture-insensitive performance, the angle-dependent spectral response is integrated over the lens aperture using:

$$M(\lambda, F) = 8F^2 \int_0^{\tan^{-1}(\frac{1}{2F})} \frac{\sin\theta}{\cos^3\theta} M(\lambda, \theta) d\theta \quad (16)$$

Here, $M'(\lambda, F)$ is the integrated transmittance distribution considering the aperture of the focusing lens, F is the F-number of the focusing lens, which controls the aperture size, θ is the angle of incidence, and $M(\lambda, \theta)$ is the transmittance distribution as a function of wavelength and angle of incidence.

Wang et al. demonstrated a compact multi-foci metalens that performs wavelength splitting and light focusing simultaneously. By encoding wavelength and phase information into a single ultrathin metasurface, the system directly maps different spectral components to distinct focal points in a single imaging plane (Figure 19b). This approach effectively condenses the dispersion functionality of a traditional spectrometer into a metalens with sub-millimeter thickness, rendering it highly attractive for on-chip photonic integration. The reported resolution approaches 1 nm, validating the system's capability for high-precision spectral discrimination while preserving a highly compact footprint. The strength of their system is its balance of spectral resolution and physical compactness. It integrates seamlessly with photonic circuits and is suitable for chip-scale implementations.^[181] Nevertheless, metalenses tend to be sensitive to fabrication errors and may experience reduced efficiency due to losses in high-index materials or imperfect phase control, especially under broadband illumination or off-axis incident light.

In the mid-infrared range, Liu et al. demonstrated a compact spectrometer based on metal-insulator-metal (MIM) metasurfaces, designed for high-resolution computational spectroscopy. Their architecture consists of an array of MIM resonators, each acting as a narrowband spectral filter. Each pixel is divided into multiple subpixels, each with a distinct spectral response. Light transmitted through the metasurface is captured by an image sensor and decoded using a neural network, trained on synthetic spectral data. The sparsity of the acquired data enables compressed sensing, which, when combined with machine learning, allows rapid and accurate reconstruction of the full input spectrum with minimal measurements.^[182]

One of the most significant advantages of metasurface-based spectrometers is their compact design with functionalities within a fraction of the volume and weight.^[165] This level of miniaturization makes metasurface-based spectrometers highly attractive for integration into portable, wearable, and space-constrained platforms, enabling new possibilities in remote sensing, environmental monitoring, and point-of-care diagnostics. Another key strength of metasurface-based spectrometers lies in their design flexibility. Metasurfaces can be engineered to perform a wide variety of optical functions, ranging from beam steering and wavelength dispersion to polarization manipulation and spectral filtering—all within a single, flat, and compact layer.^[6] This

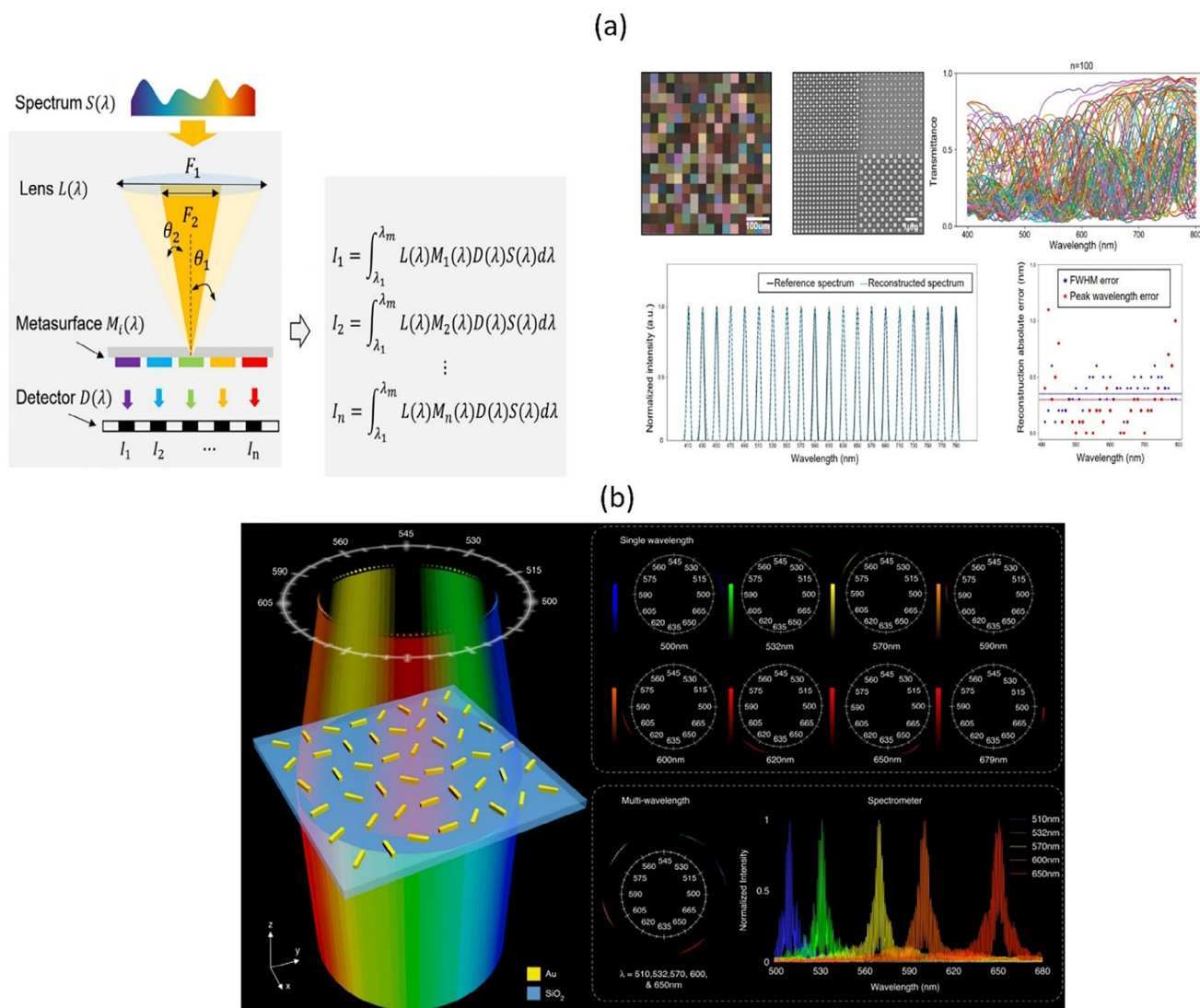


Figure 19. a) The proposed spectrometer operating based on the principle of spectral coding, including the SEM image of the fabricated filter, and the spectral characteristics of the device. Reproduced with permission.^[180] Copyright 2024, AIP Publishing. b) Schematic of a metasurface spectrometer utilizing a multi-foci metalens including, illustration of the proposed design, in which the metalens produces distinct focal points whose positions vary with wavelength, the simulated field intensity on the focal plane for monochromatic illumination at different wavelengths, showing that each color of light is focused to a unique location; the azimuthal angle of peak intensity maps directly to the input wavelength, and response to a polychromatic beam (left), here comprising five example wavelengths (510 nm, 532 nm, 570 nm, 600 nm, 650 nm), and the corresponding normalized intensity profile sampled around the lens's focal ring (right), from which the individual spectral components are extracted. Reproduced under the terms of the CC-BY 4.0 license.^[181] Copyright 2023, Springer Nature.

tunability allows researchers to customize spectral responses for specific applications, facilitating the development of task-specific and multifunctional spectroscopic devices. Moreover, metasurface spectrometers are highly compatible with photonic integrated circuits (PICs). This integration can allow the development of fully integrated, low-power, and chip-scale spectrometers with multiple functions co-located on a single substrate. Such systems are critical for applications in on-chip spectroscopy, lab-on-a-chip diagnostics, and integrated photonic sensing, where size, power, and performance are tightly constrained. In addition, from a performance standpoint, metasurface spectrometers also excel in sensitivity. When combined with highly sensitive detec-

tors, such as superconducting nanowire single-photon detectors (SNSPDs), these systems can achieve single-photon-level detection, which is essential for fields like quantum optics, fluorescence microscopy, and low-light spectroscopy.^[173]

Despite the strength and significant advancements in metasurface-based spectrometers, several challenges remain. For instance, the fabrication of metasurface-based spectrometers involves techniques, such as electron-beam lithography and reactive-ion etching which are not only time-consuming and costly but also require high precision to achieve the desired optical properties. The scalability of these fabrication methods remains a significant challenge for metasurface-based

spectrometers to be commercially viable.^[141,183] The transition from lab-scale fabrication to mass production requires the development of scalable, high-throughput techniques that can deliver consistent optical quality at lower cost.^[143]

In addition, there is a well-known trade-off between spectral resolution and optical efficiency. Designs that maximize spectral resolution—by increasing dispersion or spatial sampling—often limit the total amount of light transmitted or reflected, thereby reducing efficiency. Striking a balance between these competing factors remains a central challenge in metasurface spectrometer design. Environmental robustness is another concern, particularly for portable applications. Thermal fluctuations, for instance, can induce spectral drift by altering the refractive index of the metasurface material. To mitigate this, researchers are developing hybrid metasurfaces that combine materials with opposing thermo-optic coefficients, thereby stabilizing the spectral response under varying temperatures.^[183]

Looking ahead, advances in fabrication technologies will be critical to the next generation of metasurface based spectrometers. For instance, techniques such as nanoimprint, deep UV lithography, and self-assembly fabrication hold promise to scale up production while maintaining high fidelity and low cost.^[183]

In parallel, progress in computational methods—including compressed sensing, machine learning, and neural network-based reconstruction—will continue to enhance spectral recovery accuracy, speed, and calibration. These techniques not only improve performance but also facilitate design optimization and adaptive correction of fabrication imperfections. A particularly exciting direction involves the development of dynamically tunable metasurfaces—devices that can modulate their optical properties in response to external stimuli such as electric fields, temperature, or mechanical strain. These reconfigurable metasurfaces offer real-time control over spectral characteristics, enabling adaptive spectrometers suited for in-situ monitoring and on-the-fly spectral tuning. Ultimately, the goal is to transition metasurface-based spectrometers from laboratory prototypes to commercial products.^[184,185] This will require addressing issues of cost, reliability, manufacturability, and user accessibility. Realizing this vision will depend on interdisciplinary collaboration between academia and industry stakeholders to drive innovation, standardization, and widespread adoption across sectors.

3.4.3. Metasurface Spectral Imaging System

The field of SIS design has undergone rapid evolution over the past few decades, driven by a combination of technological breakthroughs, increasing demands for high-resolution and real-time imaging, and the expanding range of applications across various industries. This dynamic transformation is rooted in several key developments. Foremost among these is the continuous advancement of optical and computational technologies. Innovations in sensors, miniaturized optics, and data processing algorithms have significantly expanded the performance envelope of SIS. The rise of nanotechnology and microfabrication techniques has enabled unprecedented levels of miniaturization, giving rise to portable, wearable, and even chip-scale SIS. These technologies now play a critical role in fields ranging from biomedical di-

agnostics and environmental monitoring to quantum photonics and remote sensing.

Among the most significant advances in this domain design has been the emergence and rapid progress in metasurface-based designs, which have redefined the landscape of compact SIS enable extreme miniaturization. Traditional compact spectral imaging systems discussed earlier such as the DIY designs often rely on bulky components such as prisms, gratings, or interferometers, limiting their potential for integration into miniature and compact devices. In contrast, metasurfaces can be fabricated on flat substrates and integrated directly onto sensors or photonic circuits, drastically reducing the size and weight of the system. The trend towards metasurfaces-based designs has led to significant breakthroughs in both performance and application scopes. Their inherent advantages in miniaturization and compatibility with other photonic and electronic components make them ideal for portable and on-chip systems.^[6] Moreover, metasurfaces can be fabricated using processes compatible with standard semiconductor manufacturing, facilitating mass production, reducing costs, and accelerating widespread adoption. Beyond compactness, metasurfaces have introduced new capabilities to SIS design, such as polarization-sensitive and hyperspectral video imaging. A landmark example of this potential is the 2016 work by Khorasaninejad et al., which demonstrated how metasurfaces can revolutionize optical imaging by integrating multiple functionalities into a single compact device.^[186] They introduced a single planar lens capable of performing high-resolution chiral imaging and multispectral analysis. The metalens, termed the MCHL (Multispectral Chiral Lens), leverages a structure based on interlaced arrays of TiO₂ nanofins functioning as half-wave plates in order to maximize the focusing efficiency. By rotating these nanofins, the device imparts the necessary phase shifts based on the Pancharatnam-Berry phase principle to focus light with opposite helicity. This capability eliminates the need for complex setups involving multiple optical components traditionally required for chiroptical spectroscopy and imaging. The proposed device successfully imaged chiral and achiral objects, showing distinct images under right – and left – circularly polarized light and mapping circular dichroism in a chiral beetle. Their proposed MCHL's ability to resolve chiral properties makes it especially suitable for biochemical studies, such as detecting enantio-specific concentrations of amino acids and nutrients. In addition, the compact and multifunctional nature of the MCHL can be utilized for advanced remote sensing and environmental monitoring applications. The broader impact of this work lies in its demonstration that a single metasurface can replace multiple conventional optical components, all while providing novel imaging modalities. It served as a compelling proof-of-concept that metasurfaces are not just passive diffractive elements, but powerful enablers of multifunctional, polarization-sensitive, and wavelength-selective optical systems—a paradigm shift in the field of flat optics.

Building on this foundation, Faraji-Dana et al. (2019) introduced a compact HSI system based on folded metasurface optical design. This system integrates multiple optical functions—dispersing, focusing, and imaging—within a 1 mm-thick glass substrate, using a combination of reflective and transmissive metasurfaces. These metasurfaces were fabricated in a single lithographic step, with reflective elements composed of α -Si

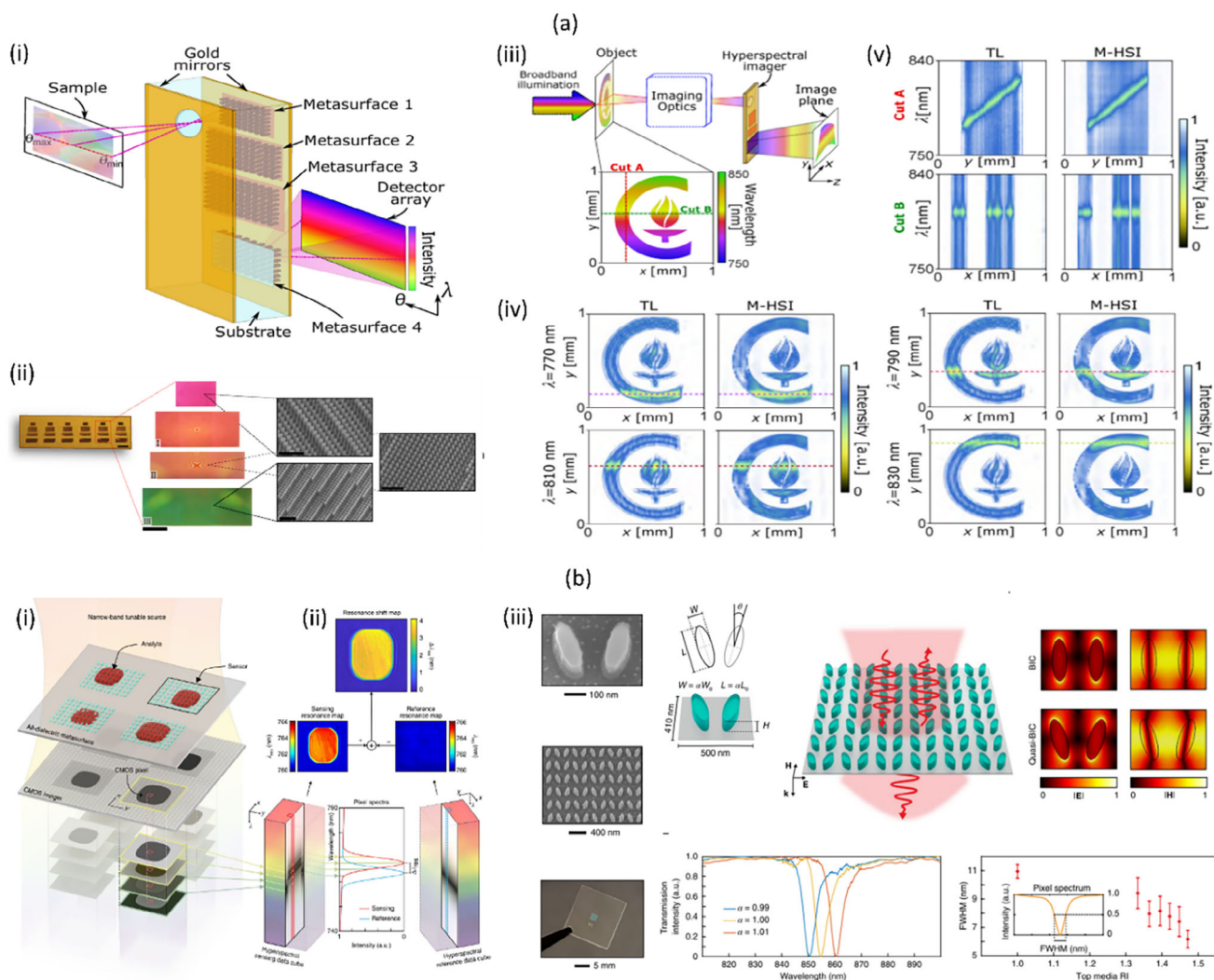


Figure 20. a) A folded HSI scheme based on metasurface optics, including the schematic illustration of the proposed scheme, the optical microscope image illustrating six integrated HSIs on a single chip (left), with magnified views detailing the intricate metasurface structures forming an individual HSI, and the fabricated metasurface (right), and the HSI acquisition of a target with the folded scheme, where i) depicts the HS imaging process, ii) Comparative analysis of the Transmitted intensity of the object measured at four different wavelengths using the proposed metasurface HSI and the one measured from a tunable laser, and iii) Intensity profiles of the transmitted light through the object along paths A and B, compared with the corresponding measurements taken using a tunable laser for comparison. Reproduced with permission.^[187] Copyright 2019, American Chemical Society. b) Mechanism of detecting biomolecules through hyperspectral imaging using all-dielectric metasurface technology, including, i) illustration of the hyperspectral imaging process, where a dielectric metasurface sensor array is illuminated by a narrow-band tunable laser, where, for each wavelength, a CMOS camera (1608 × 1608 pixels) captures images, forming a hyperspectral data cube, with each pixel recording precise spectral data. ii) The spatial resonance maps for each sensor after the hyperspectral data cube is processed. iii) The proposed High-Q dielectric metasurfaces with tunable geometry, utilizing quasi-BIC modes, include SEM images of the individual unit cells, the fabricated metasurface, and a photograph showcasing the chip containing a 5 × 5 sensor array, the overhead view of the SEM images of unit cell, featuring two elliptical dielectric nano-resonators aligned symmetrically along the major vertical axis, the schematic illustration of how the metasurface scattered an incident light, the field distribution in the unit cell, illustrating both BIC and quasi-BIC modes, the extracted transmission spectra from the HS data cubes with narrow resonance dips in the NIR, and the measured spectral linewidth of the resonance dip. The error bars in the inset of bottom right figure reflect the standard deviation in FWHM measurements taken from 2.5×10^5 pixel spectra across four separate sensors. Reproduced with permission.^[188] Copyright 2019, Springer Nature.

nanoposts, a SU-8 spacer, and gold mirrors, while transmissive elements omitted the mirror to allow light transmission. Operating in the 750–850 nm spectral range, the system achieved impressive performance: resolving over 70 spectral channels and 400 angular points, with ~ 1.5 nm spectral and 0.075° angular resolution—comparable to much larger conventional HSI systems. It employed a push-broom acquisition technique, captur-

ing spectral data line-by-line as the object moved across the imager (Figure 20a). The folded design enabled a reduction in volume (8.5 mm^3) and weight ($<20 \text{ mg}$), offering a highly miniaturized alternative ideal for applications in aerospace, consumer electronics, and mobile sensing. While limitations such as $\sim 10\%$ focusing efficiency and slower scanning speed remain, this work highlighted the immense potential of metasur-

face optics for building high-performance, ultra-compact HSI systems.^[187]

In the same year, another significant advancement was reported through the development of a biosensing platform that integrates dielectric metasurfaces with HSI.^[188] This system utilized CMOS-compatible, all-dielectric metasurfaces engineered to support BIC—non-radiating resonant modes with high Q-factors, enabling strong light confinement and sharp spectral features. The metasurface consisted of silicon nanobar pairs arranged with a 17.5° tilt to break mirror symmetry and achieve quasi-BIC resonances in the near-infrared (750–850 nm). This configuration allowed for label-free, highly sensitive detection of biomolecular interactions via shifts in the resonant wavelength. By combining this with hyperspectral imaging, the system produced spatially resolved resonance shift maps, enabling the visualization of molecular binding events across millions of pixels (Figure 20b). Remarkably, the platform demonstrated the ability to detect ultralow biomolecule concentrations, outperforming traditional sensing methods. Additional features included multi-resonance barcode sensing and the capability for studying 2D materials, positioning it as a versatile analytical tool. Although the proposed technique offers a highly innovative platform for biosensing with excellent sensitivity and throughput as demonstrated in one of their subsequent works.^[192]

Despite its advantages, the system also presents practical challenges: complex fabrication, environmental sensitivity, and the need for advanced optical components (e.g., supercontinuum lasers) may limit its immediate scalability for field-deployable or low-cost applications. Nonetheless, these works collectively highlight a transformative shift in SIS designs—driven by metasurfaces—toward multifunctional, compact, and highly sensitive systems capable of addressing the growing demands in biomedicine, sensing, and beyond. Continuing the trend toward miniaturized, real-time spectral imaging, McClung et al. presented a compact, lightweight snapshot HSI system based on parallel metasystems.^[189] Their design integrates metasurface arrays, including doublet lenses and bandpass filters on a small glass substrate, forming a multi-aperture spectral imager (Figure 21a). This metasystem captured 20 spectral channels across the near-infrared range (795 to 980 nm) simultaneously, enabling snapshot HSI. Each channel corresponds to a specific spectral band, and the system uses narrowband filters to mitigate chromatic aberrations while capturing high-resolution spectral data. Their design allows for real-time imaging of both still images and video with high spectral resolution. With an overall volume of < 8 mm³, the device is well-suited for portable consumer electronics and remote sensing. However, its narrow spectral range and fabrication complexity present challenges for broader application.

To further enhance snapshot multispectral imaging, Wen et al. proposed a metasurface-based approach based on Huygens' multilayer metasurface absorber. Their system used multiple layers of silicon nano-resonators that support Huygens' modes, achieving high absorption efficiency across different wavelengths. The design uses a quasi-analytical transfer matrix model to simulate the spectral response, ensuring high spectral resolution and compact integration (Figure 21b). The multilayer metasurface absorber are stacked, and each layer targets a narrow spectral band, allowing for the collection of a multispectral data cube in a single shot. The number of absorber layers can be flexibly changed,

and the spectral bands can be tuned based on specific application needs, making it modular and customizable. The design offers a promising avenue for improving snapshot multispectral imaging, with high potential for practical applications. However, challenges in fabrication and angle sensitivity may limit its current usability, making it an exciting but still developmental technology.^[190]

Billuart et al. demonstrated a numerical analysis through full-wave simulations, which is supported by analytical approaches such as the Sommerfeld integral for designing a single metasurfaces optimized for pushbroom HSI systems. Their design use a 1D cylindrical metasurface composed of two-level TiO₂ nano-ridges to achieve both dispersion and focusing of a continuum of wavelengths. The prototype design with a diameter of 175 μm, achieves a spectral resolution of 8.5 nm and an angular field of view of 8° (Figure 21c). Their optimization relied on full-wave simulations and analytical modeling via the Sommerfeld integral, guided by a phase-based design strategy. The phase shift imparted at normal incidence was expressed by:^[191]

$$\Phi_F(x, y, \lambda) = \Phi_F(x, y, \lambda) - \frac{2\pi}{\lambda} \times \left\{ \sqrt{[x - f_x - (a\lambda + b) \cos \alpha]^2 + y^2 + [f_z + (a\lambda + b) \sin \alpha]^2} - \sqrt{[f_x + (a\lambda + b) \cos \alpha]^2 + [f_z + (a\lambda + b) \sin \alpha]^2} \right\} \quad (17)$$

Here, each square-root expression is a geometric optical path between the point $D(x, y)$ on the metasurface and the wavelength-dependent focal point $f_{0,\lambda}$ after accounting for the detector tilt. The first root is the path length from D to $f_{0,\lambda}$; the second is the path length from the metasurface origin to $f_{0,\lambda}$. Their difference is the optical-path difference that the metasurface must exactly compensate so that all rays interfere constructively at $f_{0,\lambda}$. In particular, $\Phi_F(x, y, \lambda)$ is the phase delay the metasurface must impose at the in-plane coordinate (x, y) for normally incident light of free-space wavelength λ . The Cartesian coordinates x and y lie in the metasurface plane $z = 0$. The x -axis is the dispersive (spectral) direction, while y is the along-slit direction that preserves the scene's one-dimensional spatial information. The vacuum wavenumber $k = \frac{2\pi}{\lambda}$; multiplying an optical-path difference by k converts it into a phase term. f_x, f_z are the cartesian coordinates of the detector-plane centre $f_{0,\lambda}$ expressed in the metasurface reference frame: f_z is the axial distance (along z) from the metasurface to the detector and f_x is a lateral shift along x that allows the detector to sit off-axis. They are design parameters chosen to satisfy manufacturability and numerical-aperture constraints. a – is the linear dispersion coefficient that converts wavelength into a lateral coordinate u on the detector according to $u(\lambda) = a\lambda + b$; it fixes the spectral scale of the instrument. b – is the constant offset in the linear wavelength-to-position mapping $u(\lambda) = a\lambda + b$; it determines where a chosen reference wavelength lands on the detector. α – is the tilt angle of the detector plane with respect to the metasurface plane, measured in the $x - z$ plane, and $\alpha = 90^\circ$ (detector perpendicular to the metasurface) minimizes chromatic dispersion inside the nano-element library and is adopted in their example. Thus, Equation 17 prescribes for every point

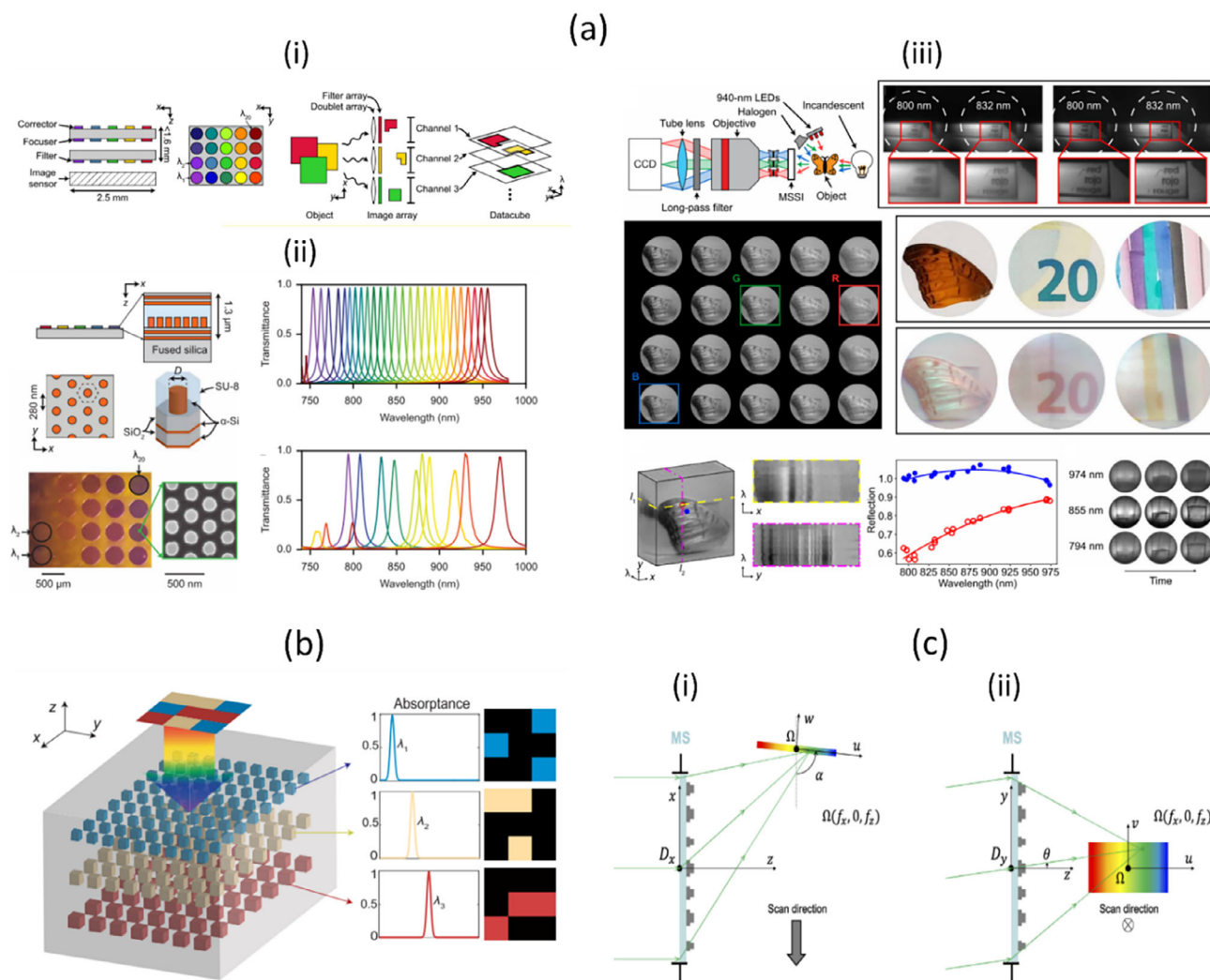


Figure 21. a) The snapshot spectral imaging scheme based on parallel metasystems outlines the design that captures and processes multispectral data in a single shot without mechanical scanning. It includes ii) a metasurface-based bandpass filter, where each filter consists of two Distributed Bragg Reflectors separated by an SU-8 polymer layer, forming a Fabry-Pérot cavity. A uniform array of amorphous silicon nanoposts on the lower DBR controls the passband. The simulated and measured transmission passbands of the filters are achieved by varying the nanopost diameters, as shown in the optical micrograph of the fabricated filter array. iii) The schematic of the proposed configuration for spectral imaging and video capture, and the imaging/video capturing characterization of the system. Reproduced under the terms of the CC-BY 4.0 license.^[189] Copyright 2020, American Association for the Advancement of Science. b) A schematic representation of the proposed multilayer absorber depicting how each layer captures incident light at specific wavelengths in order to generate the data cubes. Reproduced with permission.^[190] Copyright 2021, IOP Science. c) i) The schematic diagram of the optical configuration for the proposed hyperspectral imaging system^[191], including the illustration of the incident light focusing on the detector, and ii) the illustration of the metasurface along y-z plane illuminated at θ incident. Reproduced under the terms of the CC-BY 4.0 license.^[191] Copyright 2021, Optica Publishing Group.

on the metasurface and every wavelength in the design band, the phase profile needed to redirect normally incident light so that each wavelength focuses at the correct lateral coordinate $u(\lambda)$ on a tilted detector, realising a compact push-broom hyperspectral imager. The integration of dispersion and focusing functionalities in a single metasurface component as proposed offers significant potential for reducing the size and complexity of optical systems. However, challenges remain in extending the bandwidth, improving the numerical aperture, and ensuring consistent fabrication. Despite these limitations, the proposed metasurface design methodology provides a solid founda-

tion for future advancements in metasurface-based imaging technologies.

In 2022, Makarenko et al. introduced Hyplex, a metasurface-based HSI platform designed with artificial intelligence. Hyplex eliminates the need for traditional optics such as lenses, filters, and spectrometers, replacing them with CMOS-compatible metasurface projectors that encode spectral information directly. This configuration enables real-time hyperspectral data acquisition and segmentation, making it highly suitable for applications like object detection and scene classification.^[195] Its compact, cost-effective design and compatibility with AI-based pro-

cessing mark a significant advancement in metasurface-enabled imaging.^[196] However, its reliance on linear encoders and challenges in metasurface fabrication scalability remain areas for future research. The work by Yang et al. represented significant advancements in SIS utilizing metasurfaces.^[193,194] While both works share the common goal of enhancing metasurface-based spectral imaging capabilities, they adopt different methods and techniques. In particular, their 2022 study illustrates the limitations of conventional metasurfaces with regular geometric shapes, such as circles and squares, which restrict their ability to achieve complex light manipulation.^[193] These conventional designs restrict the metasurface's potential in applications requiring intricate light control, as they fall short in delivering the broad optical functionalities essential for high-resolution SIS. To overcome these limitations, they proposed an advanced ultraspectral imaging system using metasurfaces with freeform-shaped meta-atoms, offering design flexibility. This approach allows for more intricate manipulation of light, facilitated through the use of Bloch modes, resulting in enhanced spectral responses. The proposed design is depicted in **Figure 22a**, shows a microlens array placed between the metasurface and the image sensor. The configuration focuses the incident beam onto the image sensor after passing through the metasurface. The reconstruction of the incident beam $f(\lambda)$ from the detected signals I_i using the transmission spectra $T_i(\lambda)$ of the metasurface units, is governed by:

$$I_i = \int_{\lambda_1}^{\lambda_2} T_i(\lambda) f(\lambda) + e_i$$

$$\approx \sum_{j=1}^M T_j(\lambda_j) f(\lambda_j) + e_i, \quad i = 1, \dots, N \quad (18)$$

where e_i is the measured noised, λ_1 and λ_2 are the operational wavelength range limits, and M is the number of spectral points. Equation (18) highlights the spectral modulation capability of the proposed metasurface and the process of reconstructing the full spectrum from the modulated light signals. Their method enables high spatial and spectral resolution without mechanical scanning, overcoming the constraints posed by some traditional metasurface designs. Despite its potential, challenges such as fabrication complexity and noise sensitivity remain critical obstacles to broader applicability. Moreover, the generation of freeform-shaped metasurfaces, particularly at nanoscale feature sizes, presents significant challenges in manufacturing, which may limit scalability and increase costs. Conversely, their subsequent work demonstrated a novel approach to spectral imaging by integrating deep learning with a metasurface-based imaging system. This proposed method aims to address the limitations of traditional HSI systems, particularly in speed and spatial resolution, often constraint by scanning-based methods or complex optical systems. The proposed system integrates a metasurface with a CMOS image sensor (CIS) and employs deep learning to reconstruct the spectral data in real-time (Figure 22b). A major contribution of this work is its emphasis on speed, real-time capability, and the ability to reconstruct spectral data through a deep-learning algorithm called ADMM-net, which accelerates the spectral data reconstruction process significantly compared to traditional iterative methods.^[194] The spectral imaging pro-

cess is modeled by a system of linear equations, represented as follows:

$$y = \Phi x + e \quad (19)$$

where y is the measured signal pixel. Φ is the modulation matrix corresponding to the metasurface's transmission spectrum, x is the target spectral data cube to be reconstructed, and e represents the noise. However, for the deep-learning-based reconstruction, the Alternating Direction Method of Multipliers (ADMM)-net solves the unknown spectral data x and the key update steps for each iteration k are:

$$x^{(k)} = (\Phi^T \Phi + \gamma^{(k)} I)^{-1} (\Phi^T y + (v^{(k-1)} + u^{(k-1)})) \quad (20)$$

$$v^{(k)} = \mathfrak{D}_k(x^{(k)} + u^{(k-1)}) \quad (21)$$

$$u^{(k)} = u^{(k-1)} - (x^{(k)} + v^{(k)}) \quad (22)$$

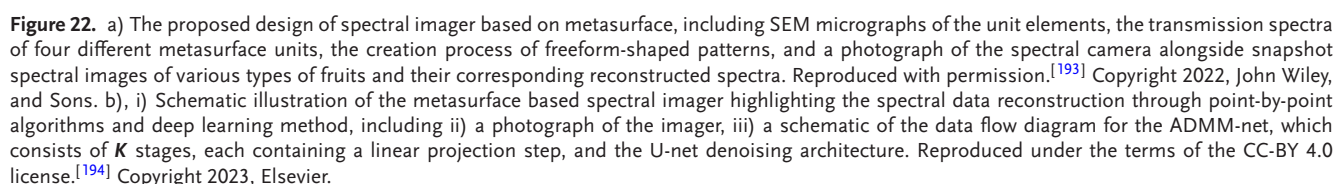
Equations (20–22) represent the linear projection, denoising process (Figure 22b), and the dual update, respectively. The parameter $\gamma^{(k)} > 0$ is a trainable variable, v denotes the estimated desired signal, and $\mathfrak{D}_k$ is a CNN-based denoiser that is applied at each iteration to enhance the reconstructed data. The proposed design allows on-chip integration and significantly improves the speed of spectral data reconstruction by combining metasurface-based spectral imaging with deep learning. While the system shows great promise for applications such as autonomous driving and real-time spectral analysis, challenges related to fabrication, noise modeling, and scalability remain. Nevertheless, the significant speed and accuracy improvements over traditional methods make this approach a valuable contribution to the field of spectral imaging.

Lin et al. proposed a novel snapshot HSI system that integrates a metasurface with machine learning and convex optimization to function effectively with small datasets. The system incorporates a custom-designed metasurface chip embedded with multi-resonant plasmonic meta-atoms, enabling multi-wavelength imaging in a compact and efficient form (Figure 23a-i). This approach allows for single-shot capture of data across multiple wavelengths, which is then processed using a small-data learning framework called CODE (Convex Optimization and Deep Learning).^[197]

The CODE technique reconstructs an 18-band HS data cube from only a 4-band input, by leveraging advanced deep learning techniques combine with convex. As illustrated in the Figure 23a-ii, the relationship between the captured multispectral image Y and the HS image X is expressed by the following equation:

$$Y = f_B(DX) \quad (23)$$

where Y is the captured multispectral image, X is the desired hyperspectral images, D is the spectral downsampling matrix that reduces the number of spectral bands from the HS image, and f_B is the blurring caused by the optical system, which distorts the spectral information captured by the metasurface. The multispectral image Y is a “blurred” and “downsampled” version of the target hyperspectral image X . The spectral downsampling matrix



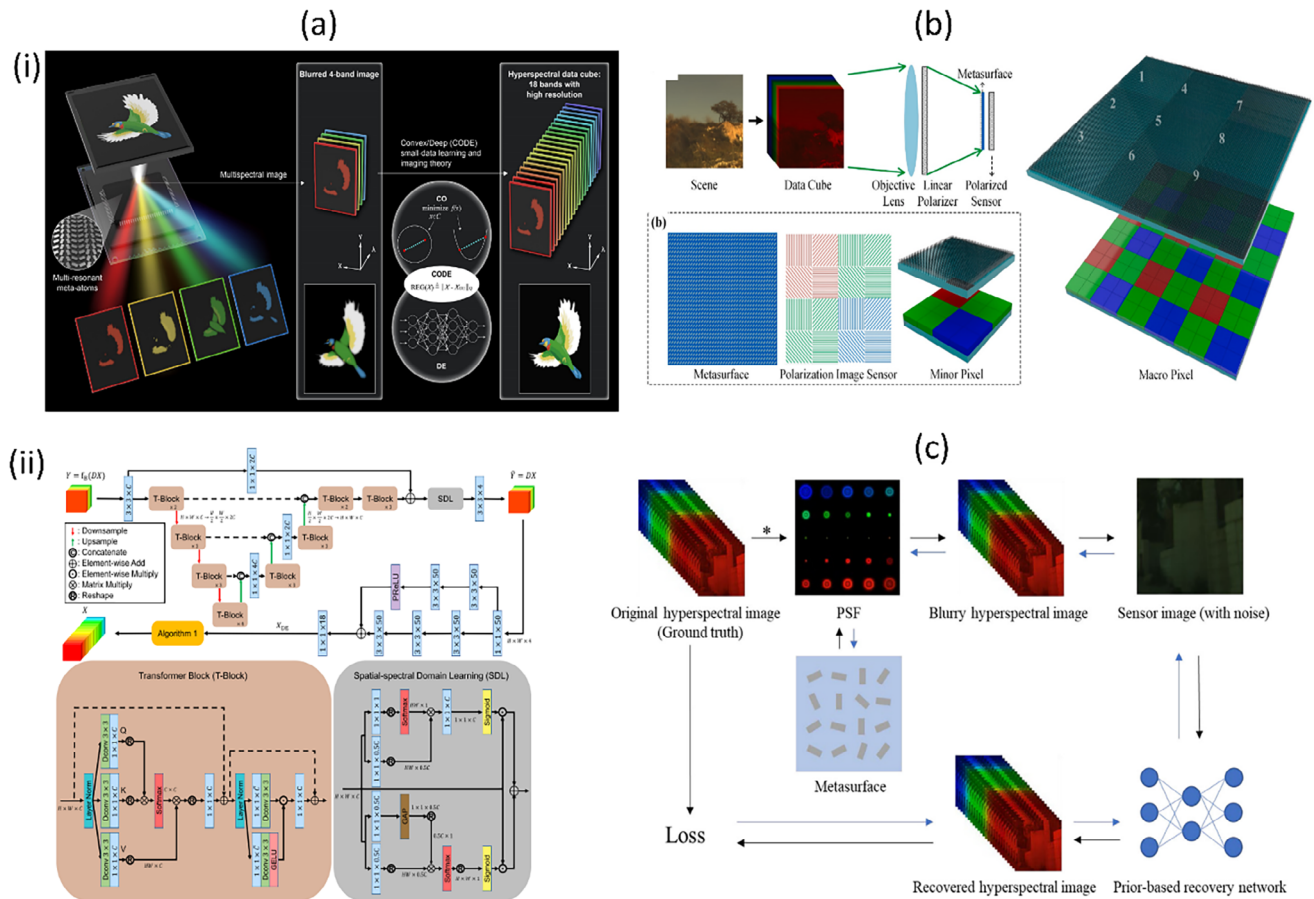


Figure 23. a) i) Metasurface-based HSI scheme, including a schematic showing the off-axis focusing meta-mirror simultaneously imaging a color object while separating it into four wavelength channels. The 4-band multispectral image is then expanded into an 18-band hyperspectral data cube using the proposed CODE small-data learning framework, combining convex optimization and deep learning. ii) schematic illustrating how the off axis focusing meta-mirror simultaneously images a color object and separates it into four wavelength channels. Using the proposed CODE small-data learning with convex optimization and deep learning, this 4-band multispectral image is expanded into an 18-band hyperspectral data cube. Reproduced under the terms of the CC-BY 4.0 license.^[197] Copyright 2023, Springer Nature. b) The proposed PCM-BCCHI scheme highlighting a polarization conversion metasurface with identical MBB arrays and a polarization image sensor pixel, and the macro pixel made up of 3 × 3 minor pixels, repeated to form the complete imager. Reproduced with permission.^[198] Copyright 2023, Elsevier. c) An overview of the HSI reconstruction system, with black and blue arrows representing forward and back propagation, respectively. Reproduced with permission.^[199] Copyright 2022, Elsevier.

D reduces the spectral resolution, while the blurring function f_B accounts for imperfections introduced by the metasurface. To recover the hyperspectral image X from the multispectral image Y and undo the blurring effect caused by the metasurface, they proposed the use of deep learning model (a Transformer) to approximate the inverse of the blurring function f_B^{-1} and de-blur the image Y as given in Equation 24 below:

$$\hat{Y} = f_B^{-1}(Y) \quad (24)$$

where f_B^{-1} is the inverse of the blurring function learned using a deep Transformer model, and $\hat{Y}$ is the de-blurred multispectral image. In addition, the Frobenius norm is used to quantify how well the reconstructed HS image fits the observed multispectral image, and Q-norm is used to regularize the solution and avoid overfitting the small amount of available data. The proposed work presents a groundbreaking approach to HSI by combining metasurfaces with an innovative machine learning theory that can op-

erate efficiently with minimal data. It has strong potential for real-time applications in compact settings, though improvements in fabrication and spectral range may be necessary for broader adoption. What is more is that the proposed small-data learning theory can be a powerful approach for certain applications, but it may not generalize well when exposed to noisy or incomplete data, as it has less information to draw from compared to traditional data-rich machine learning models.

Kang et al. demonstrated a metasurface-based platform for MSI using a planar cavity structure with potential applications in optical security. The proposed design is based on a multilayered cavity consisting of a color control unit (CCU) and an emission control unit (ECU), designed to simultaneously record mutually independent visible and infrared images on solid surfaces. The structure relies on controlling the visible colors through the CCU and adjusting IR emissions via a phase-changing layer of $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) in the ECU, providing independent control of visible and IR images in a single structure. While the approach

is scalable and flexible, its limitations in resolution and dynamic reconfiguration are present for further improvement.^[200]

Zhang et al. proposed HSI system that leverages metasurfaces and compressive sensing to improve efficiency and reduce the size of traditional imaging systems (Figure 23b). The proposed system, named PCM-BCCHI, uses polarization conversion metasurfaces (PCM) to enable simultaneous capture of two channels of hyperspectral data with orthogonal polarizations. The proposed PCM-BCCHI system represents a significant advancement in HSI by offering a compact and miniaturized solution with improved efficiency, particularly due to the implementation of the macro-pixel segmentation (MPS) method. The MPS approach simplifies the observation matrix by creating periodic submatrices for each patch of the image, reducing the computational complexity of the reconstruction process. However, challenges related to light efficiency, fabrication complexity, and sensitivity to errors must be addressed to fully realize the potential of this technology in practical applications.^[198]

In a similar effort to minimize the overall size of the HSI system while maintaining high image reconstruction quality, Y. Zhang et al. demonstrated a joint optimization framework that combines a geometric metasurface with a neural network. The metasurface exhibits strong dispersion properties, which are used to encode spectral information, while the neural network is designed to reconstruct high-quality hyperspectral images based on that encoded data (Figure 23c). The phase distribution of the proposed metasurface is defined as a combination of a hyperboloidal phase and a polynomial phase given below:^[199]

$$\Phi(r) = 2\pi \frac{f - \sqrt{r^2 + f^2}}{\lambda} + \sum_{i=0}^n a_i \left(\frac{r}{R}\right)^{2i} \quad (25)$$

The first term represents the hyperboloidal phase, which generates a spherical wavefront. The second term is the polynomial phase, where the coefficients a_i are optimized jointly with the neural network to improve hyperspectral image recovery. Their design represents a significant step forward in HSI technology, particularly for compact and portable applications. By co-optimizing the optical and computational components, the system delivers high-quality image reconstructions in a small form factor. However, challenges related to fabrication complexity, light efficiency, and the exploitation of metasurface capabilities remain areas for future work.

C. Zanhg et al., presented a method for high-resolution UV spectral imaging and bio-detection by leveraging quasi-BIC resonant modes in a carefully structured dielectric metasurface. The proposed design is based on a diamond-based ring-shaped metasurface designed to operate in the UV spectrum. By breaking the structural symmetry, they achieve high-Q quasi-BIC resonance, which improves local electromagnetic field intensities, making it highly responsive to external changes, such as the presence of biomolecules on the surface.^[201] The reported work opens up promising avenues for future research and applications in high-sensitivity UV spectral imaging and biomolecule analysis, particularly in medical diagnostics and biological research. However, the potential complexity of its fabrication process and noise sensitivity are areas that may need further exploration.

The work of Shaik et al., demonstrated an approach to long-wave infrared (LWIR) MSI by integrating plasmonic filters onto

a single thermal image sensor. The proposed approach provides a compact, and cost-effective solution capable of high-resolution multispectral imaging in the LWIR spectrum (8-14 μm), making it applicable to fields such as surveillance, medical diagnostics, and gas detection. A key aspect of their system is the super-pixel approach where each spectral band in the mosaic covers multiple pixels, allowing for a higher collection efficiency and facilitating data acquisition at specific wavelengths without sacrificing spatial resolution. They also developed a novel demosaicing algorithm to reconstruct multispectral images from the mosaic-patterned data, which utilizes a graph-based interpolation method to estimate missing spectral information and enhance image resolution. While the method offers high transmission efficiency and robust image reconstruction, limitations such as noise sensitivity, alignment precision, and limited spectral tunability remain challenges to be addressed.^[202]

In 2024, Yang et al. introduced an approach to angle insensitive spectral imaging using a combination of plasmonic structures and a topology-optimization technique. The proposed approach incorporates a novel metasurface design, leveraging a combination of freeform shape optimization and a topology-optimization technique to achieve angle insensitivity in the visible spectrum (Figure 24a). The proposed optimization process differs from conventional methods that rely on regular, symmetrical shapes, which are typically more angle-sensitive. The metasurface's response is modeled as a matrix equation that relates the detected signals, transmission spectra, and measurement noise. Specifically, the equation is given as:^[203]

$$I = Tf + e \quad (26)$$

where I is the vector of the measured intensities across different metasurface units, T is the transmission matrix characterizing each metasurface unit's response across the spectrum, f is the vector representing the unknown to be recovered, and e denotes the measurement noise. The proposed system reconstructs the spectral information by solving for f using data from multiple angle-robust metasurface units, and the objective includes an optimization function aimed at ensuring angle insensitivity and minimizing spectral overlap among metasurface units. The optimization function is expressed as:

$$\text{minimize } \alpha \cdot \frac{1}{N} \sum_{i=1}^N |T_{i1} - T_{i2}| + \frac{1}{N(N-1)} \sum_{i \neq j}^N |T_i - T_j| \quad (27)$$

where α is the weight factor, N is the number of transmission spectra, T_{i1} and T_{i2} are the transmission spectra of a given metasurface unit at incident angle of 0° and 15° , and T_i and T_j are the normalized transmission spectra of different metasurface units under normal incidence. Equation 27 seeks to reduce the angular dependency by minimizing the spectral differences between different incident angles, while also maximizing the spectral diversity across metasurface units. This combined approach allows the metasurface to achieve angle-insensitive spectral imaging by balancing angle robustness and spectral resolution. While challenges in fabrication and noise resilience remain, the approach represents a significant step forward in developing practical, portable spectral imaging systems with a broader field of view.

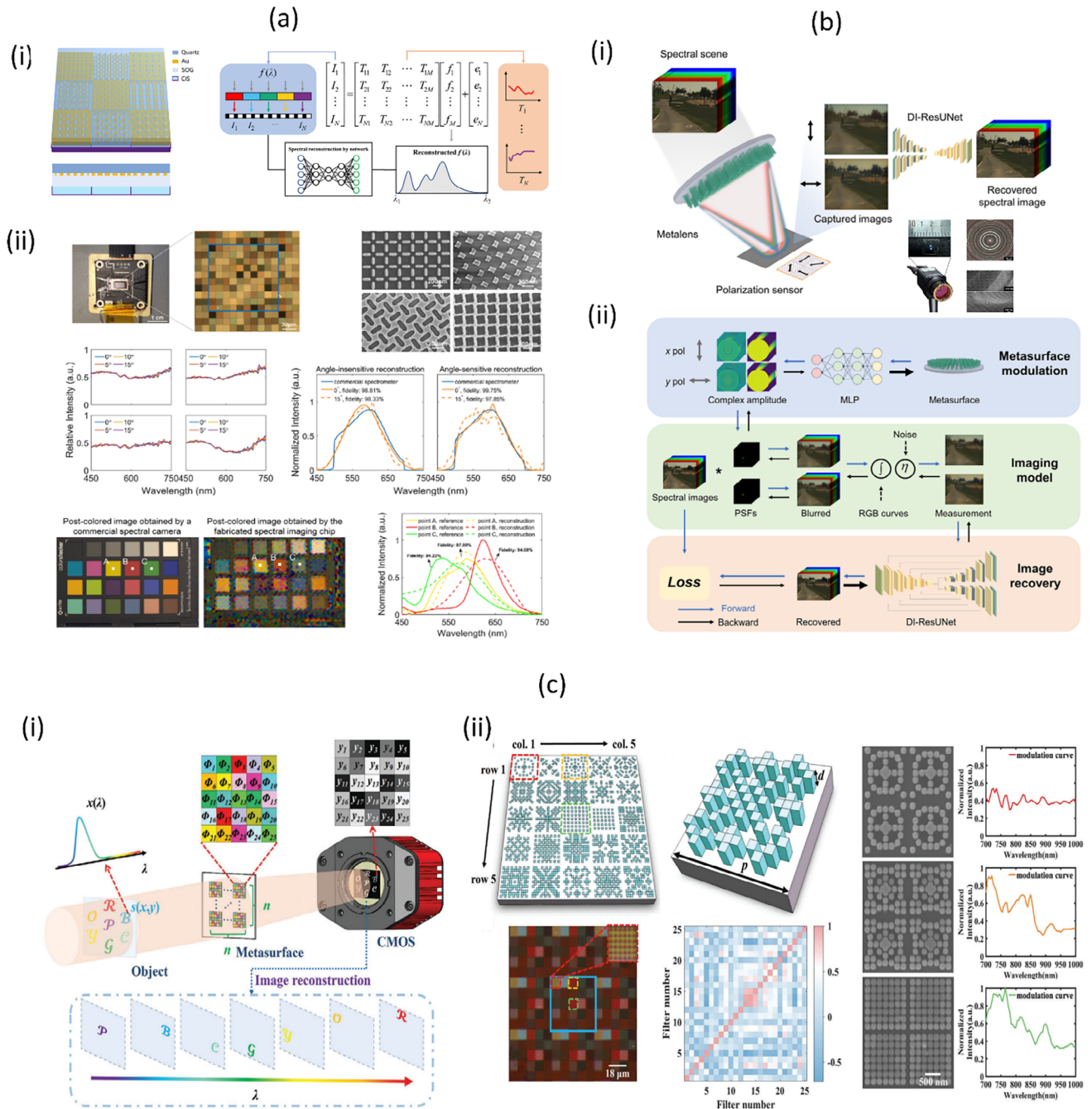


Figure 24. a) Metasurface-based HSI scheme including i) a schematic showing the metasurface integrated with a CMOS sensor and the fundamentals of the reconstruction technique, and ii) the micrograph of the fabricated device, SEM image metasurface unit, the transmission characterization of the four metasurface units, comparison between commercial spectral camera and their proposed device, and the reconstructed spectra. Reproduced with permission.^[203] Copyright 2024, John Wiley, and Sons. b) A co-designed neural-optic system for polarization-multiplexed spectral imaging, highlighting i) a 3D model of the proposed device, and ii) the complete spectral imaging process including three key stages: metasurface modulation, imaging model, and image reconstruction. Reproduced with permission.^[204] Copyright 2024, John Wiley, and Sons. c) i) Overview of snapshot spectral imaging using a C4 symmetric metasurface. A colorful object is modulated through the metasurface containing an array of $n \times n$ spectrometer pixels and captured by the sensor as a grayscale image. ii) design and evaluation of the proposed metasurface unit structure. Reproduced with permission.^[205] Copyright 2024, John Wiley, and Sons.

An approach combining a polarization-multiplexed metalens with a neural network framework, resulting in an end-to-end computational spectral imaging system was demonstrated by Zhang et al.^[204] The metalens encodes both the spectral and spatial information of a scene by modulating the light's polarization channels. Specifically, it manipulates orthogonal polarization states (x - and y -polarized channels), enabling distinct spectral data capture through a single optical element. The system reconstructs the full spectral image using a Dual-Input Residual U-Net (DI-ResUNet) neural network, which decodes the polarization-encoded RGB images back into a spectral image (Figure 24b). This fully differentiable system allows for joint optimization of both the metalens structure and the neural network parameters, thereby enhancing the optical encoding process and computational spectral recovery. The metalens modulates the incident light to produce a polarization- and wavelength-dependent PSF, which serves as the encoding kernel for spectral reconstruction, and is described mathematically as:

$$\text{PSF}_{\text{pol},\lambda} = \int F(P_{\text{pol}}, \rho, \lambda, f) d\rho \quad (28)$$

where $\text{PSF}_{\text{pol},\lambda}$ is the polarization-specific point spread function for a given wavelength (λ), P_{pol} represents the complex amplitude response of the metalens under different polarization, ρ is the radial coordinates, and f is the focal length of the metalens. The RGB image formation process is described below, which models the convolution of the spectral image with the polarization-specific PSF and the camera's sensitivity:

$$J_{\text{pol},\text{RGB}} = \int F(I_{\lambda} * \text{PSF}_{\text{pol},\lambda}) \cdot S_{\lambda,\text{RGB}} d\lambda + n \quad (29)$$

Here, $J_{\text{pol},\text{RGB}}$ is the captured RGB image under a specific polarization state, I_{λ} is the original spectral image at wavelength λ , $S_{\lambda,\text{RGB}}$ is the spectral sensitivity function of the camera sensor, and n denotes the noise. The neural network reconstructs the spectral image using a loss function that minimizes the difference between the ground truth spectral image I_{λ} and the reconstructed spectral image $\hat{I}_{\lambda}$ given as:

$$\text{Loss} = \|I_{\lambda} - \hat{I}_{\lambda}\|_2^2 \quad (30)$$

While their approach offers significant benefits in terms of system miniaturization, efficiency, and spectral fidelity, challenges such as complex metalens fabrication and noise sensitivity persist. Despite these limitations, this technology marks a significant advancement in the miniaturization of SIS and holds promise for diverse applications, ranging from mobile devices to augmented reality.

Similarly, He et al., demonstrated NIR-SIS that combines a metasurface with a Meta-Attention Network (MAN) for spectral reconstruction.^[205] The proposed system is designed by utilizing a metasurface with 25 C4-symmetric structures that modulate light and capture spectral information, offering lightweight solution to traditional bulky spectral imaging systems (Figure 24c). The MAN with Iterative Denoising Reconstruction (MAN-IDR) algorithm allows for the reconstruction of high-quality NIR spectral images from a single grayscale image, significantly reducing hardware size while maintaining spectral accuracy. The system

has promising applications in mobile devices and areas such as health diagnostics and aerospace. The imaging process is summarized in Equation (31), where the spectral information $x(\lambda)$ at a spatial point $s(x, y)$ is modulated by a set of 25 filters $\phi_i(\lambda)$. The result is captured as a grayscale value y_i on the CMOS sensor, where $\eta(\lambda)$ represents the quantum efficiency of the sensor, and n_i is noise during the measurement.

$$y_i = \int_{\lambda_1}^{\lambda_2} x(\lambda) \phi_i(\lambda) \eta(\lambda) d\lambda + n_i \quad (31)$$

$$y = \Phi x + c \quad (32)$$

Equation (31) delineates the relationship between the captured grayscale image, the system's modulation curves, and the spectral image to be reconstructed. The vectorized version of the grayscale measurements y , the modulation curves Φ , and the spectrum x are represented in matrix form in Equation (32). Despite some limitations related to incident angle tolerance and fabrication complexity, the system holds considerable promise for a wide range of applications.

Yang et al., demonstrated an approach to a reconfigurable HSI system using metallic metasurfaces with gradient geometry that operates over a broad wavelength range of 400–800 nm. The proposed metasurfaces consist of periodic nanostructures with varying dimensions (Figure 25a), enabling them to encode spectral information into transmitted light patterns, and the gradient in the geometry of these structures allows for capturing different light intensities, which can be processed to reconstruct the spectral images. The captured images are processed using a monochromatic pattern matching algorithm combined with post-calibration, enabling fast spectral reconstruction while reducing computational complexity and facilitating easy calibration. Their work presents a promising approach for compact HSI using gradient geometry metasurfaces, focusing on miniaturization, cost reduction, and defect tolerance. However, mitigating challenges such as alignment sensitivity and noise susceptibility requires further exploration.^[206]

The work of Du et al. demonstrated an approach to achieve full-Stokes polarizing-spectral imaging (PSI) using a metasurface-based SIS. They designed a metasurface with a metal-insulator-metal (MIM) structure featuring anisotropic detour phases to generate a polarizing-spectrally dependent point spread function (PSF) (Figure 25b). This enables simultaneous spectral and polarization information capture in a compact setup, offering advantages over traditional polarimetry systems which require larger, time-divided elements. The output electric field (Jones vector) based on the input electric field and the transformation matrix induced by the metasurface can be modelled as:^[207]

$$\begin{aligned} \begin{bmatrix} E_{\text{out}, x(r)} \\ E_{\text{out}, y(r)} \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} F_x(f) & 0 \\ 0 & F_y(f) \end{bmatrix} \begin{bmatrix} E_{\text{in}, x} \\ E_{\text{in}, y} \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} F_x(f) \cdot E_{\text{in}, x} + F_y(f) \cdot E_{\text{in}, y} \\ F_x(f) \cdot E_{\text{in}, x} + F_y(f) \cdot E_{\text{in}, y} \end{bmatrix} \Bigg|_{f=\frac{r}{\lambda \cdot f_{\text{focus}}}} \end{aligned} \quad (33)$$

$$I_{\text{image}}(r) = \sum_{j=1}^n \sum_{i=0}^3 F_i(r, \lambda_j) * S_i(r, \lambda_j) \quad (34)$$

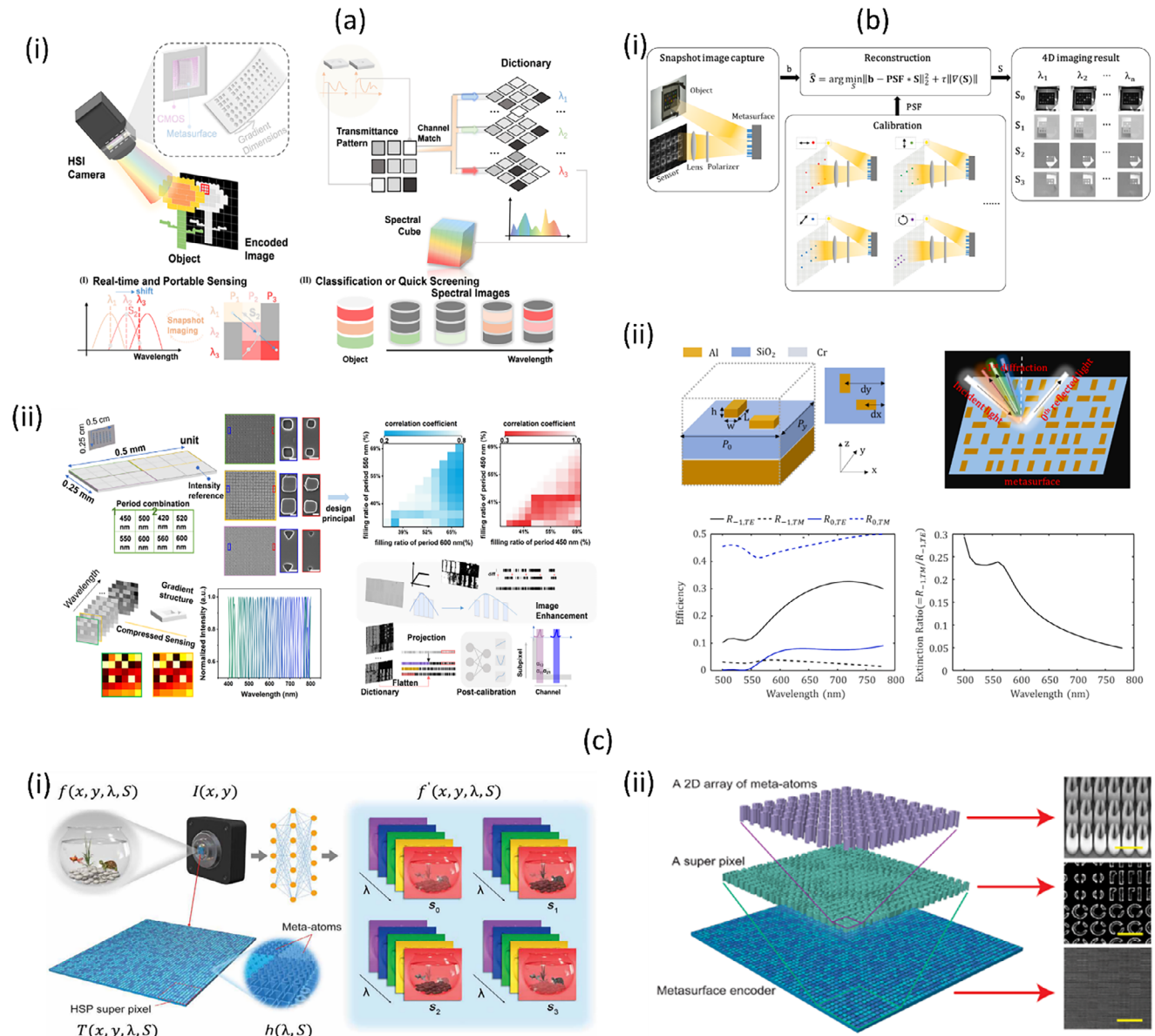


Figure 25. a) Reconfigurable metasurface-based HSI, including i) a reconfigurable snapshot hyperspectral imaging sensor which combines a monochromatic camera with a gradient geometry metasurface, enabling hyperspectral imaging. This system supports real-time spectral mapping for portable sensing and surface imaging, useful for rapid screening and classification based on light reflection differences at various wavelengths. ii) The gradient geometry metasurface design involves silicon templates with varied periodic structures, achieving different transmission spectra by adjusting nanohole dimensions and filling ratios. Spectral reconstruction uses simulated patterns and an orthogonal projection algorithm for accurate spectral data extraction under monochromatic light. Reproduced with permission.^[206] Copyright 2024, American Chemical Society. b) The Full-Stokes PSI using a metasurface, including i) The proposed system captures light diffracted by a metasurface, passing it through a polarizer and lens to form an image on a sensor. Calibration and reconstruction yield a 4D data cube, with each pixel represented by its Full-Stokes parameters and spectral data, obtained by solving a nonlinear inverse problem with sparsity constraints. ii) Their meta-atom design consisting of two perpendicular nanorods that manipulate the incident electric field based on their displacement, enabling phase modulation within the meta period. The -1 st diffraction order carries the detour phase, showing high diffraction efficiency and extinction ratio for TE polarization, allowing effective polarization control. Reproduced with permission.^[207] Copyright 2024, Elsevier. c) The proposed HSP camera i) encoding a 4D scene with spectral and polarization details into a 2D intensity map, which is then decoded back to 4D using machine learning. This system combines encoding metasurfaces to capture wavelength and polarization and an ML algorithm to reconstruct the full image data. ii) Schematic of the metasurface and the SEM image of the fabricated metasurfaces. Reproduced under the terms of the CC-BY 4.0 license.^[208] Copyright 2024, American Association for Advancement of Science.

Here, $F_x(f)$ and $F_y(f)$ are the Fourier transforms of the detour phase carried by the meta-atoms aligned along the x and y directions, respectively. This describes how the metasurface modulates the incident light based on its polarization state. The final 2D image captured by the system is modeled by Equation (34), which is a convolution of the PSFs (F_i) and the Stokes parameters (S_i) over multiple spectral channels (λ_j). This forward model forms the basis for reconstructing the high-dimensional data cube containing spatial, spectral, and polarimetric information.

Recent advancements in multifunctional metasurfaces have significantly enhanced the capabilities of metasurface-based SIS, enabling the capture of detailed spectral and polarization data within a single and compact platform.^[211,212] These metasurfaces can be engineered and tuned to respond selectively to specific wavelengths, polarization states, or phase shifts. This sophisticated control is facilitated by innovations in nanofabrication, allowing each meta-atom to simultaneously encode multiple optical properties. For example, a single metasurface can generate distinct spatial intensity distributions for various wavelengths and polarization states, effectively compressing multidimensional light information into a single, easily analyzable snapshots. Moreover, the integration of computational approaches, particularly machine learning, has further expanded the potential of multifunctional metasurfaces.

Algorithms can decode the complex intensity patterns produced by the metasurface, reconstructing high-resolution hyperspectral and polarimetric images by identifying the encoded spectral and polarization information. For instance, Zhang et al. present an approach to real-time hyperspectral-polarimetric (HSP) imaging, leveraging a compact, machine-learning-enhanced metasurface integrated with a conventional camera. This system encodes spectral and polarization data from a single snapshot, allowing a neural network (NN) decoder to reconstruct HSP images at an impressive rate of 28 frames per second.^[208] The proposed metasurface design incorporates arrays of chiral meta-atoms and spectro-polarimetric superpixels, which uniquely encode intensity distributions for various wavelengths and polarization states into distinct patterns (Figure 25c). This encoded data is then reconstructed by the NN into the four Stokes parameters $S = [S_0, S_1, S_2, S_3]$, providing complete spectral and polarization information. The transmitted intensity I_i captured by the camera pixels behind the metasurface is given as:

$$I_i = \sum_{k=0}^3 \int_{\lambda_{\min}}^{\lambda_{\max}} T_i^{S_k}(\lambda) f^{S_k}(\lambda) d\lambda \quad (35)$$

where $T_i^{S_k}(\lambda)$ are wavelength-dependent transmission coefficients that modulate the spectral-polarimetric data. Through training, the NN learns to map the encoded snapshot to its underlying spectral and polarization characteristics, with initial retrieval of these features optimized by linear regression and least-squares method. Similarly, Chen *et al.* demonstrated a single multifunctional metasurface capable of simultaneously capturing spectral, polarization, and spatial information. Their design employs rectangular silicon nanopillars that function as half-wave plates, where it generates distinct focal points and vortex beams, mapping spectral and polarization information onto

these focal points for simplified simultaneous detection. The proposed device achieves an imaging resolution of up to 1.4 times the wavelength, making it effective for edge detection and high-resolution imaging tasks.^[209]

Recently, Audhkhasi et al. presented a compact and low-latency multispectral imaging system that avoids the computational complexity typical of conventional snapshot systems. The core concept relies on using engineered optical components to carry out spectral discrimination directly in the optical domain, thus enabling real-time imaging across eight spectral channels within the 400–900 nm range. The system architecture consists of two major components: a multiresonant transmission filter and an array of strongly chromatic metalenses. The transmission filter is fabricated using chirped $\text{Si}_3\text{N}_4/\text{SiO}_2$ DBRs, designed to produce narrowband transmission peaks. The filter isolates specific wavelength bands, and each band corresponds to one of the eight metalenses in the array. These metalenses, designed with a hyperbolic phase profile and made from square Si_3N_4 nanopillars on a SiO_2 substrate, individually focus light at their designated wavelengths, spatially separating the spectral slices of the scene at the sensor plane. Their experimental validation was conducted using two test objects: a standard color calibration chart and a scene of flowers and leaves. For both cases, the monochrome sensor captured spatially distinct multispectral images, which were later processed using modest post-processing steps like Wiener deconvolution and haze removal. Reconstructed RGB images based on the multispectral slices showed strong qualitative agreement with phone-captured photographs of the same scenes, thereby demonstrating the viability of the optical encoding approach. Their proposed system demonstrates a compact, real-time MSI approach that performs spectral separation entirely in the optical domain, eliminating the need for intensive computation. This method offers a scalable pathway to real-time MSI. It is also CMOS-compatible, alignment-free, and operates across a broad range, making it well-suited for portable applications. However, it suffers from reduced signal-to-noise ratio due to spectral power splitting, limited spatial resolution per channel due to parallel imaging on a single sensor, and some spectral crosstalk. Additionally, the system is fixed to predefined wavelengths and may require complex nanofabrication. Despite these limitations, it offers a promising low-latency alternative to conventional multispectral imagers.^[210]

Table 3 offers a comprehensive overview of the advancements in metasurface-based SIS over the last decade. As discussed earlier, these systems leverage the unique properties of metasurfaces to replacing bulky traditional optics with compact, multifunctional structures capable of advanced SIS for applications like environmental monitoring and biomedical diagnostics. The evolution of these systems is critically examined across several design parameters, including size, spectral range, spectral resolution, angular resolution, acquisition techniques, and system complexity. Each of these parameters plays a vital role in determining the system's performance and application potential, and these advancements highlights both the technical challenges and the progress made within this field. As seen from the table, miniaturization remained a cornerstone of metasurface-based SIS designs. The sizes of the systems ranges from $1 \times 1 \text{ mm}^2$ to $8 \times 6.4 \text{ mm}^2$, indicating a clear trend towards compact, portable systems that retain high-performance imaging capabilities. This reduc-

Table 3. Comparative overview of the advances in metasurface-based SIS designs.

Ref.	Weight [kg]	Size	NC	Spectral Range [μm]	SR[nm]	Spatial Resolution	AR [°]	Acquisition Techniques	Complexity
[186]	–	3 mm	–	~0.4–0.7	–	–	–	Snapshot	Moderate (interlaced nanofin arrays, metasurface fabrication)
[187]	0.2	8.5 mm ²	70	0.75–0.85	1.5	400 (SP)	0.075	Pushbroom	Moderate (single lithography, metasurfaces, four layers)
[188]	–	–	–	0.75–0.85	2	0.5 μm ² per pixel	–	Snapshot	High (requires high-Q resonators, fine-tuned parameters, and computational post-processing)
[189]	0.2	8 mm ³	20	~0.8–0.1	7	100 μm (EPS)	–	Snapshot	Medium to high (requires fabrication of multiple metasurface layers, precision alignment, and computational assembly of spectral data cubes)
[190]	–	–	–	0.54–0.75	14–16	~100 μm	–	Snapshot	High (requires complex stacking of layers, precision alignment, and computational processing for data extraction)
[191]	–	175 μm	–	–	8.5	–	0.2	Pushbroom	High (requires complex dispersion engineering, full-wave simulation, and phase optimization)
[195]	–	–	–	0.4–1	~5	–	0.4	Snapshot	High (requires metasurface design, end-to-end training, and real-time processing of high-dimensional data)
[193]	–	8 × 6.4 mm ²	–	0.45–0.75	0.5	356 × 436	–	Snapshot	High (requires complex fabrication, freeform pattern generation, and computational reconstruction)
[194]	–	8 × 6.4 mm ²	26	0.45–0.75	0.5	–	0.25	Snapshot	High (requires deep-learning algorithm training, high-speed reconstruction, and noise management)
[197]	–	6 × 6 mm	18	0.5–0.65	10	–	0.1	Snapshot	High (requires computational learning algorithms, metasurface design, and real-time processing with small data)
[200]	–	–	–	0.4–0.7 and 2.5–15	–	100 × 500	–	–	Medium to high (requires phase-change material manipulation, laser patterning, and multispectral modulation)
[198]	–	–	25	0.4–0.64	10	~3.45 μm	–	Snapshot	High (PCM, polarization camera, computation)
[199]	–	8 × 6 mm ²	–	0.5–1.0	–	–	0.5	–	High (requires joint optimization of hardware and software, computational complexity, and metasurface phase tuning)
[201]	–	6 × 6 mm ²	10	0.2–0.4	1	–	0.2	–	High (requires BIC mode design, material-specific tuning, and high-Q resonance optimization)
[202]	–	–	3	8–14	1000	68 × 68 μm per pixel	–	Staring	Medium to high (requires fabrication of plasmonic filter mosaics and algorithmic image reconstruction)
[203]	–	8 × 6.4 mm ²	6	0.45–0.75	5	–	0.2	Snapshot	High (requires optimization via genetic algorithms, computational processing, and fabrication)
[204]	–	–	6	0.55–0.75	10	–	0.2	Snapshot	High (requires neural network-based optimization and polarization multiplexing in compact setup)
[205]	–	–	25	0.85–1.00	0.05	1.5 μm per pixel	–	Snapshot	High (requires deep-learning-based algorithm optimization and integration of metasurfaces with compact sensors)
[206]	–	–	–	0.4–0.8	1.99	75 × 135 super-px	–	Snapshot	Moderate (simplified calibration, lightweight computing, fast reconstruction)
[208]	–	1.8 × 1.8 mm ²	5–7	0.7–1.15	0.23	42 × 50 super-px	–	Snapshot	High (requires ML training, complex NN architecture, but enables real-time 28 FPS video)
[209]	–	–	6	1.52–1.62	2.5	–	–	Snapshot	Moderate (requires intensity analysis of focal points and vortex beams, no ML, but complex coupling limits full polarization-spectral extraction)
[207]	–	1 × 1 mm ²	12	0.565–0.685	2.4	–	–	Snapshot	Moderate (requires processor for reconstruction, but uses lightweight GS algorithm)
[210]	–	–	8	0.4–0.9	–	–	–	Snapshot	Low (minimal computation, Wiener deconvolution and haze reduction, no ML)

NC: Number of channels, SR: Spectral Resolution, AR: Angular Resolution, SP: Spatial Pixels, EPS: Effective Pixel Size, PX: Pixels.

tion in size is a direct consequence of the utilization of metasurfaces, which enable the integration of highly functional optical elements in a small footprint. Such systems are well-suited for applications in handheld devices, drones, and mobile sensing platforms, where space constraints and portability are paramount. Despite the small size, these systems are able to maintain impressive spectral imaging performance, which would otherwise require much larger, bulkier components. The spectral range of these systems predominantly covers the visible and NIR regions (approximately 0.4 to 1.0 μm), optimal for a broad range of imaging and sensing applications. Some of these systems, however, extend beyond the traditional visible spectrum into the mid-infrared and far-infrared ranges, which covers 0.4–0.7 μm and extends up to 2.5–15 μm .^[200] The ability to manipulate and optimize metasurfaces across such extended spectral ranges greatly enhances the capability of these systems, enabling them to be applied to chemical sensing, material characterization, and even thermal imaging.

The spectral resolution, which defines the system's ability to distinguish closely spaced wavelengths, varies substantially across the system designs, from 0.05 nm to 1000 nm. Systems that achieve higher spectral resolution are often characterized by the integration of high-Q resonators and fine-tuned metasurfaces, which require intricate fabrication techniques and advanced computational post-processing. These systems excel in applications where precise wavelength differentiation is required, such as in detailed chemical analysis or spectral imaging of biological tissues. In contrast, designs that prioritize faster data acquisition,^[210] trade-off some spectral precision for broader spectral coverage and faster imaging speeds, making them suitable for real-time monitoring and environmental sensing. Angular resolution plays a critical role in determining the system's ability to resolve fine spatial details in angular space. The systems exhibit angular resolution values ranging from 0.075° to 0.5°, with higher angular resolution (AR) values typically achieved by systems requiring joint optimization of hardware and software. This includes metasurface design, phase tuning, end-to-end training, and real-time processing of high-dimensional data, all contributing to increased computational complexity.^[195,199]

Achieving high AR is crucial for applications in remote sensing, optical tomography, and target tracking, where small angular discrepancies must be discerned to ensure high accuracy. However, these high AR designs often come at the cost of increased system complexity, presenting significant challenges in scaling up these technologies for practical use. In terms of acquisition techniques, the majority of the systems utilize snapshot imaging, which enables instantaneous capture of full spectral images without the need for scanning. This approach is particularly beneficial in dynamic environments, where speed is critical, and real-time processing is required. Snapshot systems are increasingly employed in many applications, where rapid data capture is necessary for immediate analysis. Conversely, pushbroom techniques involve sequential line-by-line scanning of the scene.^[187] While slower than snapshot systems, pushbroom systems are advantageous for applications requiring high spatial fidelity over large areas, such as in satellite-based imaging or large-scale environmental surveys.^[191]

The complexity of metasurface-based SIS designs has increased alongside advancements in performance. Systems re-

quiring high spectral resolution, angular precision, and real-time processing often necessitate advanced fabrication processes and sophisticated computational algorithms. For instance, designs employing high-Q metasurfaces, advanced material manipulation, and machine learning-based post-processing to achieve optimal performance have been demonstrated.^{[188],[197]} These complex systems enable precise imaging but involve trade-offs in terms of fabrication difficulty, computational requirements, and cost. Other designs utilize simpler processing algorithms, such as Wiener deconvolution, for noise reduction and image reconstruction, which makes them easier to implement and computationally efficient.^[210] These simpler systems are often preferred for applications prioritizing real-time performance and low computational demand over the highest possible precision.

In summary, over the past decade, metasurface-based compact SIS have evolved towards ever greater miniaturization, precision, and real-time performance. As device footprints shrink to the millimeter scale, the volume and complexity of the data they generate have grown in parallel, necessitating increasingly sophisticated computational frameworks. In particular, the integration of machine learning and AI-driven algorithms—for tasks such as optical optimization, spectral and spatial reconstruction, and adaptive decision making—has become essential, as exemplified by recent approaches.^[194,205]

These data-centric methods enable on-the-fly calibration and correction, thereby extending the practical utility of spectral imaging in dynamic environments. Looking forward, continued advances in both metasurface fabrication and computational processing will be critical for unlocking the next generation of SIS performance. Improvements in meta-atom design and nanofabrication precision promise higher spectral and angular resolutions without a proportional increase in device complexity. At the same time, more efficient machine-learning architectures—perhaps leveraging physics-aware priors or lightweight neural networks—can mitigate the computational burden associated with real-time spectral reconstruction. Recent work by Zhang et al. combines tailored metasurface layouts with deep-learning-based signal processing, illustrating how such hybrid approaches can enhance both resolution and processing speed.^[208]

Despite this progress, significant challenges remain. Scalable manufacturing of high-quality metasurfaces continues to be costly, and variations in fabrication tolerances can introduce noise and systematic errors that hinder system performance. Moreover, real-time computational demands, particularly for high-frame-rate applications, can exceed the capabilities of onboard processors, limiting deployment in resource-constrained settings. Addressing these limitations will require a concerted effort to develop cost-effective lithography techniques, robust design-for-manufacturing strategies, and specialized hardware accelerators for spectral data processing. Thus, the synergy between metasurface innovation and advanced computational methods is propelling compact SIS toward unparalleled capabilities in fields as diverse as environmental monitoring, medical diagnostics, and remote sensing. By refining both the physical design of metasurfaces and the algorithms that interpret their data, researchers can overcome current cost and performance barriers, paving the way for ubiquitous, multidimensional optical imaging solutions.

4. Discussion and Future Outlook

The preceding sections have traced the migration of SIS technology from laboratory benches to millimetre-scale payloads, enabled by breakthroughs in nanophotonics, precision fabrications and computational optics. Four hardware paradigms now coexist—DIY assemblies, free-form optics, filter-on-chip architectures and metasurface platforms—each optimised for a different balance of cost, volume, optical fidelity and on-board processing demand. Parallel advances in machine-learning pipelines increasingly absorb the burdens of calibration, reconstruction and decision-making, turning raw spectral data cubes into actionable information in real time. This section first compares the strengths and limitations of the competing hardware strategies—from the accessibility of 3-D-printed prototypes to the functional density of metasurface based spectral imagers—and then identifies the technological inflection points likely to shape the next decade. For instance, DIY platforms have opened the field to non-specialists by exploiting 3-D printing and off-the-shelf optics. Their principal advantages are very low cost, design flexibility and strong educational value, which together accelerate rapid prototyping and community-driven innovation. However, mechanical tolerances of printed parts, labor-intensive calibration, and limited robustness under harsh environmental conditions reduce measurement accuracy and long-term stability. Consequently, DIY systems excel as proof-of-concept or teaching tools, but require additional engineering before they can meet the repeatability standards of industrial or clinical deployment. On the other hand, freeform optics spectral imagers consolidate dispersion, aberration correction and focusing into a single non-rotational surface. They deliver high image fidelity across wide fields while respecting stringent payload limits for cubesats and drones. As a result, they provide high image fidelity across wide fields of view while meeting stringent payload constraints, particularly in CubeSats and drones. Nevertheless, these benefits come at the cost of elevated fabrication expenses, extended lead times, and tight alignment tolerances that complicate field deployment. Therefore, such designs are most attractive in scenarios where maximizing performance per gram is essential and production volumes remain modest. Filter-on-Chip Architectures take a different approach by monolithically integrating Fabry–Pérot, photonic crystal, or plasmonic filters directly onto CMOS sensors. This eliminates moving parts and enables true snapshot acquisition. As a result, these systems offer exceptional SWaP efficiency, making them particularly well-suited for wearables and point-of-care diagnostics. However, challenges persist in scaling up nanofabrication over large areas, maintaining uniformity across layers, and suppressing second-order diffraction. Additionally, post-fabrication tuning is generally infeasible, limiting adaptability once the device is manufactured. Finally, Metasurface-Based Systems represent the cutting edge of optical miniaturization. Their subwavelength meta-atoms enable multifunctional operations—including spectral splitting, polarization control, and computational preprocessing—within an ultrathin form factor. When coupled with data-driven reconstruction algorithms, such systems can achieve video-rate, high-resolution spectral imaging. Nonetheless, they are constrained by fabrication challenges: electron-beam or nanoimprint lithography must maintain nanometre-scale accuracy

across centimetre-scale apertures. Moreover, fabrication-induced phase errors can introduce spectral noise, often necessitating more intensive computational correction at the device level. In summary, no single architecture is universally optimal. DIY platforms maximize accessibility and flexibility; freeform optics optimize optical quality per unit mass; filter-on-chip systems excel in mechanical simplicity and integration; and metasurfaces offer the highest functional density in the smallest footprint. Thus, selecting the most appropriate architecture ultimately depends on application-specific constraints—such as system size, cost, data throughput, and required calibration effort.

Looking ahead, the future of compact SIS lies at the intersection of hardware innovation and computational sophistication. Emerging developments are poised to redefine the capabilities and deployment contexts of spectral imaging. Computational spectral reconstruction is a major area of ongoing advancement. For instance, ML algorithms have already shown significant promise in spectral data interpretation, providing automated tools for classification, anomaly detection, and predictive modeling.^[213–218] Deeper integration of ML into SIS could lead to systems capable of real-time spectral analysis, potentially enabling “smart” imaging devices. For example, a handheld SIS device could use onboard ML models to analyze plant health or detect pollutants on the spot, offering instant feedback without needing post-processing on a separate system. Progress in real-time reconstruction algorithms will be crucial for dynamic environments. This shift toward edge intelligence will require co-optimization of hardware and software, especially for platforms with limited power and computing capacity such as UAVs and CubeSats. Tunable and Programmable Metasurfaces offer another promising avenue.

Future designs could adapt their spectral range and resolution in real time, responding dynamically to environmental conditions or task-specific demands—such as switching between broadband and narrowband modes for different medical or environmental applications. Embedding MEMS, electro-optic components, or tunable materials—combined with real-time computational control—could enable this adaptability with minimal increase in size or power consumption.^[219,220]

Expanding spectral coverage beyond the visible and near-infrared is also a key frontier. While current metasurfaces are largely confined to these bands, future work may extend their operation into the ultraviolet and mid-infrared, unlocking new applications in astronomy, chemical sensing, and security. Achieving this will depend on material innovations that deliver high optical efficiency across broad wavelength ranges while preserving nanoscale structural fidelity. As fabrication technologies continue to evolve, freeform optics could be increasingly customized to application-specific requirements, boosting resolution and efficiency in areas like precision agriculture, biomedical imaging, and remote sensing. Finally, the growing ecosystem of IoT and edge computing presents an opportunity for SIS integration into distributed networks. For instance, fleets of SIS-enabled drones or in-field sensors could capture and locally process spectral data, transmitting only essential insights to a central hub. This decentralized approach would facilitate scalable environmental monitoring, providing real-time feedback on soil health, crop conditions, or water quality across large areas.

5. Conclusion

In conclusion, this review has highlighted the substantial progress in compact SIS, focusing on advancements that have made the technology more accessible, affordable, and versatile. The transformation from traditional, bulky designs to miniaturized, portable solutions has paved the way for widespread applications, including environmental monitoring, precision agriculture, and medical diagnostics. Four principal hardware paradigms—DIY assemblies, free-form optical elements, filter-on-chip architectures and metasurface platforms—have each addressed specific bottlenecks in miniaturization, optical performance and cost. DIY devices have democratized rapid prototyping; free-form optics now consolidate dispersion and focusing in a single surface for weight-critical platforms such as CubeSats and UAVs; on-chip filter arrays embed spectral discrimination directly on CMOS sensors, offering unmatched mechanical robustness; and metasurfaces compress dispersion, beam shaping and even polarimetric analysis into ultrathin, multifunctional layers, enabling less than 1 cm³ payloads without sacrificing resolution. Moreover, advancements in computational imaging and machine learning have enhanced compact SIS by enabling high-quality image reconstruction from fewer measurements, thereby reducing hardware complexity. Machine learning further strengthens spectral analysis, offering efficient data interpretation, classification, and tailored insights for diverse applications. Notwithstanding these gains, significant challenges remain. Nanometer-tolerance fabrication of metasurfaces and filter stacks is still expensive and yield-limited, constraining commercial scalability. Trade-offs between spectral resolution, optical throughput and computational load persist across all compact designs: devices that maximise dispersion or channel count often pay a penalty in light efficiency or reconstruction latency. Noise sensitivity, calibration overhead and algorithmic robustness under real-world conditions continue to delimit performance envelopes, particularly for resource-constrained edge platforms. Looking ahead, three converging vectors are poised to redefine the SIS landscape over the next decade. First, algorithm–hardware co-design will place physics-aware neural networks in application-specific integrated circuits (ASICs), enabling sub-Watt, on-device spectral analytics for autonomous drones and handheld diagnostics. Second, dynamically reconfigurable metasurfaces that integrate MEMS actuation, phase-change materials or electro-optic tuning will permit real-time adaptation of passbands and angular response, blurring the line between sensor and processor. Third, scalable manufacturing and standardisation—via roll-to-roll nano-imprint, high-throughput deep-UV lithography and open calibration protocols—will be essential to translate laboratory prototypes into mass-market products.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

compact spectral imaging system, DIY spectral imagers, filters on chip, freeform optics, metasurface photonics, miniaturized spectrometers, SWaP-constrained sensors

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